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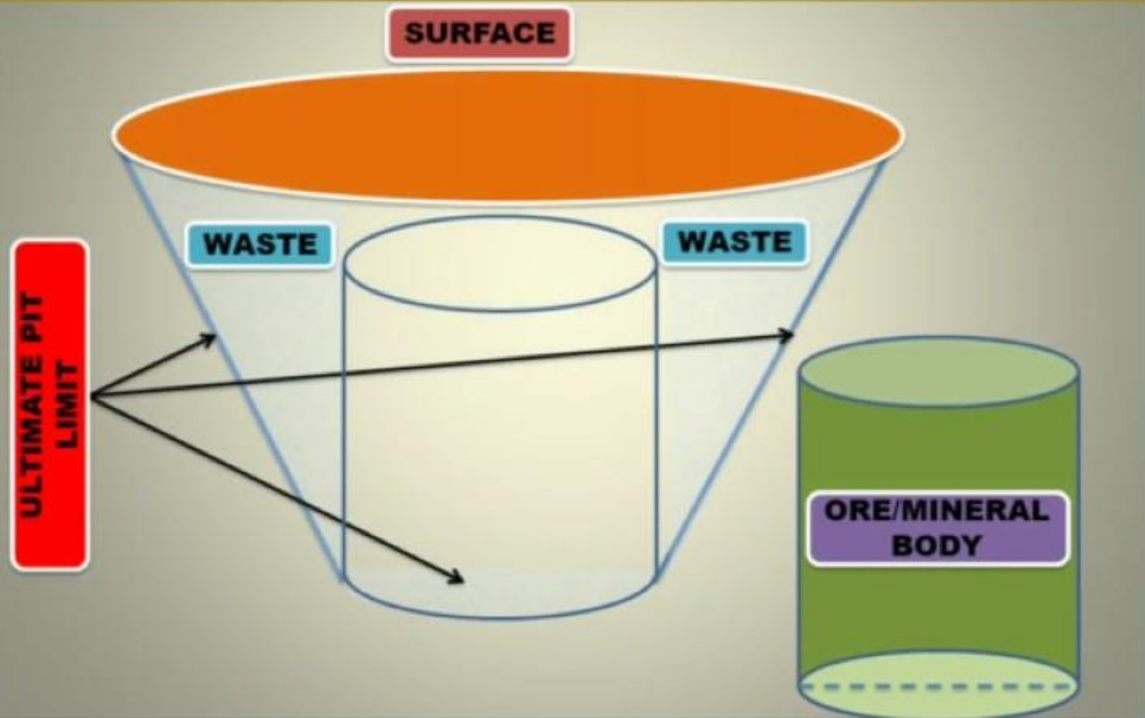
5.8.1 The RTZ open-pit generator

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TYPES OF STRIPPING RATIO

1. Volumetric Stripping Ratio
2. Thickness Stripping Ratio
3. Instantaneous Stripping Ratio
4. Average Stripping Ratio
5. Overall stripping Ratio
6. Break-Even Stripping Ratio
7. Limiting Stripping Ratio
8. Incremental Stripping Ratio

PIT LIMITS



STRIPPING RATIO

$$\text{STRIPPING RATIO} = \frac{\text{WASTE}(\text{m}^3)}{\text{ORE/MINERAL}(\text{te})}$$

$$\text{STRIPPING RATIO} = \frac{\text{WASTE}(\text{m}^3)}{\text{ORE/MINERAL}(\text{m}^3)}$$

$$\text{STRIPPING RATIO} = \frac{\text{WASTE}(\text{te})}{\text{ORE/MINERAL}(\text{te})}$$

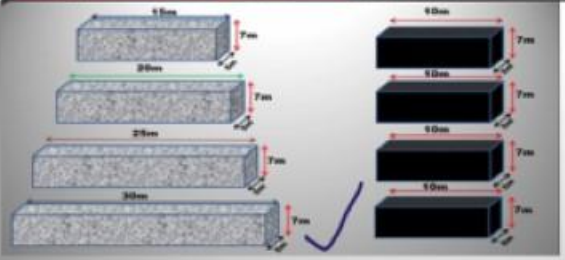
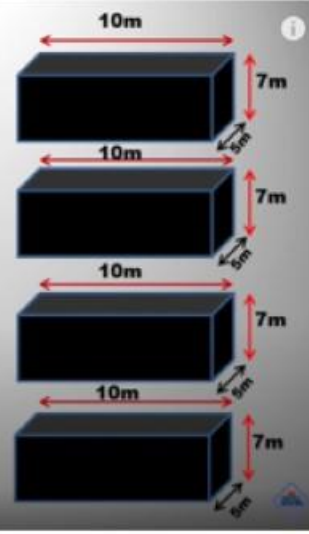
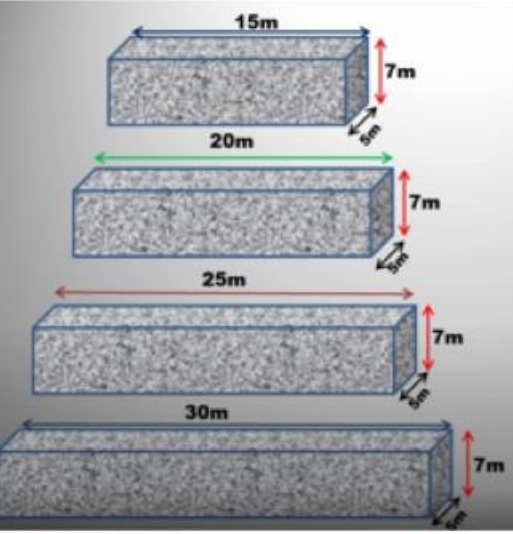
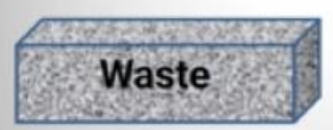
$$\text{STRIPPING RATIO} = \frac{\text{WASTE}(\text{m})}{\text{ORE/MINERAL}(\text{m})}$$

PIT LIMITS

INSTANTANEOUS STRIPPING RATIO

It is the ratio of volume of Waste in a given time duration & to the tonnage of Ore/Mineral obtained during the same giving time duration.

$$\text{INSTANTANEOUS STRIPPING RATIO} = \frac{\text{VOLUME OF WASTE(m}^3\text{) IN A GIVEN TIME DURATION}}{\text{TONNAGE OF ORE/MINERAL(te) IN A GIVEN TIME DURATION}}$$



Assume,
 T_w = Strip thickness of waste
 L_w = Strip length of waste
 W_w = Strip width of waste
 ρ_w = Density of waste
 T_c = Strip thickness of coal
 L_c = Strip length of coal
 W_c = Strip width of coal
 ρ_c = Density of coal

Let,
 $\rho_w = \rho_c = 1 \text{ Kg/m}^3$

$$\text{ISR}_{\text{STRIP 1}} = \frac{\text{WASTE(Te)}}{\text{COAL(Te)}}$$

$$\text{ISR}_{\text{STRIP 1}} = \frac{T_w \times L_w \times W_w \times \rho_w}{T_c \times L_c \times W_c \times \rho_c}$$

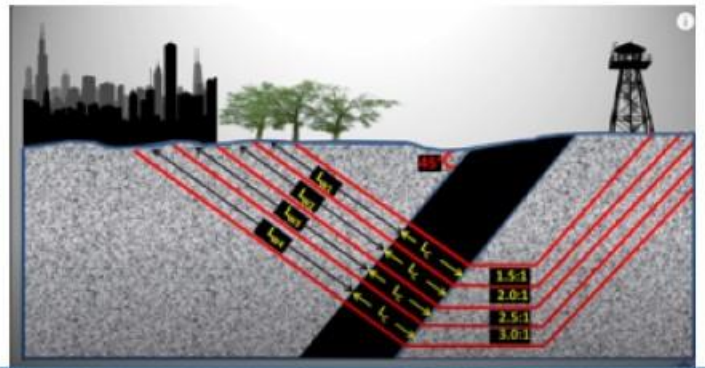
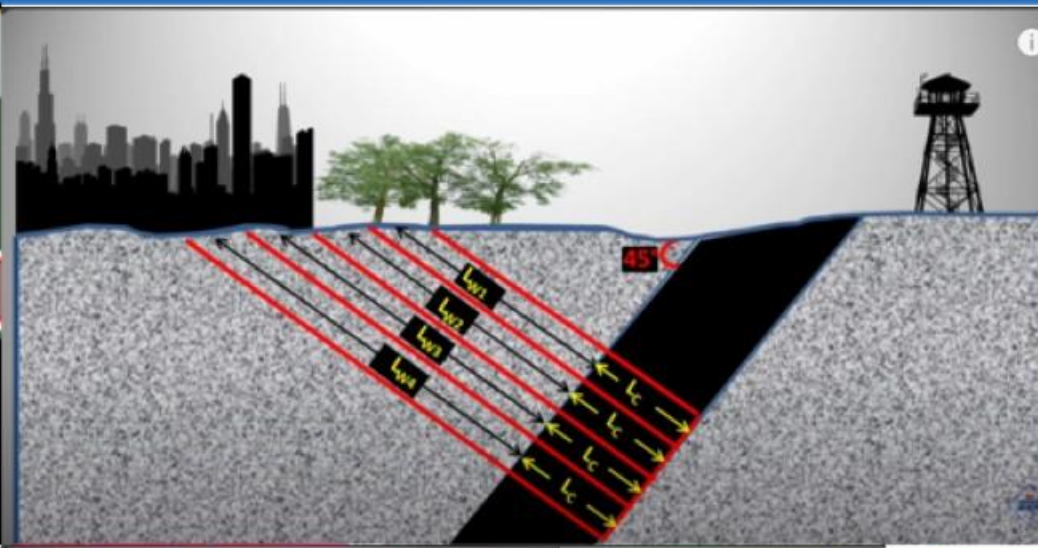
$$\text{ISR}_{\text{STRIP 1}} = \frac{7 \times 15 \times 5 \times 1}{7 \times 10 \times 5 \times 1}$$

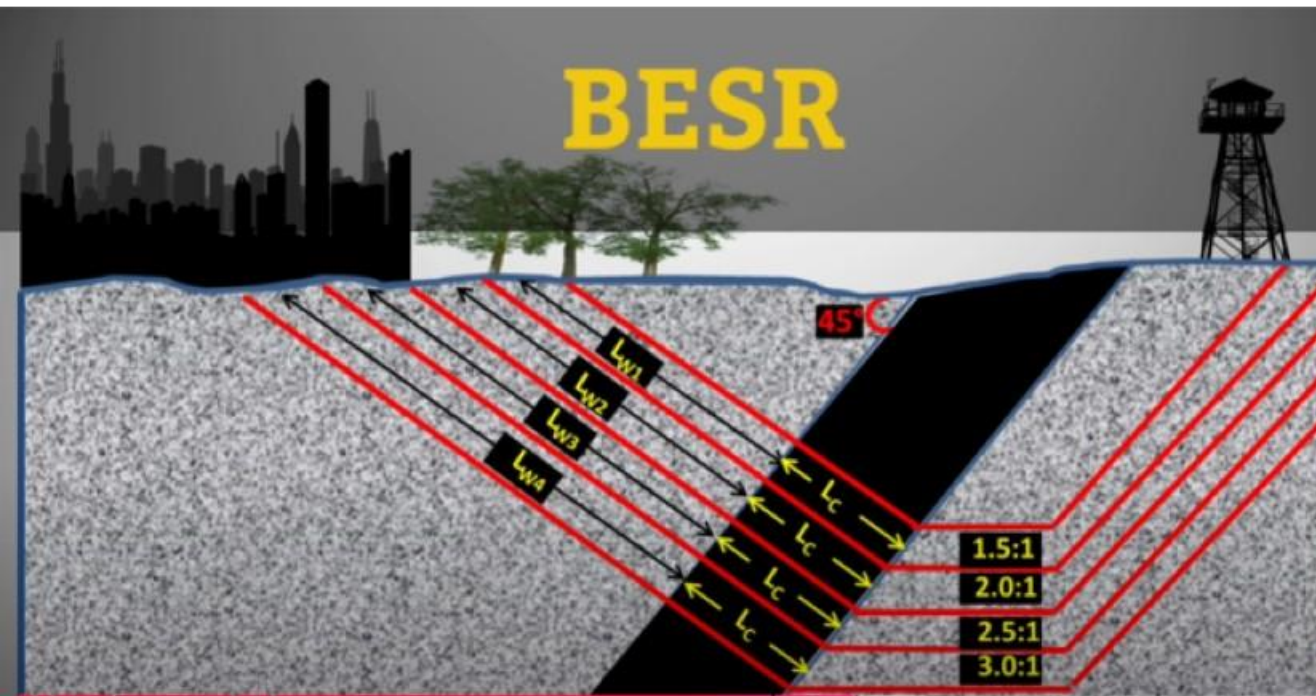
$$\text{ISR}_{\text{STRIP 1}} = \frac{L_w}{L_c}$$

ISR STRIP 1	=	$\frac{15}{10}$	=	1.5
ISR STRIP 2	=	$\frac{20}{10}$	=	2.0
ISR STRIP 3	=	$\frac{25}{10}$	=	2.5
ISR STRIP 4	=	$\frac{30}{10}$	=	3.0

$$\text{ISR}_{\text{STRIP 1}} = \frac{\sum_{i=1}^n T_{wi} \times L_{wi} \times W_{wi} \times \rho_{wi}}{T_c \times L_c \times W_c \times \rho_c}$$

i= Different waste material type





$$\text{BESR} = \frac{\text{SELLING PRICE OF ORE(COAL) PER TONNE} - \text{PRODUCTION COST PER TONNE OF ORE(COAL)}}{\text{STRIPPING COST OF REMOVAL OF WASTE ROCK PER m}^3}$$

$$\text{SR}_{\text{MAX}} = \frac{\text{SELLING PRICE} - \text{PRODUCTION COST (MINING COST)}}{\text{STRIPPING COST}}$$

* Selling Price = Production Cost + Stripping Cost

$$\text{SR}_{\text{MAX}} = \frac{\text{SELLING PRICE (₹ per te)} - \text{PRODUCTION COST (₹ per te)}}{\text{STRIPPING COST (₹ per m}^3\text{)}}$$

$\text{SR}_{\text{MAX}} > 1$ – Economical Mine
 $\text{SR}_{\text{MAX}} = 1$ – Breakeven (Marginal) Mine
 $\text{SR}_{\text{MAX}} < 1$ – Uneconomic Mine

$$\text{SR}_{\text{MAX}} = \frac{\text{₹/Te} - \text{₹/Te}}{\text{₹/ m}^3}$$

$$\text{SR}_{\text{MAX}} = \frac{\text{₹/Te}}{\text{₹/ m}^3}$$

$$\text{SR}_{\text{MAX}} = \frac{\cancel{\text{₹}}}{\text{Te}} \times \frac{\text{m}^3}{\cancel{\text{₹}}}$$

$$\text{SR}_{\text{MAX}} = \text{m}^3 / \text{te}$$

The feasibility study of an opencast mine revealed the following :

Recoverable value per tonne of ore = Rs. 900

Production cost per tonne of ore = Rs. 550

Stripping Cost per m³ of waste = Rs. 45

The break even stripping ratio will be-

$$\text{BESR} = \frac{900 - 550}{45} = 10$$

VOLUMETRIC STRIPPING RATIO

It is the ratio of volume of waste to the volume of mineral/ore.

Geometric Stripping Ratio

$$\text{VOLUMETRIC STRIPPING RATIO} = \frac{\text{VOLUME OF WASTE (m}^3\text{)}}{\text{VOLUME OF ORE/MINERAL (m}^3\text{)}}$$

THICKNESS STRIPPING RATIO

It is the ratio of thickness of waste to the thickness of mineral/ore.

$$\text{THICKNESS STRIPPING RATIO} = \frac{\text{THICKNESS OF WASTE(m)}}{\text{THICKNESS OF ORE/MINERAL(m)}}$$

INSTANTANEOUS STRIPPING RATIO

It is the ratio of volume of Waste in a given time duration & to the tonnage of Ore/Mineral obtained during the same giving time duration.

$$\text{INSTANTANEOUS STRIPPING RATIO} = \frac{\text{VOLUME OF WASTE(m}^3\text{) IN A GIVEN TIME DURATION}}{\text{TONNAGE OF ORE/MINERAL(te) IN A GIVEN TIME DURATION}}$$

AVERAGE STRIPPING RATIO

It is the ratio of volume of Waste up to a given time duration & to the tonnage of Ore/Mineral obtained up to the same giving time duration.

$$\text{AVERAGE STRIPPING RATIO} = \frac{\text{VOLUME OF WASTE(m}^3\text{)UP TO A GIVEN TIME DURATION}}{\text{TONNAGE OF ORE/MINERAL(te)UP TO A GIVEN TIME DURATION}}$$

It is also called CUMULATIVE STRIPPING RATIO.

OVERALL STRIPPING RATIO

It is the ratio of volume of Waste over the entire life of the mine & to the tonnage of Ore/Mineral obtained over the entire life of the mine.

$$\text{OVERALL STRIPPING RATIO} = \frac{\text{VOLUME OF WASTE(m}^3\text{) OVER THE ENTIRE LIFE OF THE MINE}}{\text{TONNAGE OF ORE/MINERAL(te) OVER THE ENTIRE LIFE OF THE MINE}}$$

BREAK-EVEN STRIPPING RATIO

Cutoff Stripping Ratio

It is the ratio of profit margin per tonne of ore mineral to the cost of stripping per cubic metre of waste.

$$\text{BESR} = \frac{\text{PROFIT MARGIN PER TONNE OF ORE/MINERAL (Rs./te)}}{\text{COST OF STRIPPING PER CUBIC METRE OF WASTE REMOVAL (Rs./m}^3\text{)}}$$

LIMITING STRIPPING RATIO

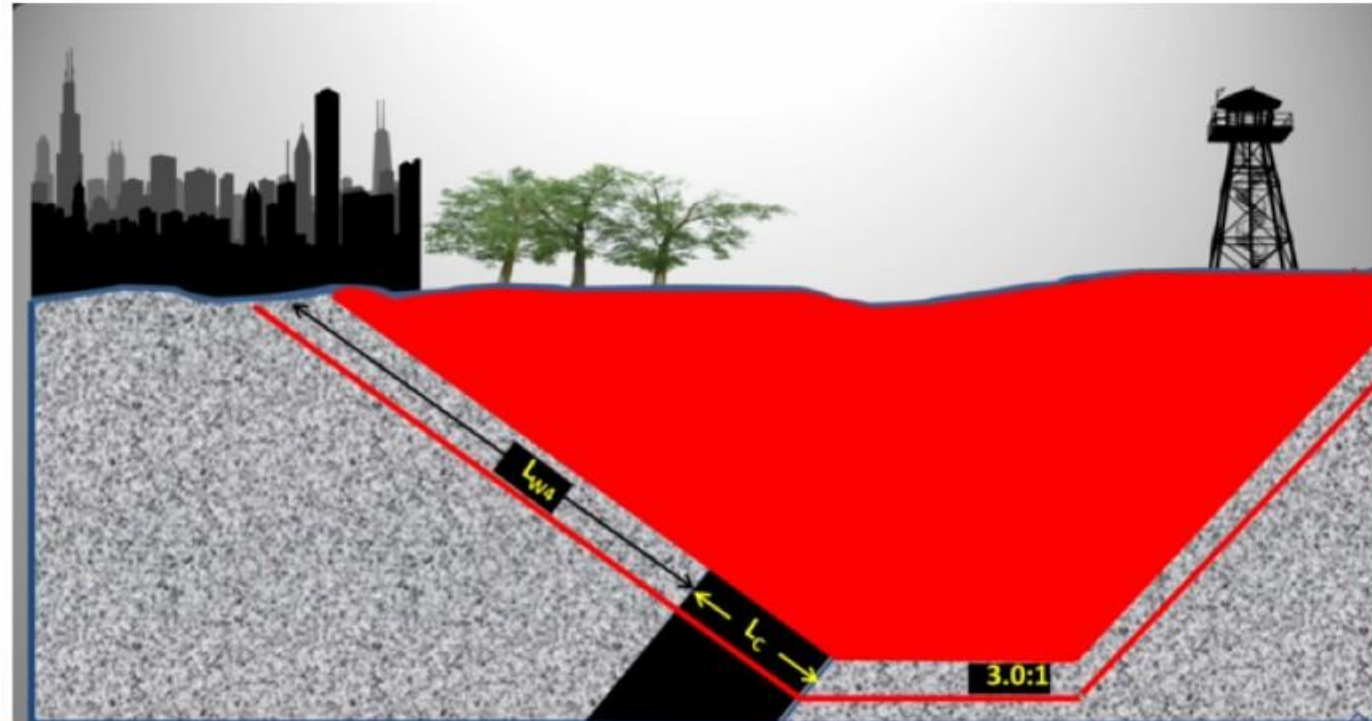
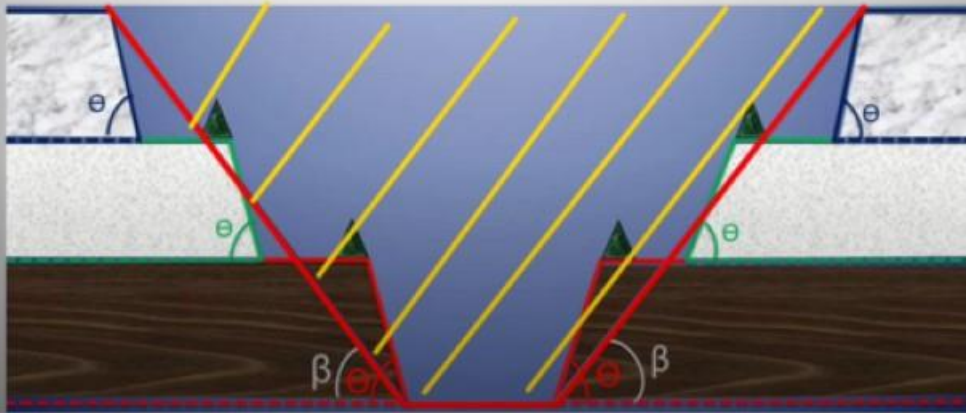
$$\text{LSR} = \frac{\text{PROFIT GAINED BY SURFACE MINING(Rs./te)} - \text{PROFIT GAINED BY UNDERGROUND MINING(Rs./te)}}{\text{COST OF STRIPPING PER CUBIC METRE OF WASTE REMOVAL(Rs./m}^3\text{)}}$$

PIT LIMITS



PIT LIMIT

Pit Limit is the maximum vertical or longitudinal length until which the mining of an ore deposit by surface mining, can be carried out with economic viability.



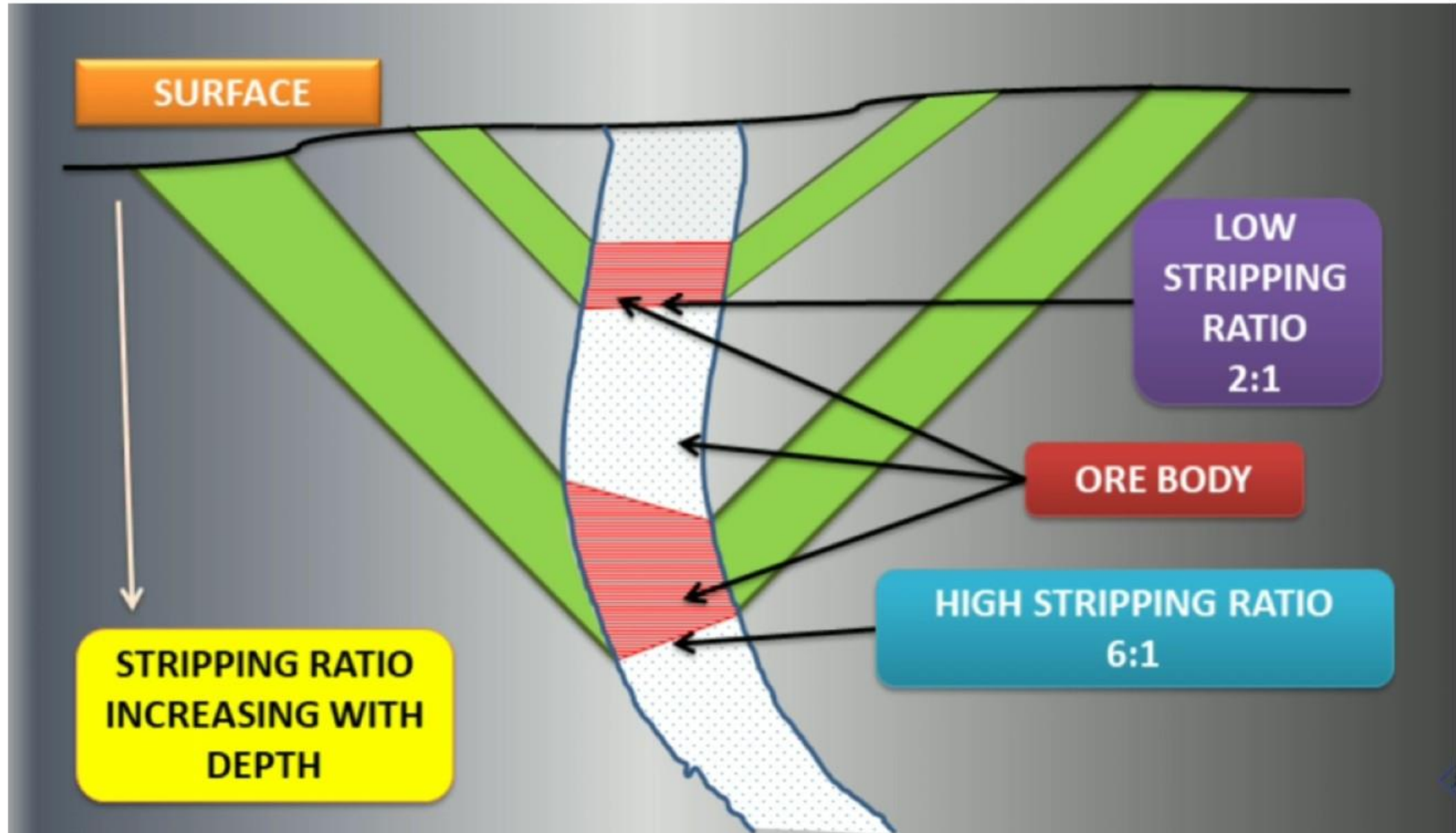
- * **Stripping ratio** gives the idea that how much quantity of waste will be removed in order to get 1 unit of ore/coal.
- * **Instantaneous Stripping ratio** gives the idea that how much quantity of waste will be removed in order to get 1 unit of ore/coal in a particular strip.
- * **Break even Stripping ratio** gives the idea that what maximum quantity of waste will be removed in order to get 1 unit of ore/coal so that no loss will occur to the firm.
- * **Ultimate pit limit** is the limit beyond which, mining will face loss.

$$ISR_{STRIP} = \frac{\sum_{i=1}^n T_{wi} \times L_{wi} \times W_{wi} \times \rho_{wi}}{T_c \times LC \times W_c \times \rho_c}$$

i= Different waste material type

$SR_{MAX} > 1$ – Economical Mine
 $SR_{MAX} = 1$ – Breakeven(Marginal) Mine
 $SR_{MAX} < 1$ – Uneconomic Mine

$$SR_{MAX} = \frac{\text{SELLING PRICE(₹ per te)} - \text{PRODUCTION COST(₹ per te)}}{\text{STRIPPING COST(₹ per m}^3\text{)}}$$



INTRODUCTION

- ✓ The time has now come to combine the economics introduced in Chapter 2 with the mineral inventory developed in Chapter 3 under the geometric constraints discussed in Chapter 4 to define the mineable portion of the overall inventory.
- ✓ The process involves the development and superposition of a geometric surface called a pit onto the mineral inventory.
- ✓ The mineable material becomes that lying within the pit boundaries.
- ✓ A vertical section taken through such a pit is shown in Figure 5.1.
- ✓ The size and shape of the pit depends upon economic factors and design/production constraints. With an increase in price the pit would expand in size assuming all other factors remained constant.
- ✓ The inverse is obviously also true.

INTRODUCTION

- ✓ The pit existing at the end of mining is called the 'final' or the 'ultimate' pit. In between the birth and the death of an open-pit mine, there are a series of 'intermediate' pits.
- ✓ This chapter will present a series of procedures based upon:
 - ✓ (1) hand methods,
 - ✓ (2) computer methods, and
 - ✓ (3) computer assisted hand methods

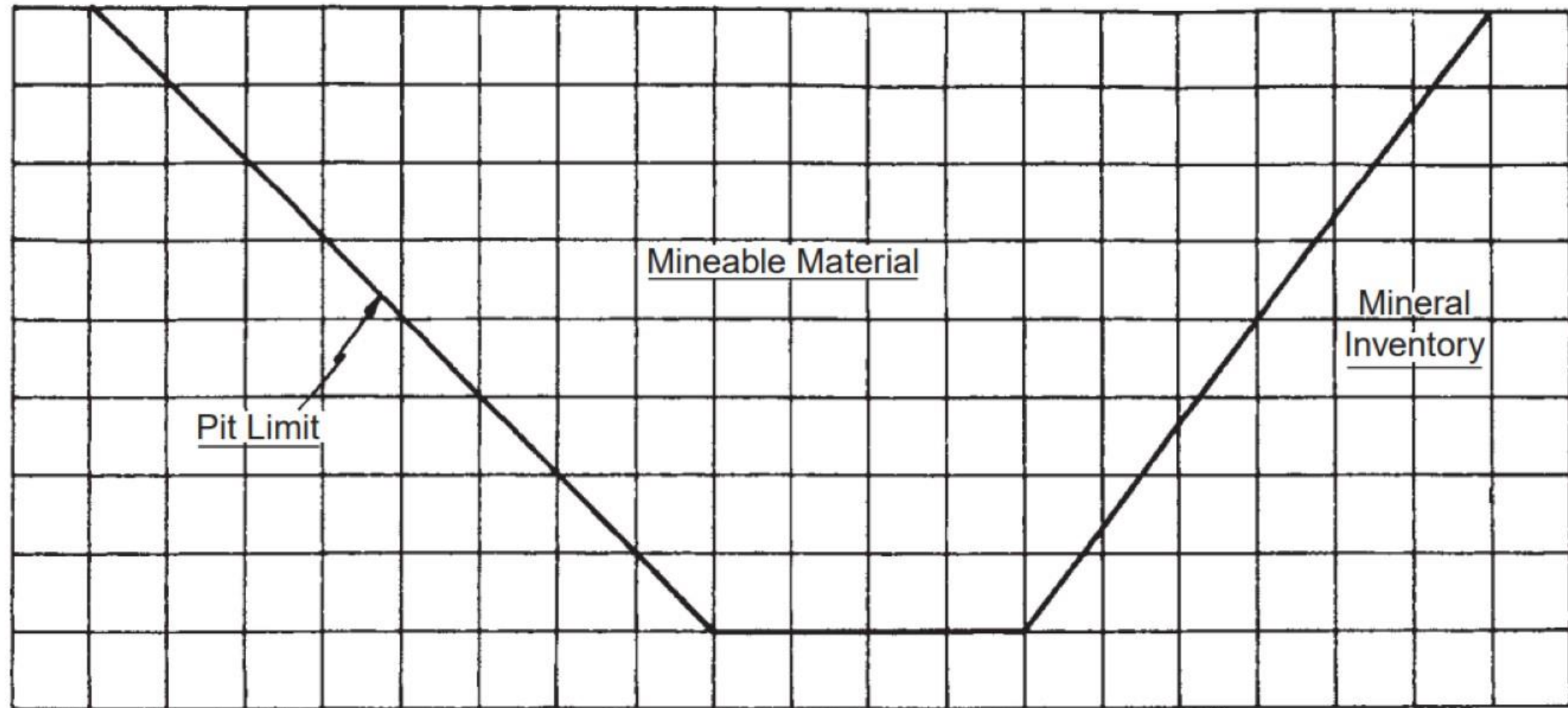


Figure 5.1. Superposition of a pit onto a mineral inventory.

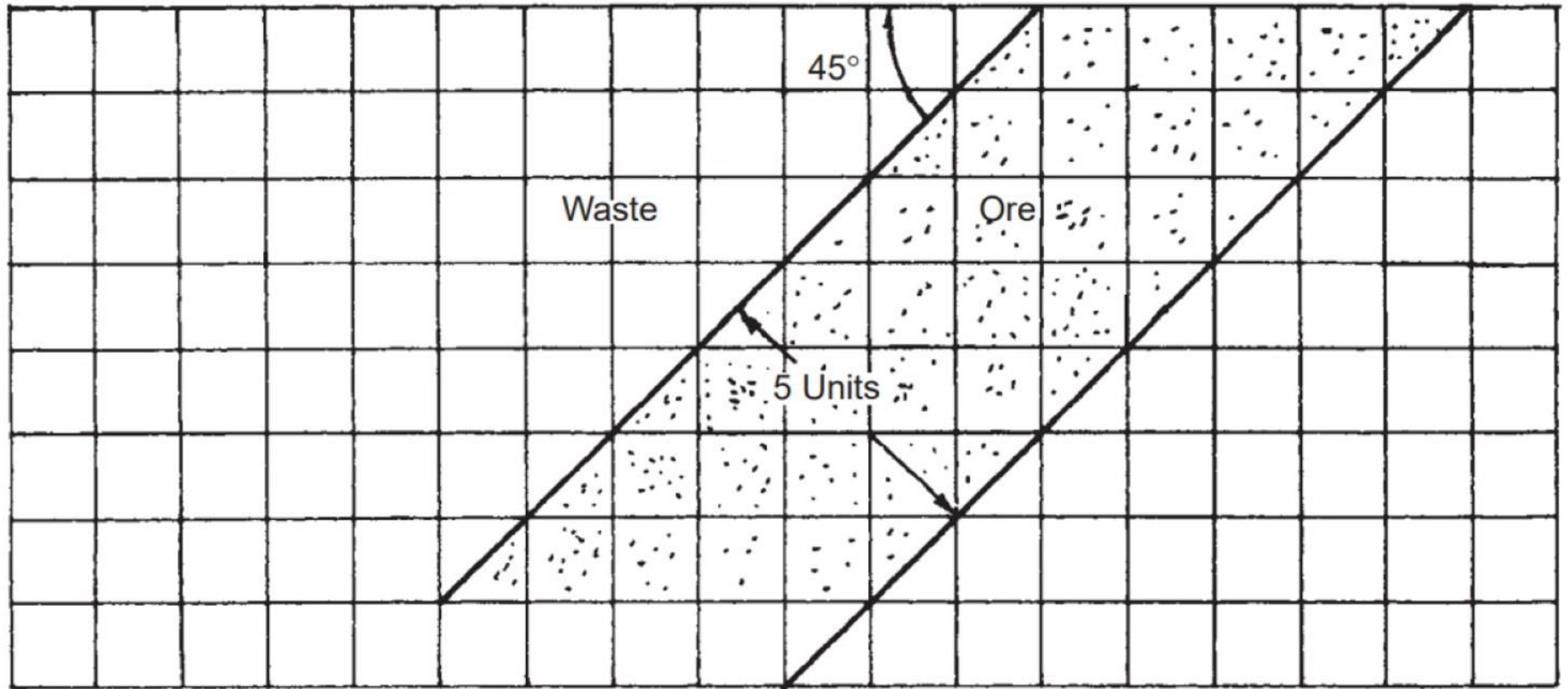


Figure 5.2. Cross-section through an idealized orebody.

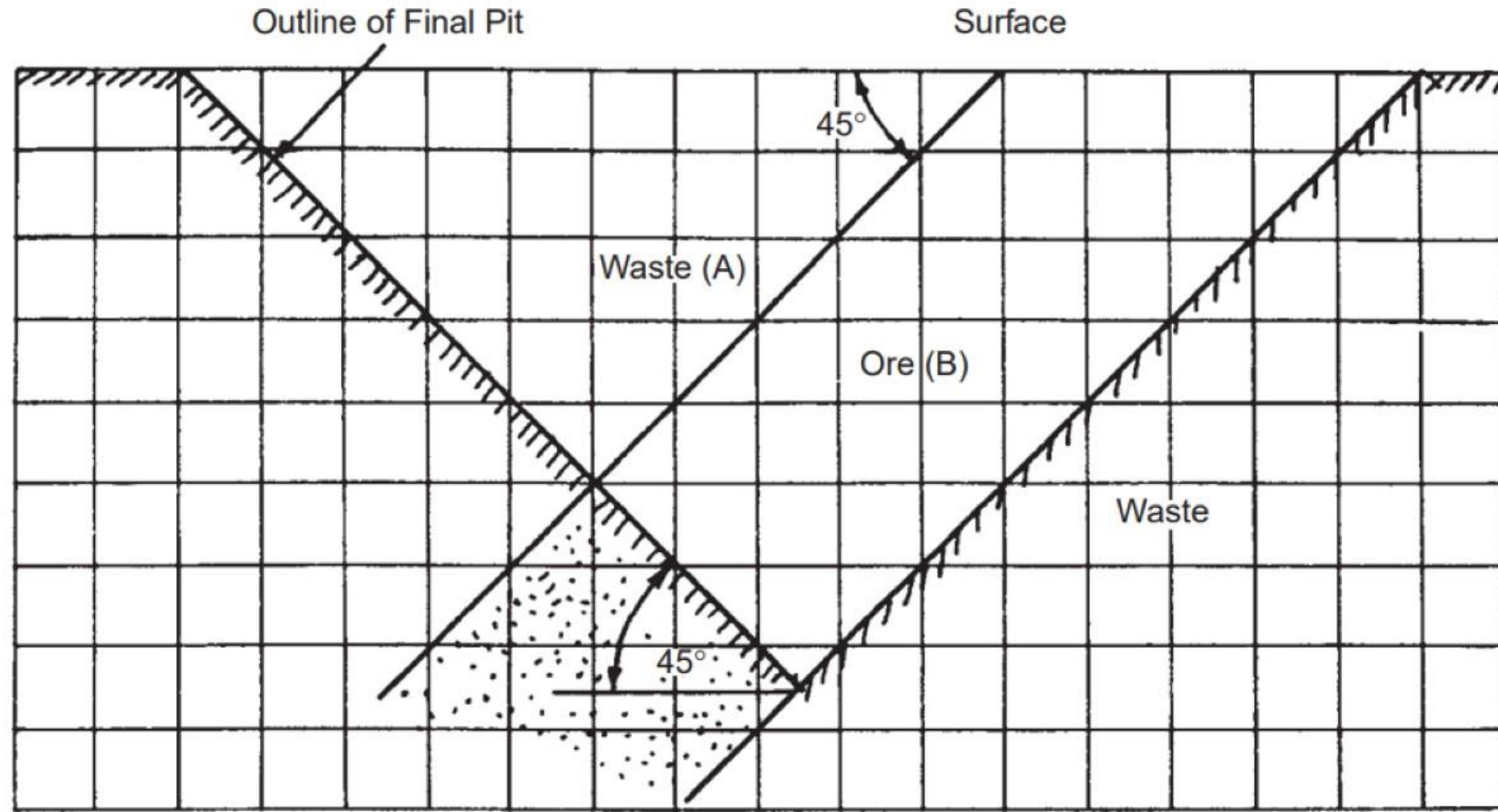


Figure 5.3. Diagrammatic representation of the final pit outline on this section.

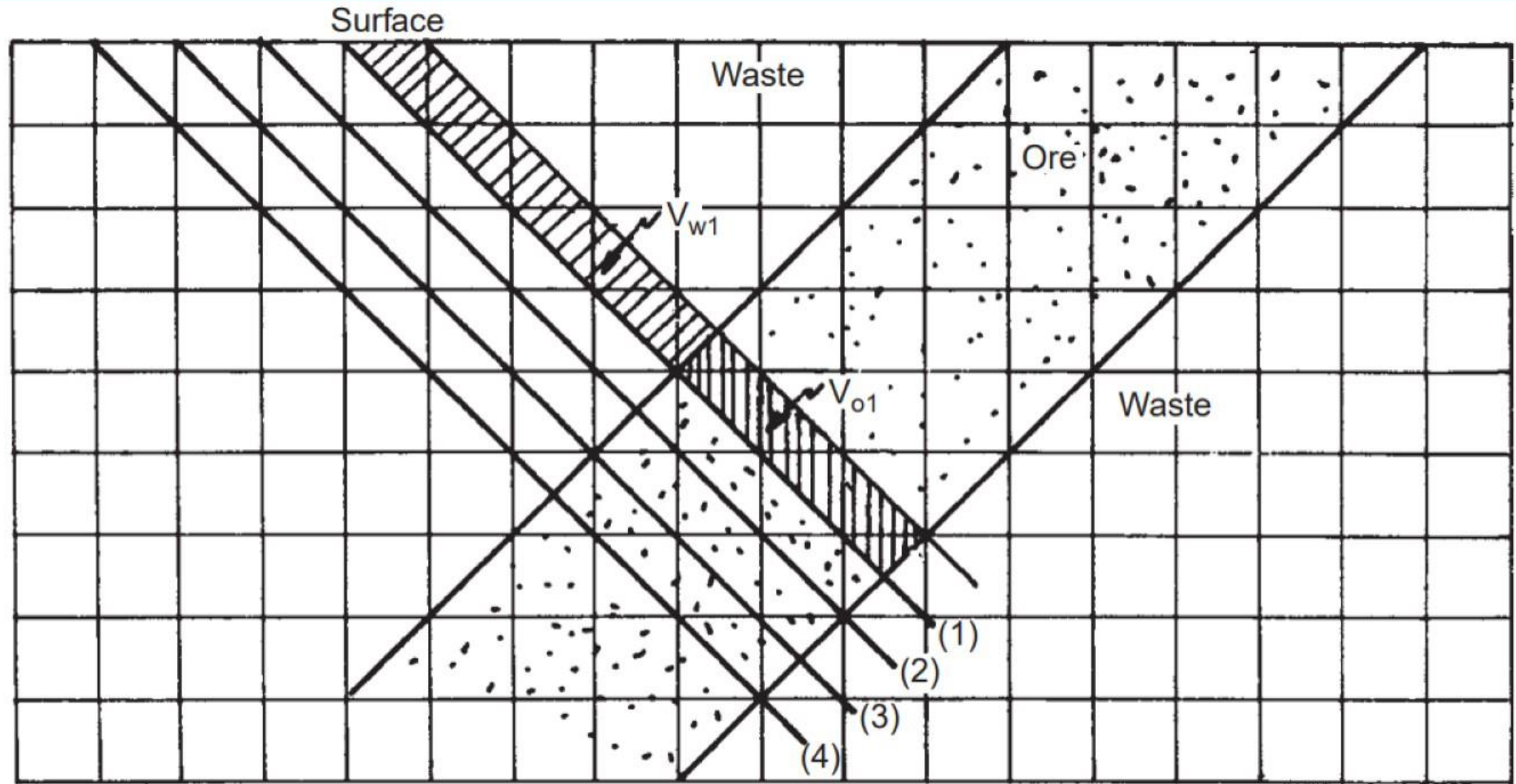


Figure 5.4. Slices used to determine final pit limits.

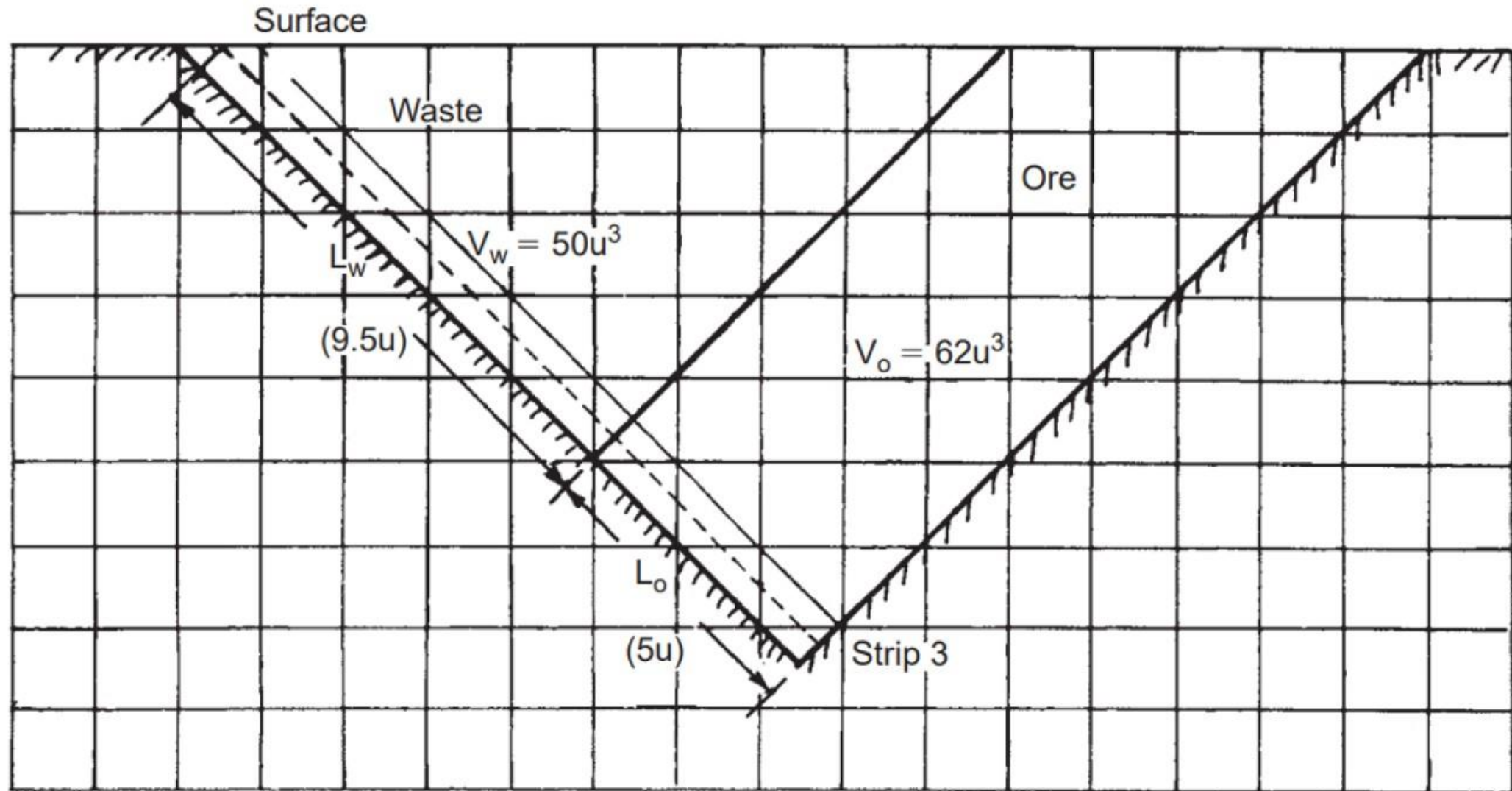


Figure 5.5. Final pit outline showing ore-waste distribution.

5.2.2 The net value calculation

- ✓ In the previous section, the nature of the deposit was such that there was no ambiguity regarding what was meant by ore and waste.
- ✓ For many deposits however the distinction is much more subtle. The term 'cutoff grades' refers to grades for which the destination of pit materials changes. It should be noted that 'grades' were used rather than 'grade' since there may be several possible destinations.
- ✓ The simplest case would be that in which there are two destinations: the mill or the waste dump. One cutoff grade is needed. For many operations today there are three possible destinations: the mill, the leach dump and the waste dump.
- ✓ Each of the decisions – mill or leach? – leach or waste? requires a cutoff grade.

5.2.2 The net value calculation

- ✓ A definition of cutoff grade which is often used (Davey, 1979):
- ✓ Cutoff grade = the grade at which the mineral resource can no longer be processed at a profit. applies to the simple ore-waste decision.
- ✓ This will be used in developing the preliminary pit limits.
- ✓ The only destinations allowed are the waste dump or further processing. With this definition the net value of material as a function of grade must be determined.
- ✓ That grade for which the net value is zero is called the breakeven cutoff grade. This calculation will be illustrated using the example provided by Davey (1979) for copper.
- ✓ The copper ore is milled thereby producing a copper concentrate. This mill concentrate is shipped to a smelter and the resulting blister copper is eventually refined.

5.2.2 The net value calculation

- ✓ In this example the following will be assumed:
- ✓ Mill recovery rate = 80%
- ✓ Mill concentrate grade = 20%
- ✓ Smelting loss = 10 lbs/st of concentrate
- ✓ Refining loss = 5 lbs/st of blister copper
- ✓ The steps of the net value computation are outlined below for an ore containing 0.55% copper.
- ✓ All of the costs and revenues will be calculated with respect to one ton of ore.

5.2.2 The net value calculation

Step 1. Compute the amount of saleable copper (lbs/st of ore).

(a) Contained copper (CC) is

$$CC = 2000 \text{ lbs/st} \times \frac{0.55}{100} = 11.0 \text{ lb}$$

(b) Copper recovered by the mill (RM) is

$$RM = 11.0 \times \frac{80}{100} = 8.8 \text{ lb}$$

(c) Concentration ratio (r). The ratio of concentration is defined as

$$r = \frac{\text{lbs Cu/st of concentrate}}{\text{lbs Cu recovered/st of ore}} \quad (5.3)$$

Since the mill product runs 20% copper there are 400 lb of copper contained in one ton of concentrate. One ton of ore contains 8.8 lb of recoverable copper. Hence

$$r = \frac{400}{8.8} = 45.45$$

This means that 45.45 tons of ore running 0.55% copper are required to produce 1 ton of concentrate running 20%.

(d) Copper recovered by the smelter (RS). The mill concentrate is sent to a smelter. Since the smelting loss is 10 lb/st of concentrate, the smelting loss (SL) per ton of ore is

$$SL = \frac{10 \text{ lb/st of concentrate}}{45.45 \text{ tons of ore/st of concentrate}} = 0.22 \text{ lb}$$

5.2.2 The net value calculation

Thus the recovered copper is

$$RS = 8.8 - 0.22 = 8.58 \text{ lb}$$

(e) Copper recovered by the refinery (RR). The number of tons of ore required to produce one ton of blister copper is

$$\frac{2,000 \text{ lb/st of blister copper}}{8.58 \text{ lb of copper/st of ore}} = 233.1$$

Since refining losses are 5 lb/st of blister copper, the refining loss (RL) per ton of ore is

$$RL = \frac{5 \text{ lb of copper/st of blister copper}}{233 \text{ tons of ore/st of blister copper}} = 0.02 \text{ lb}$$

Thus the recovered copper is

$$RR = 8.58 - 0.02 = 8.56 \text{ lb}$$

Step 2. Compute the gross value (GV) for the ore (\$/st). The copper price assumed for this calculation is \$1.00/lb. Furthermore there is a by-product credit for gold, molybdenum, etc. of \$1.77/st of ore. Thus the gross value is

$$GV = 8.56 \times \$1 + \$1.77 = \$10.33$$

5.2.2 The net value calculation

Step 6. Construct a net value – grade curve. The two points on a net value – grade curve which have been determined by the process outlined above

Point	Net value (\$/st)	Grade (% Cu)
1	\$2.50	0.55
2	–\$0.30	0.35

are plotted in Figure 5.6. Assuming that the net value – % Cu relationship is linear it is possible to find an equation of the form $y = a + bx$ relating net value (y) to grade (x). The result is

$$y = -\$5.20 + \$14.0x$$

where y is the net value (\$/st of ore) and x is the percent copper.

5.2.2 The net value calculation

Step 7. Determine the breakeven cutoff grade (for application at the pit limit). The breakeven cutoff grade is defined as that grade for which the net value is zero. One can determine that point by inspecting Figure 5.6 or by solving the equation found in Step 6 for $y = 0$. One finds that

$$x(\text{breakeven}) = 0.37\% \text{ Cu}$$

Step 8. Developing a stripping ratio – grade curve. The cutoff grade distinguishes that material which can be mined and processed with a net value greater than or equal to zero. Material with a zero net value cannot pay for any stripping. Thus it must be exposed at the surface or be overlying richer blocks which can pay for the required stripping. Assume that the cost for stripping 1 ton of waste is \$1.00. Ore with a net value of \$1.00 can pay for the stripping of 1 ton of waste. Ore with a net value of \$2.00/ton can pay for the stripping of 2 tons of waste, etc. The stripping ratio axis has been added in Figure 5.7 to show this. The net value – grade equation

$$NV = -\$5.20 + \$14.00 \times (\% \text{ Cu}) \quad (5.4)$$

can be modified to yield the stripping ratio (SR) – grade relationship

$$SR = -\$5.20 + \$14.00 \times (\% \text{ Cu}) \quad (5.5)$$

For a grade of 0.55% Cu, the breakeven stripping ratio is

$$SR(0.55\%) = 2.5$$

5.2.2 The net value calculation

- ✓ This was as expected since the net value is \$2.50 and the stripping cost is \$1.00 $SR = \$2.50 / \$1.00 = 2.50$
- ✓ Step 9. Presenting the final curves.
- ✓ The net value – grade curve should be completed by the addition of the cost of stripping line (SC). This is shown in Figure 5.8. It should be noted that no material can ever have a value less than that of waste. In this case the value of waste is $-\$1.00$. The horizontal line ($NV = -\1.00) and the NV-grade line ($NV = -5.20 + 14.00 \times (\% \text{ Cu})$) intersect at a grade of $\% \text{ Cu} = 0.30$. For grades less than 0.30%, the material is considered as waste with respect to milling. Depending upon the economics, some other treatment process such as dump leaching may be possible. When using hand methods all material having grades less than the breakeven cutoff (0.37% in this case) is considered as waste.
- ✓ The final stripping ratio-grade curve is shown in Figure 5.9. For the computer techniques discussed later, the portion of the curve lying between 0.30 and 0.37% Cu is also included.

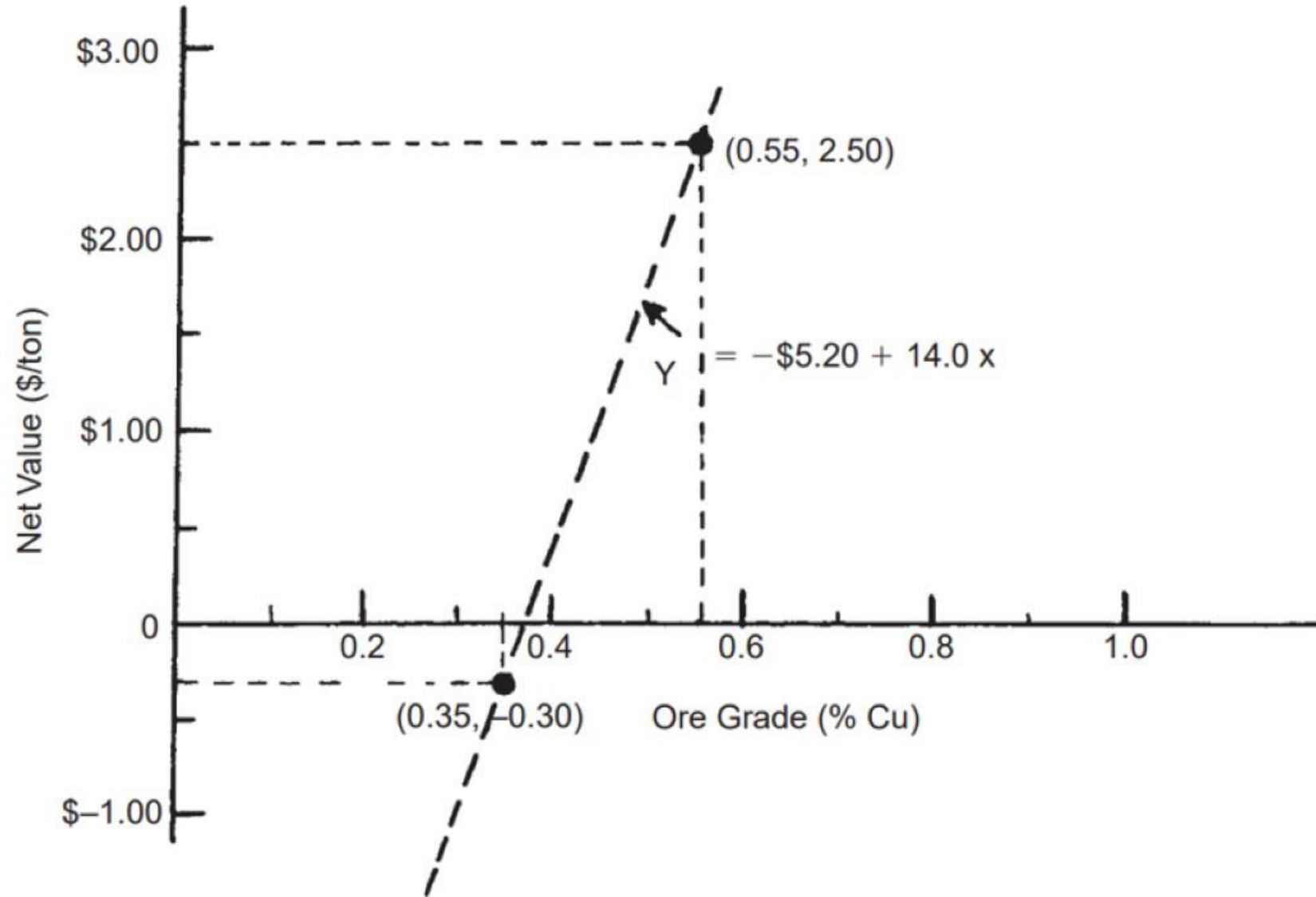


Figure 5.6. Net value – ore grade curve.

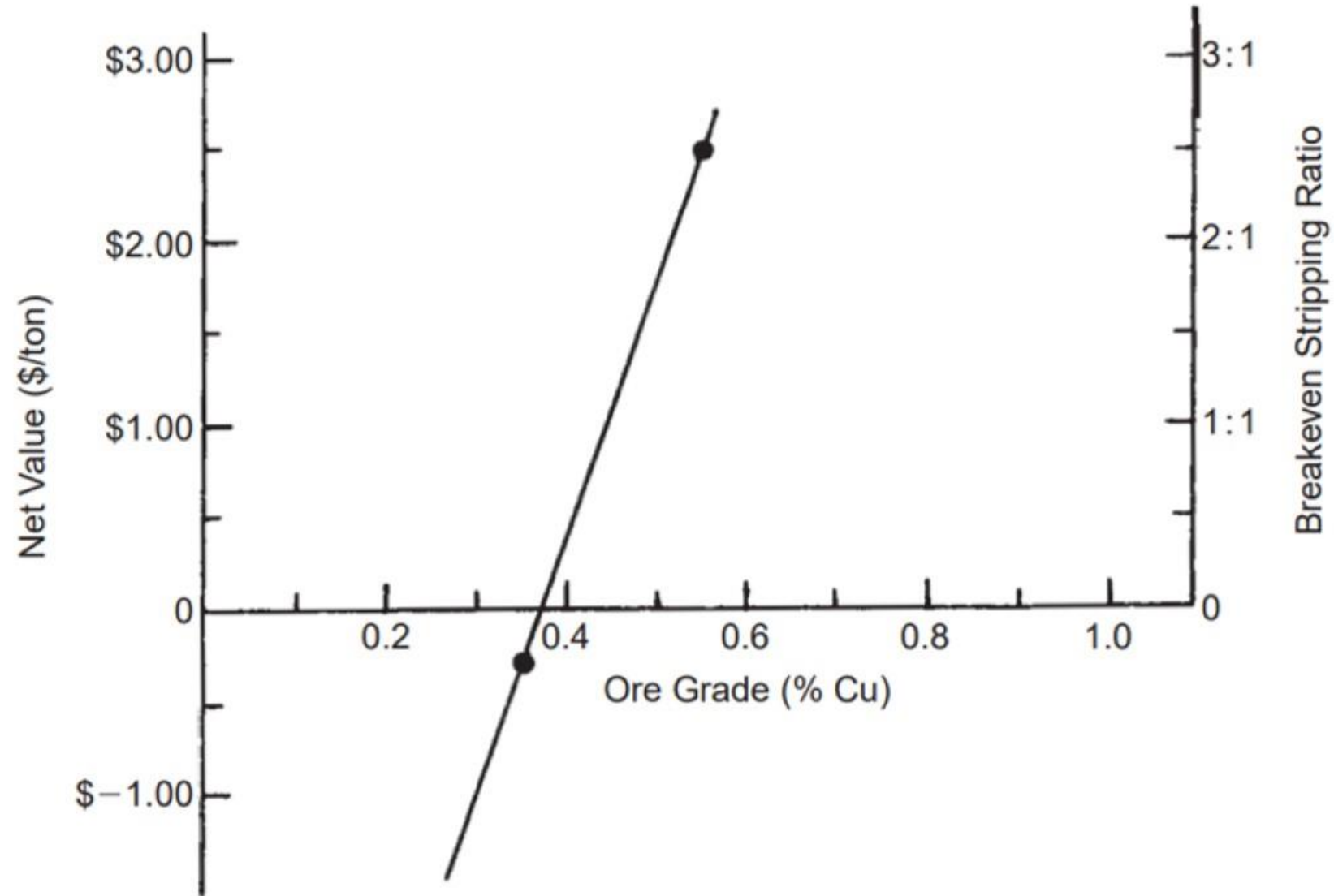


Figure 5.7. Net value and breakeven stripping ratio versus ore grade.

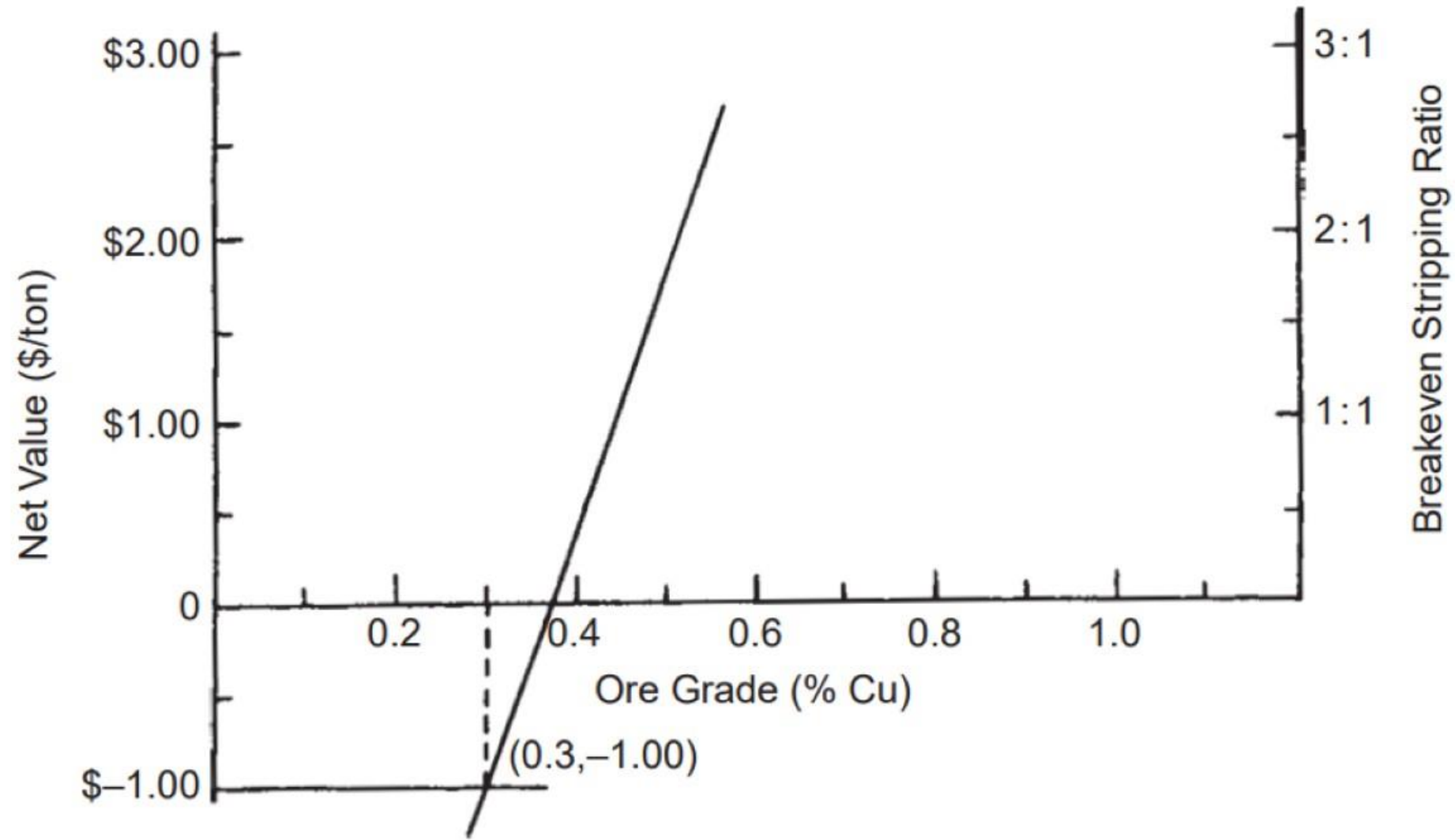


Figure 5.8. Addition of the minimum value portion.

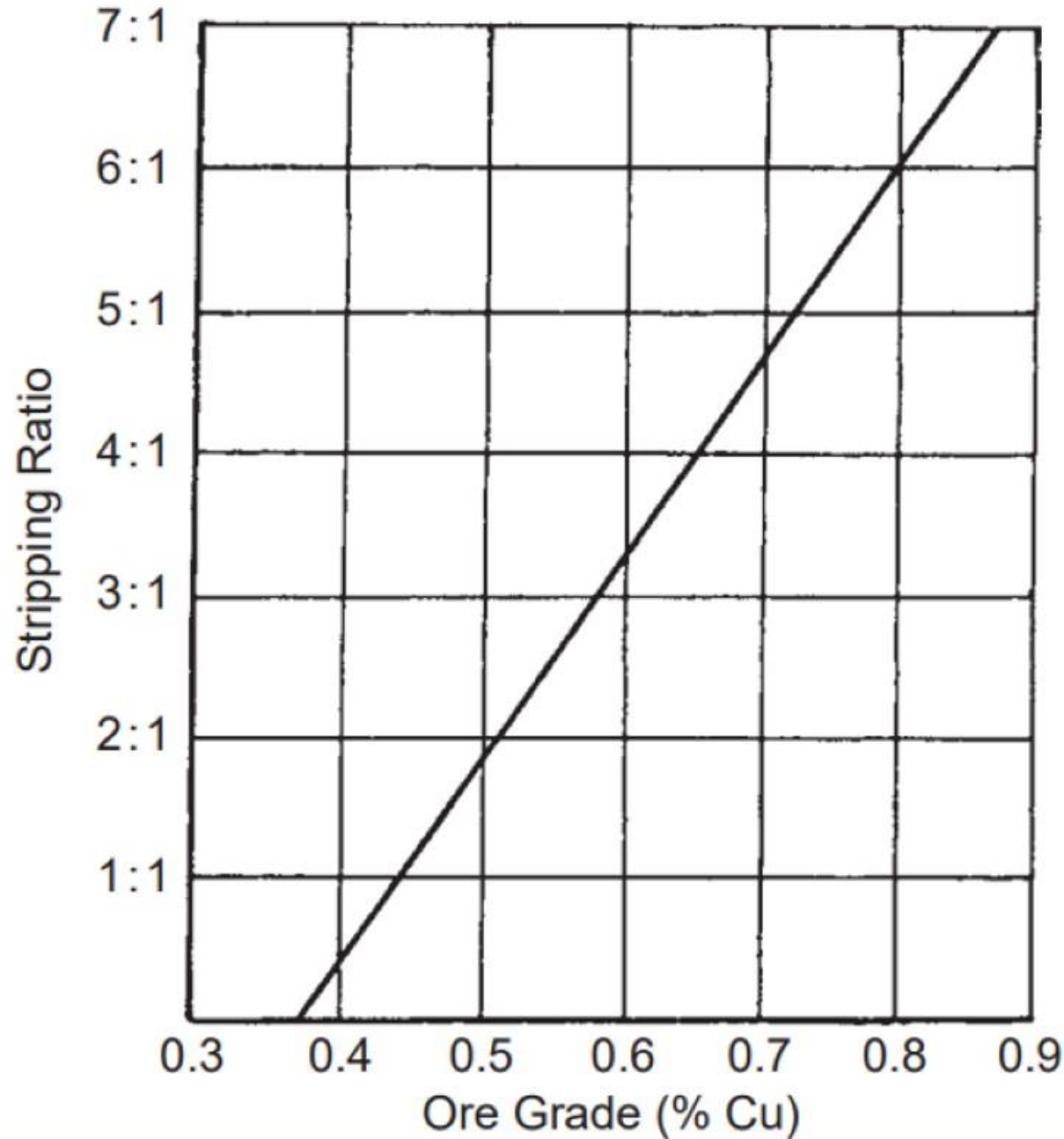


Figure 5.9. Stripping ratio – ore grade curve used for pit limit determinations (Koskineemi, 1979).

PIT LIMITS

5.2.3 Location of pit limits – pit bottom in waste

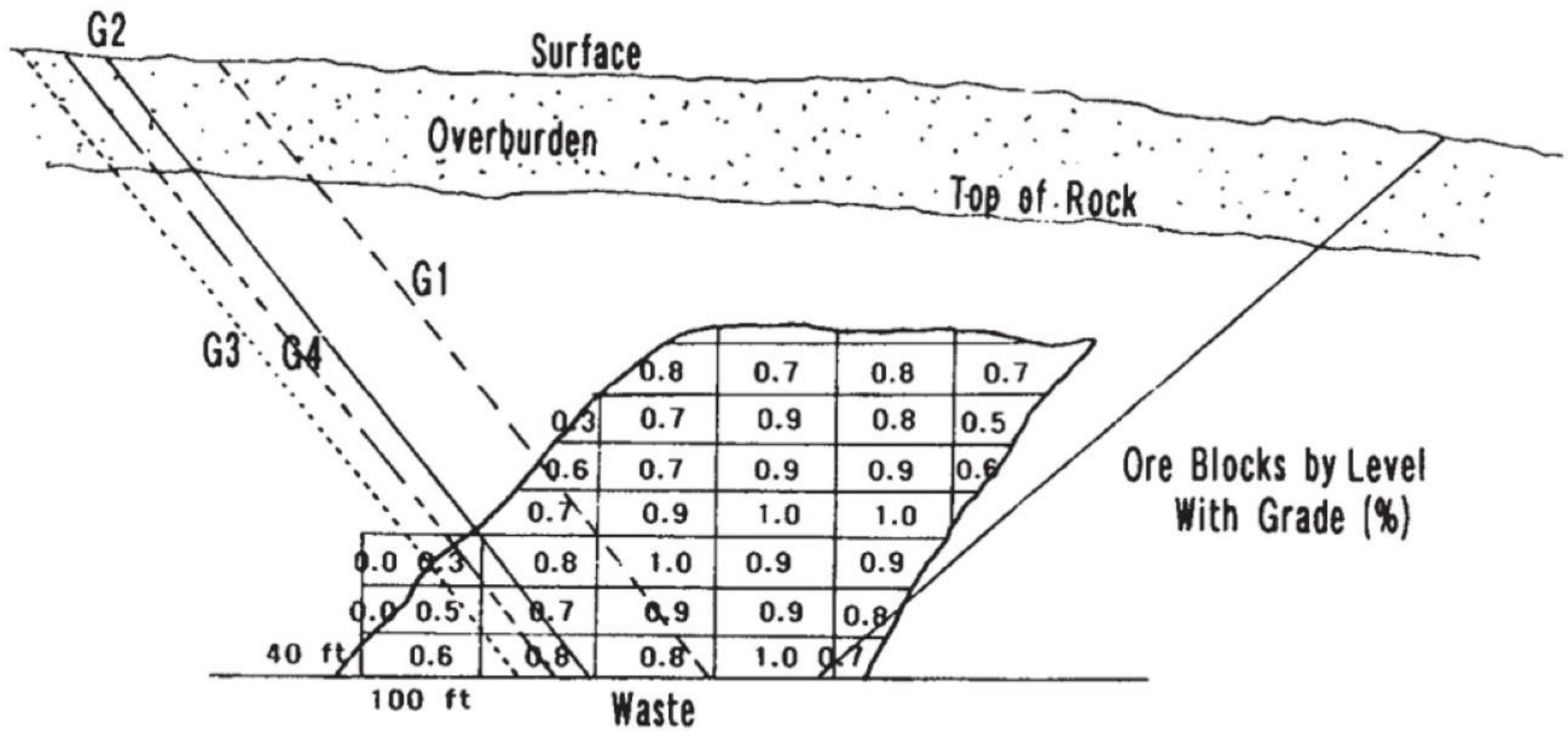


Figure 5.10. Pit limit determination with bottom in waste (Koskiniemi, 1979).



5.2.3 Location of pit limits – pit bottom in waste

The application of this curve to locating the final pit wall positions will be illustrated using the vertical section (Fig. 5.10) taken through the block model.

The basic process used has been presented by Koskiniemi (1979).

Locating the pit limit on each vertical section is a trial and error process.

5.2.3 Location of pit limits – pit bottom in waste: It will be assumed that

- Pit slopes:
 - Left hand side = 50° ;
 - Right hand side = 40° ;
- Minimum width of the pit bottom = 100 ft;
- Material densities:
 - Ore = 165 lb/ft³;
 - Waste rock = 165 lb/ft³;
 - Overburden = 165 lb/ft³;
- Relative mining characteristics:
 - Waste rock = 1;
 - Overburden = 1;
- The stripping ratio
- ore grade curve of Figure 5.9 applies;
- The pit bottom is at the ore-waste contact.

Table 5.1. Bottom in waste: pit limit guess #1 (Line G1).

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
130	296			
		30	0.6	18.0
		52	0.7	36.4
		52	1.0	52.0
		52	0.9	46.8
		52	0.8	41.6
130	296	238	$\bar{g} = 0.82$	194.8

$$SR(\text{actual}) = \frac{130+296}{238} \cong 1.79 : 1;$$

$$SR(\text{allowable}) \cong 6.2 : 1;$$

Conclusion: move to the left.

Table 5.1. Bottom in waste: pit limit guess #1 (Line G1).

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
130	296			
		30	0.6	18.0
		52	0.7	36.4
		52	1.0	52.0
		52	0.9	46.8
		52	0.8	41.6
130	296	238	$\bar{g} = 0.82$	194.8

$$SR(\text{actual}) = \frac{130+296}{238} \cong 1.79 : 1;$$

$$SR(\text{allowable}) \cong 6.2 : 1;$$

Conclusion: move to the left.

Table 5.2. Bottom in waste: pit limit guess #2 (Line G2).

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
130	385	52	0.8	41.6
		52	0.7	36.4
		52	0.8	41.6
130	385	156	$\bar{g} = 0.77$	119.6

$$SR(\text{actual}) = \frac{130 + 385}{156} \cong 33 : 1;$$

$$SR(\text{allowable}) \cong 5.6 : 1;$$

Conclusion: move to the left.

Table 5.3. Bottom in waste: pit limit guess #3 (Line G3).

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
130	443			
		52	0.5	26.0
		52	0.8	41.6
130	443	104	$\bar{g} = 0.65$	67.6

$$SR \text{ (actual)} = \frac{130 + 443}{104} \cong 5.51 : 1;$$

$$SR \text{ (allowable)} \cong 3.9 : 1;$$

Conclusion: move to the right.

Table 5.4. Bottom in waste: pit limit guess #4 (Line G4).

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
130	435			
		52	0.7	36.4
		52	0.8	41.6
130	435	104	$\bar{g} = 0.75$	78.0

$$SR(\text{actual}) = \frac{130 + 435}{104} \cong 5.43 : 1;$$

$$SR(\text{allowable}) \cong 5.4 : 1;$$

Conclusion: final limit.

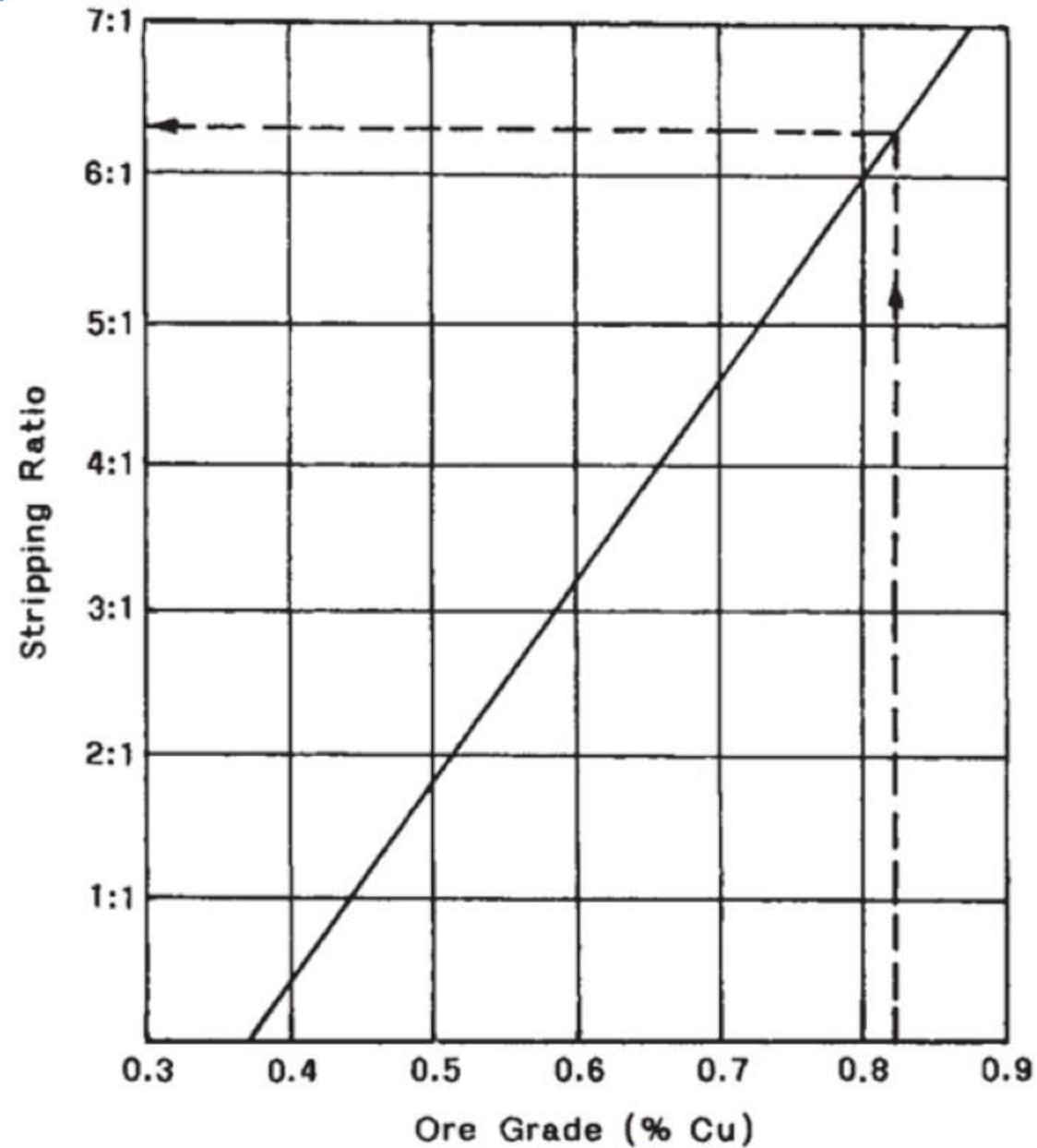


Figure 5.11. Allowable stripping ratio for a grade of 0.82%.

Table 5.5. Bottom in ore: pit limit guess #1 (Line G1).

(a) Left hand side

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
130	330			
		48	0.7	33.6
		52	0.8	41.6
		13	0.7	9.1
		39	0.9	35.1
		22	0.8	17.6
		50	0.8	40.0
130	330	224	$\bar{g} = 0.79$	177.0

$$SR(\text{actual}) = \frac{130 + 330}{224} \cong 2.05;$$

$$SR(\text{allowable}) \cong 6:1.$$

(b) Right hand side

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
143	283	48	0.6	28.8
	13	48	1.0	48.0
		70	0.9	63.0
		11	0.8	8.8
		57	0.9	51.3
		26	1.0	26.0
		39	1.0	39.0
		11	0.8	8.8
143	296	310	$\bar{g} = 0.88$	273.7

$$SR(\text{actual}) = \frac{143 + 296}{310} \cong 1.42 : 1;$$

$$SR(\text{allowable}) \cong 7 : 1;$$

Conclusion: the pit can be 'floated' considerable deeper.

Table 5.6. Bottom in ore: pit limit guess #2 (Line G2).

(a) Left hand side

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
139	626			
		52	0.5	26.0
		52	0.6	31.2
		52	0.6	31.2
		52	0.6	31.2
		52	0.6	31.2
		52	0.6	31.2
139	626	258	$\bar{g} = 0.58$	150.8

$$SR(\text{actual}) = \frac{139 + 626}{258} \cong 2.97 : 1;$$

$$SR(\text{allowable}) \cong 2.95 : 1.$$

(b) Right hand side

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
157	617			
		65	0.6	39.0
		61	0.6	36.6
		65	0.5	32.5
		61	0.5	30.5
		50	0.6	30.0
157	617	302	$\bar{g} = 0.56$	168.6

$$SR(\text{actual}) = \frac{157+617}{302} \cong 2.56 : 1;$$

$$SR(\text{allowable}) \cong 2.6 : 1;$$

Conclusion: guess #2 is close to the pit slope location.

PIT LIMITS

5.2.4 Location of pit limits – pit bottom in ore

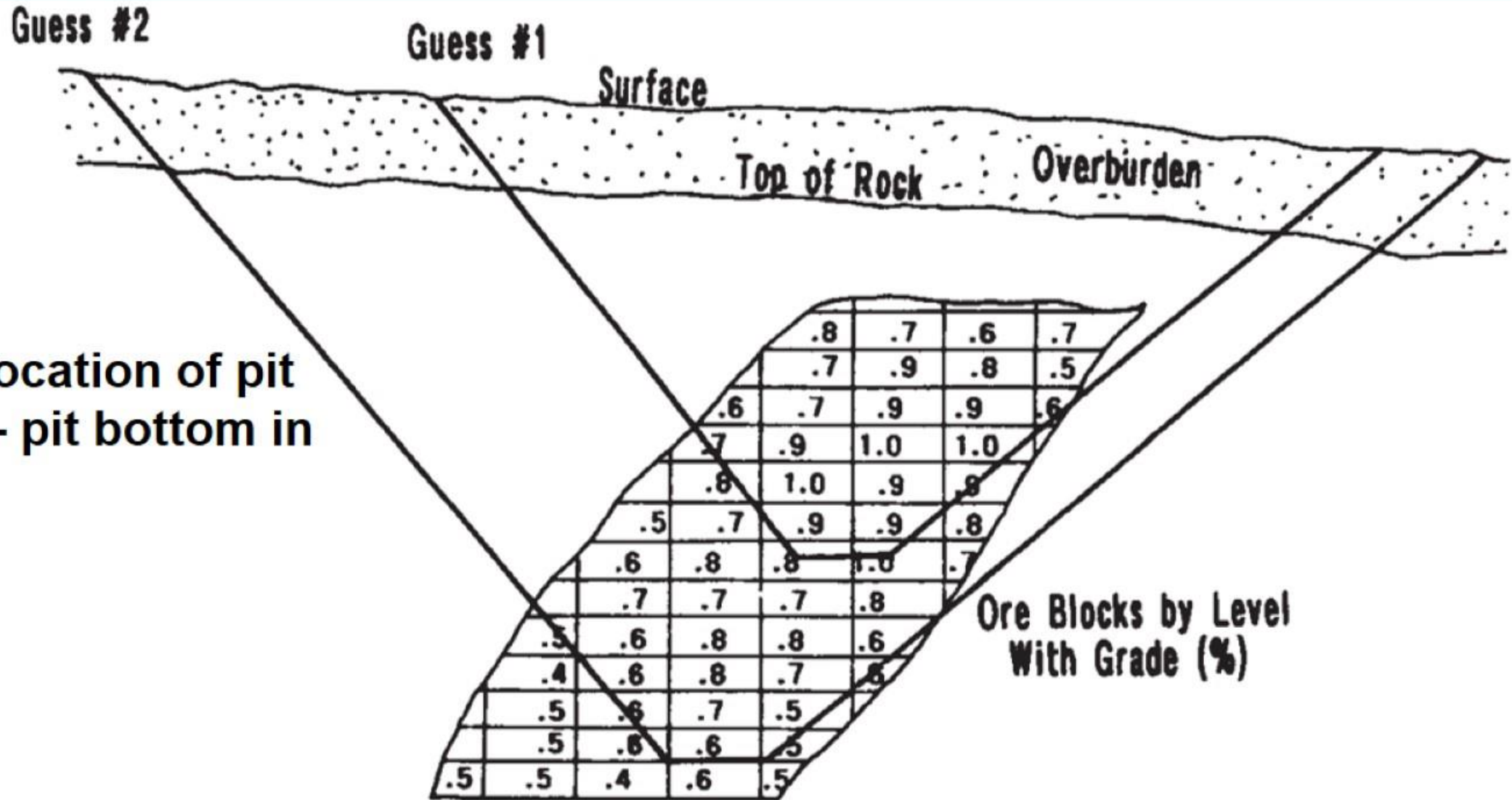


Figure 5.12. Final pit limits with bottom in ore.

5.2.5 Location of pit limits – one side plus pit bottom in ore

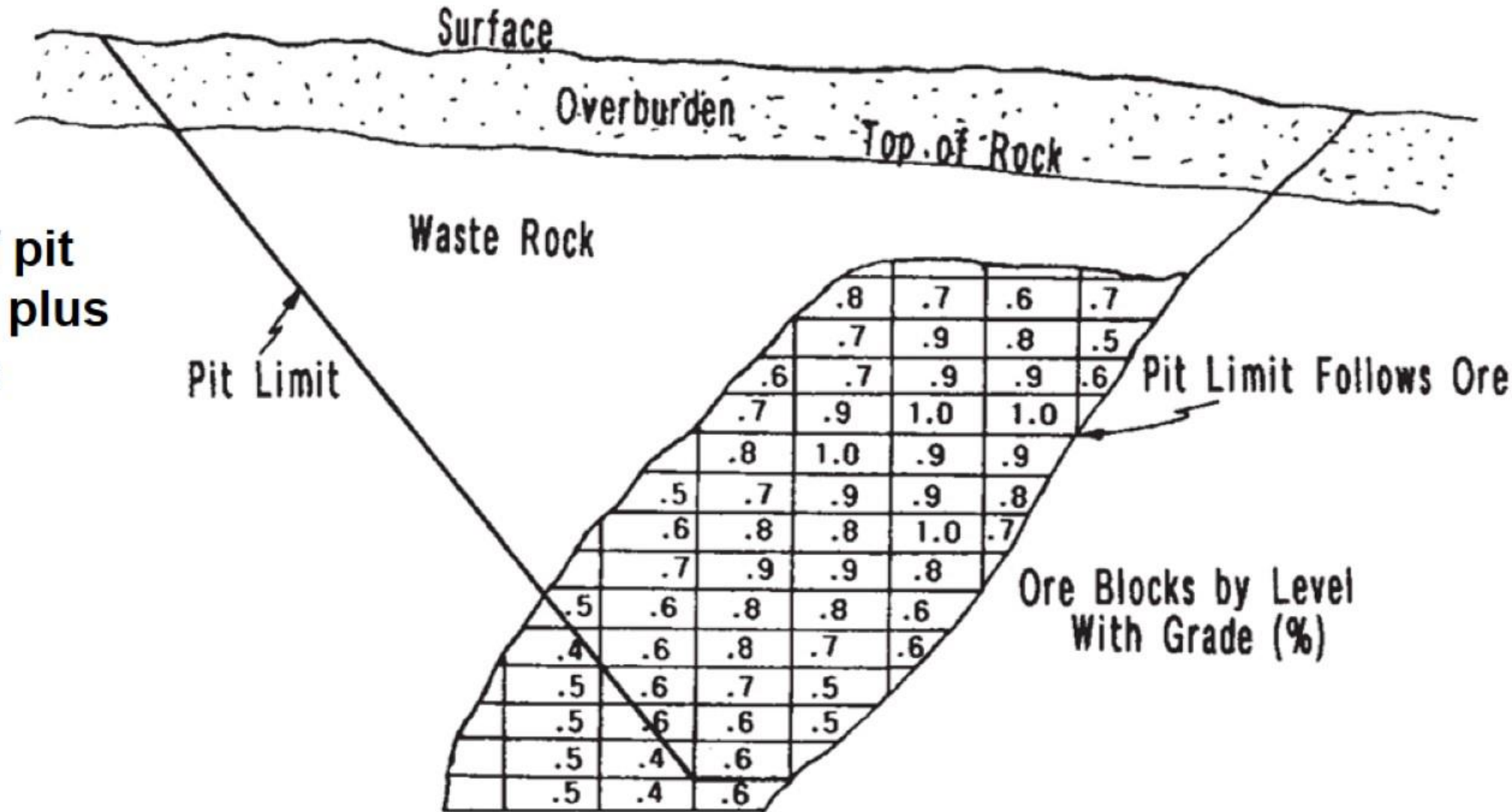


Figure 5.13. Final pit limits with right hand side in ore.

Table 5.7. Final pit limit location for one side plus pit bottom in ore (Fig. 5.13).

Length (ft)			Ore grade (g_{oi})	Ore length $\times$ ore grade ($l_{oi}g_{oi}$)
Overburden (l_{ob})	Waste (l_w)	Ore (l_{oi})		
130	626			
		43	0.5	12.5
		52	0.4	20.8
		52	0.6	31.2
		52	0.6	31.2
		52	0.4	20.8
		100	0.6	60.0
130	626	351	$\bar{g} = 0.53$	185.5

$$SR(\text{actual}) = \frac{130 + 626}{351} \cong 2.15;$$

$$SR(\text{allowable}) \cong 2.21;$$

Conclusion: this is the final pit location.

5.2.6 Radial sections

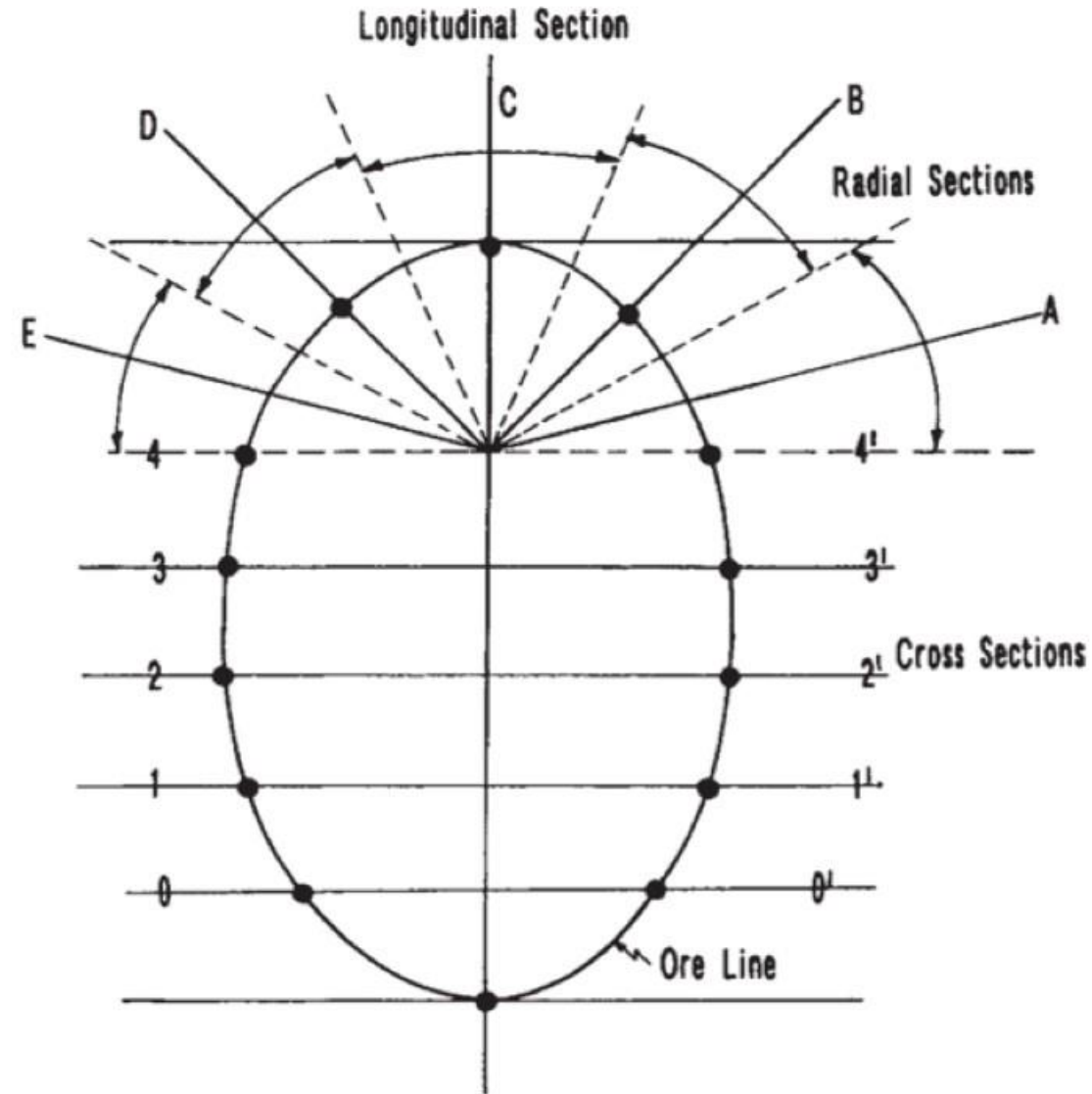


Figure 5.14. Plan view of orebody illustrating the different section types (Koskineemi, 1979).

5.2.6 Radial sections

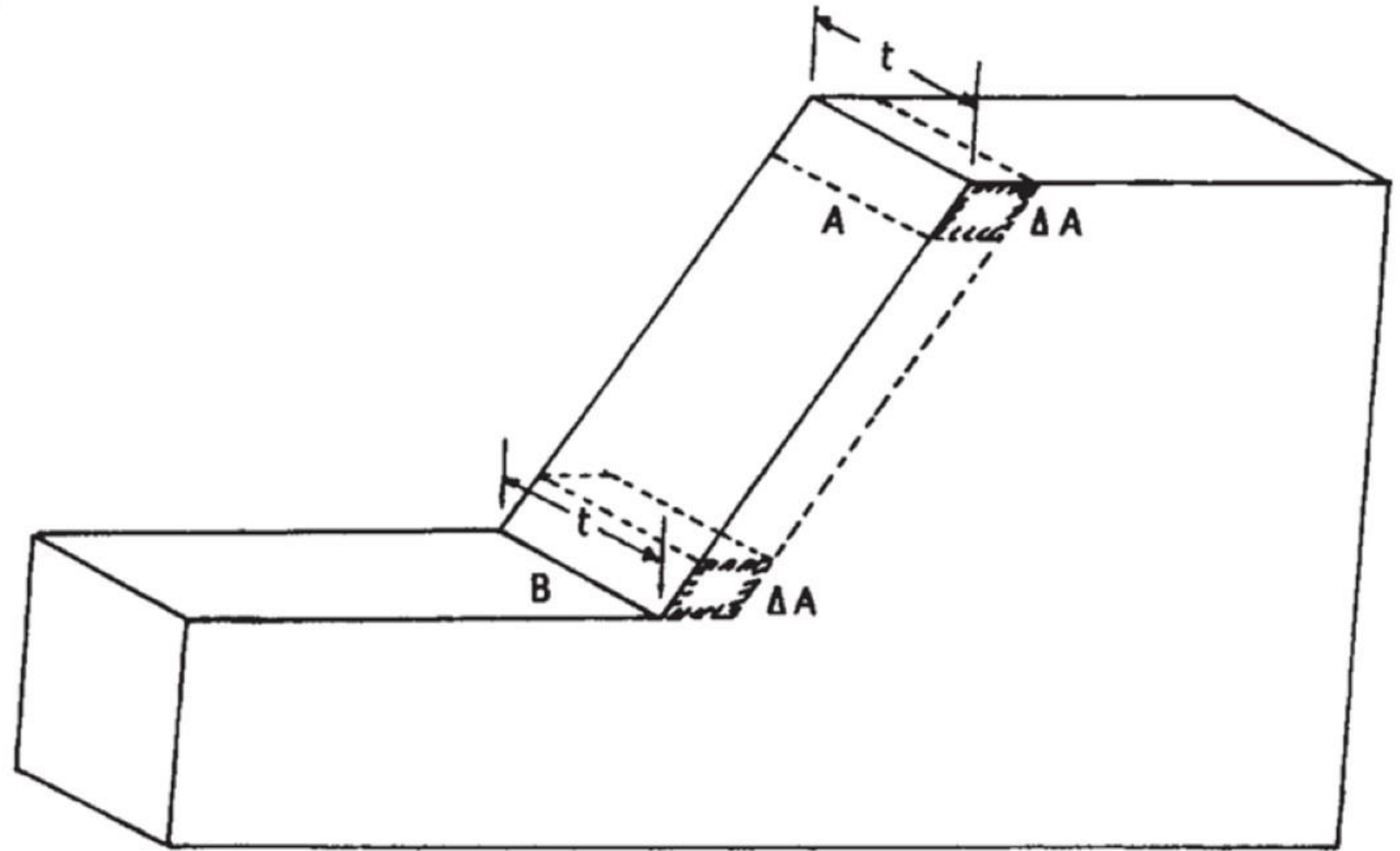


Figure 5.15. Isometric view of a parallel section.

5.2.6 Radial sections

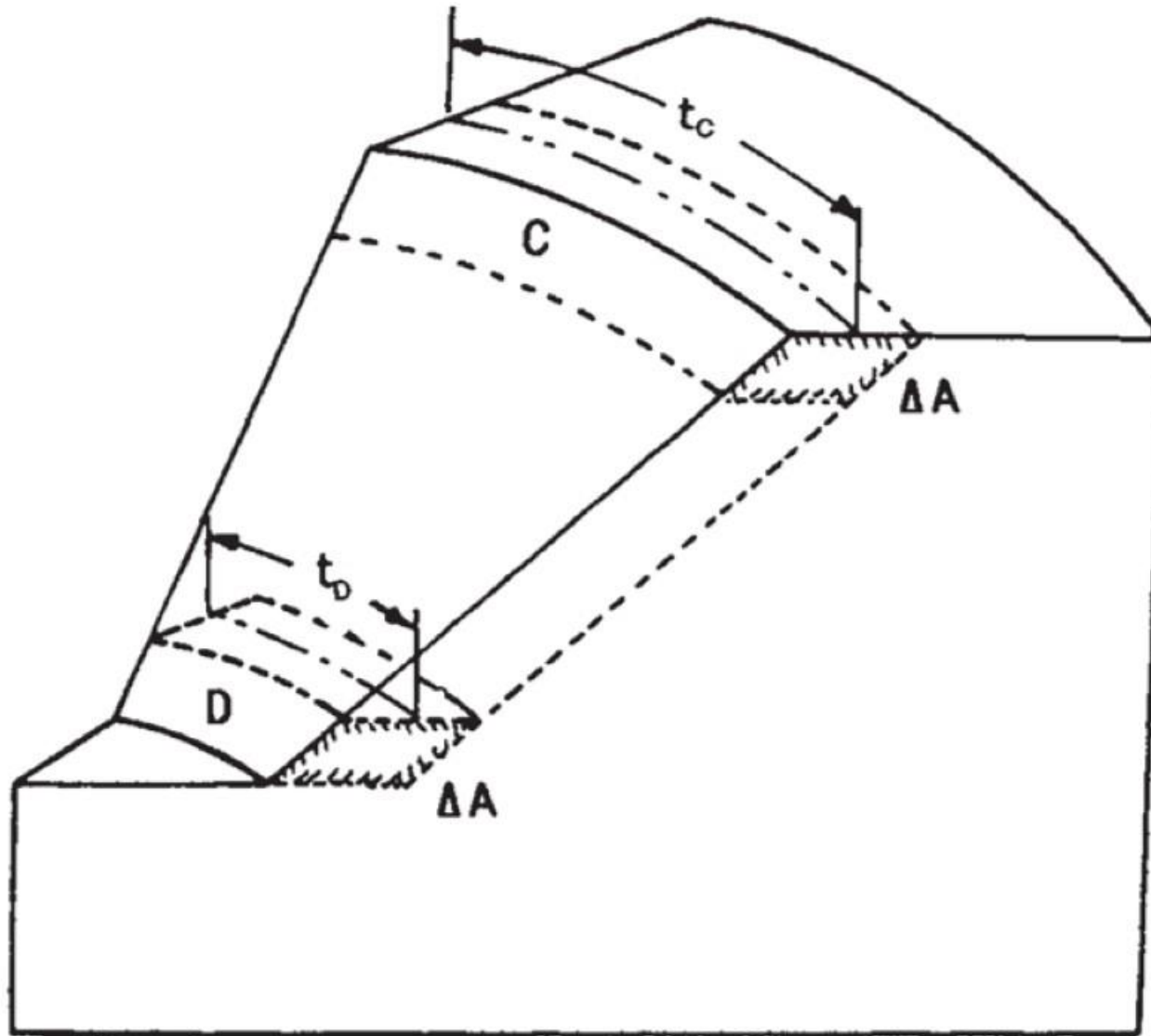


Figure 5.16. Isometric view of a radial section.

PIT LIMITS

5.2.6 Radial sections

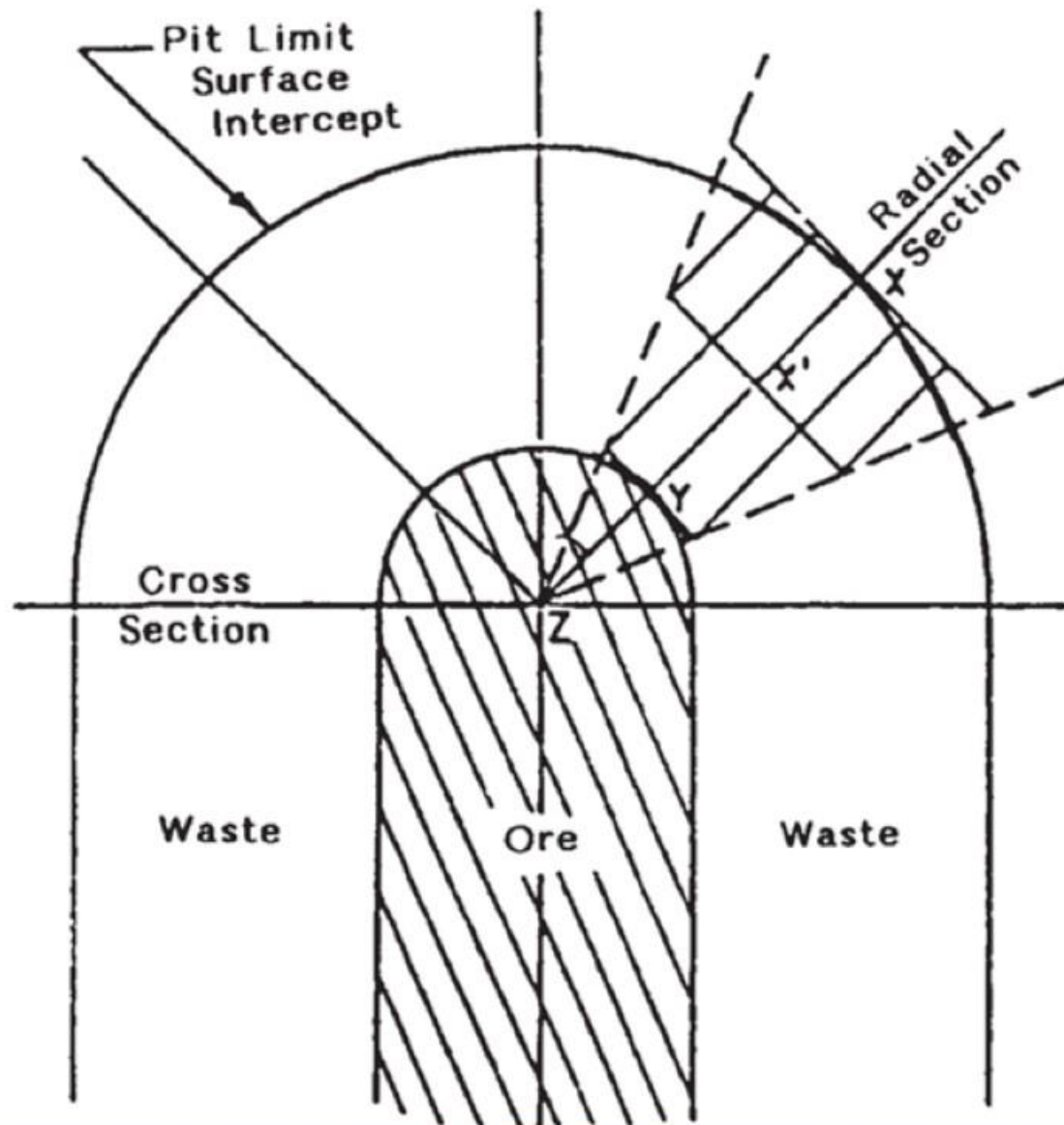


Figure 5.17. Plan representation showing the area of influence for a radial section (Koskineemi, 1979).

5.2.6 Radial sections

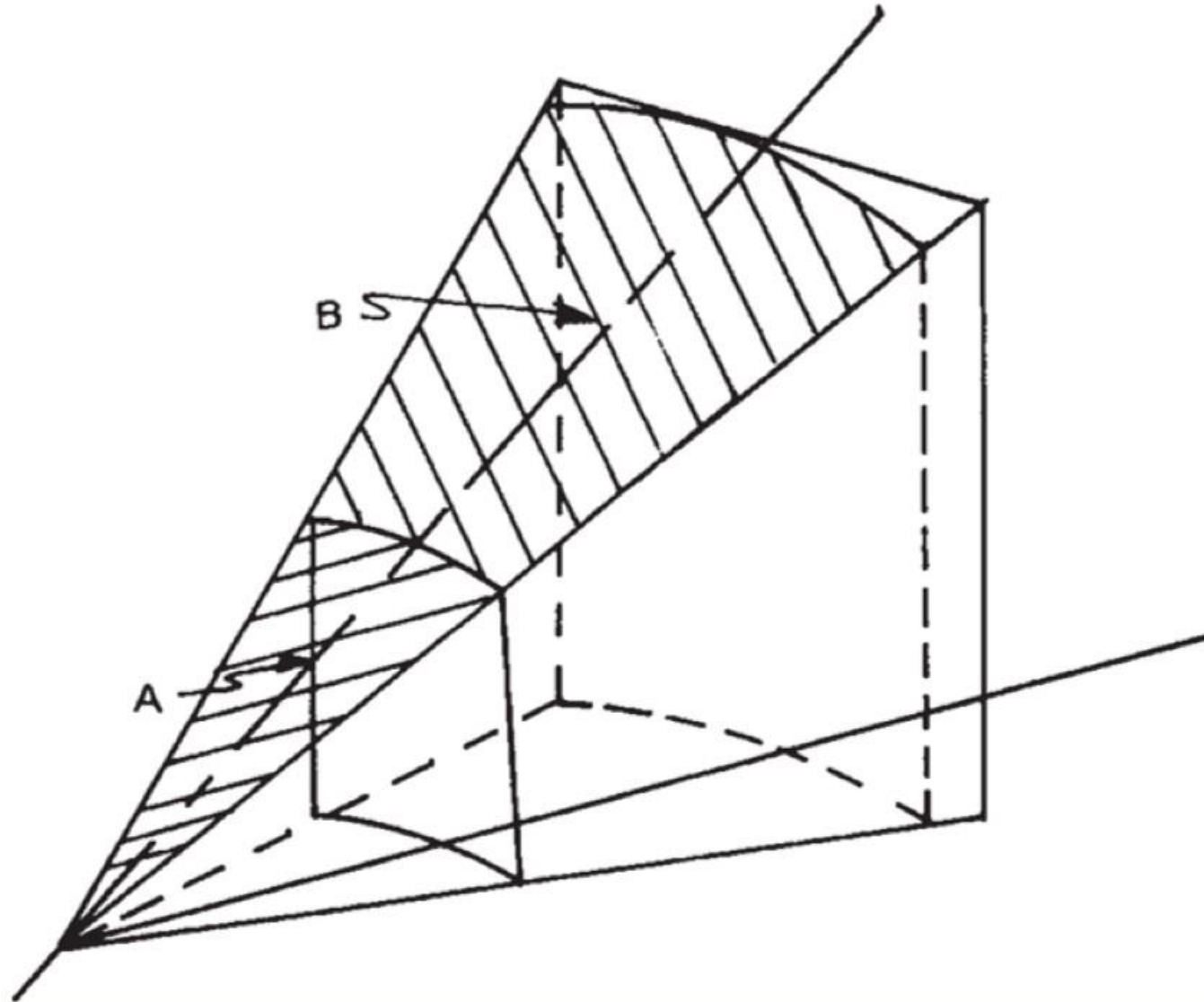


Figure 5.18. Isometric view of the radial sector.

5.2.6 Radial sections

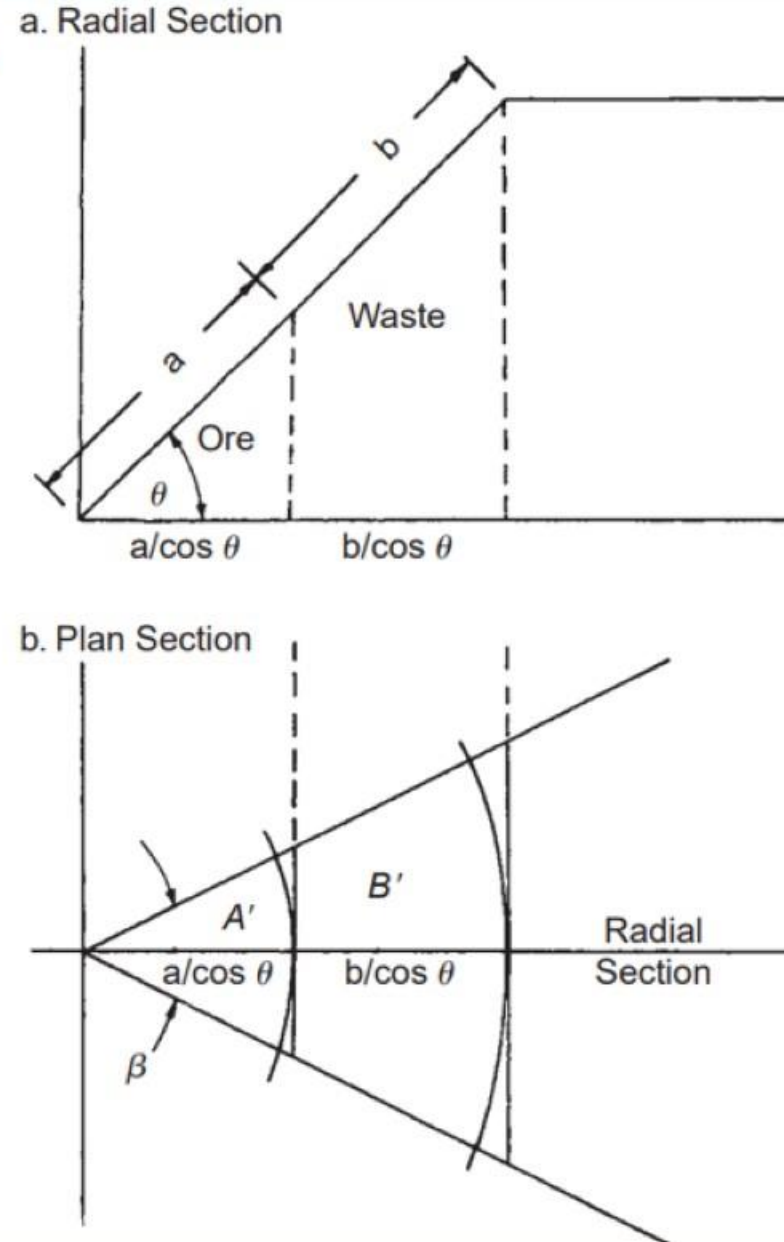
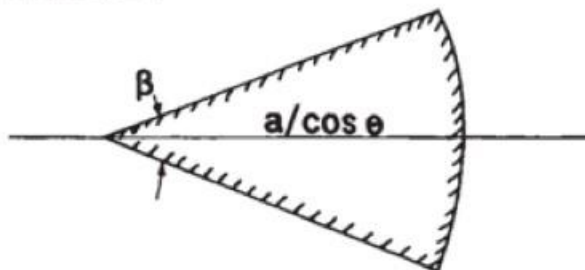


Figure 5.19. Plan and section views of the radial sector shown in Figure 5.18.

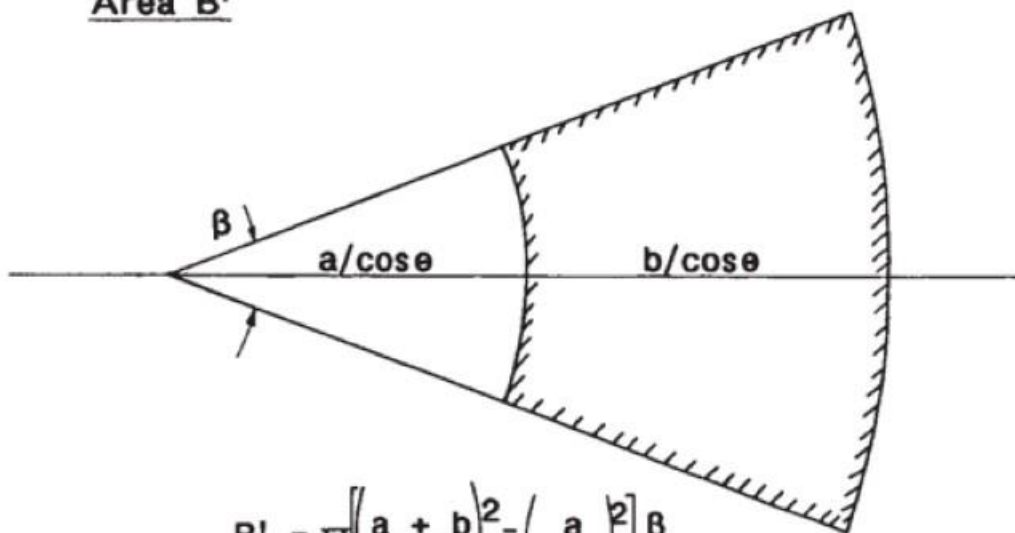
5.2.6 Radial sections

Area A'



$$A' = \pi \left(\frac{a}{\cos \theta} \right)^2 \left(\frac{\beta}{2\pi} \right)$$

Area B'



$$B' = \pi \left[\left(\frac{a+b}{\cos \theta} \right)^2 - \left(\frac{a}{\cos \theta} \right)^2 \right] \frac{\beta}{2\pi}$$

Table 5.8. Comparison of true versus measured stripping ratios for radial sections.

Stripping ratio	
Measured	True
0	0
0.25	0.56
0.50	1.25
0.75	2.06
1.0	3.0
1.25	4.06
1.5	5.25
2.0	8.0
2.5	11.25
3.0	15.0

Figure 5.20. The ore and waste areas involved.

5.2.6 Radial sections

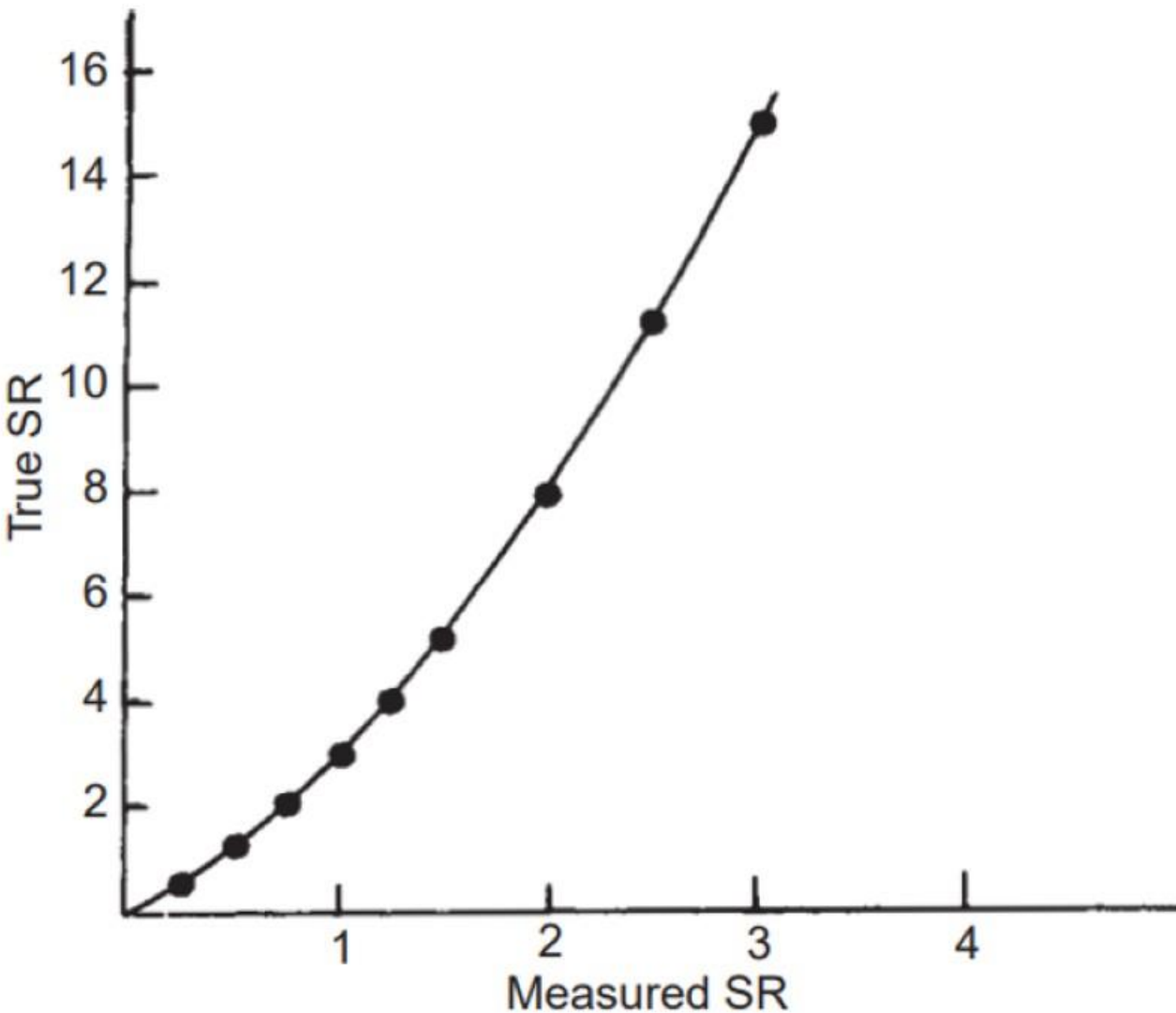


Figure 5.21. True versus measured strip-ping ratios (Koskiniemi, 1979).

PIT LIMITS

5.2.6 Radial sections

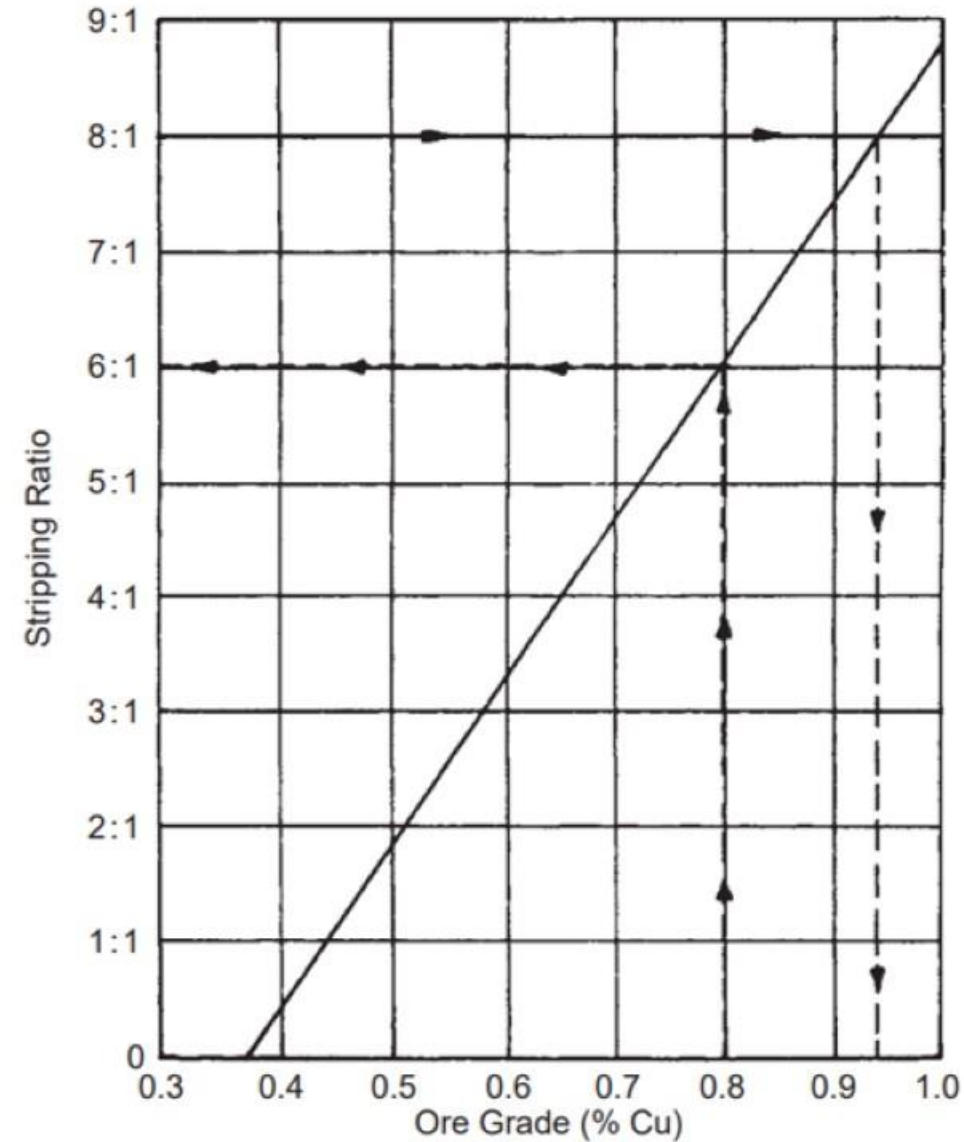


Figure 5.22. Example of stripping ratio calculation for a radial section.

PIT LIMITS

5.2.7 Generating a final pit outline

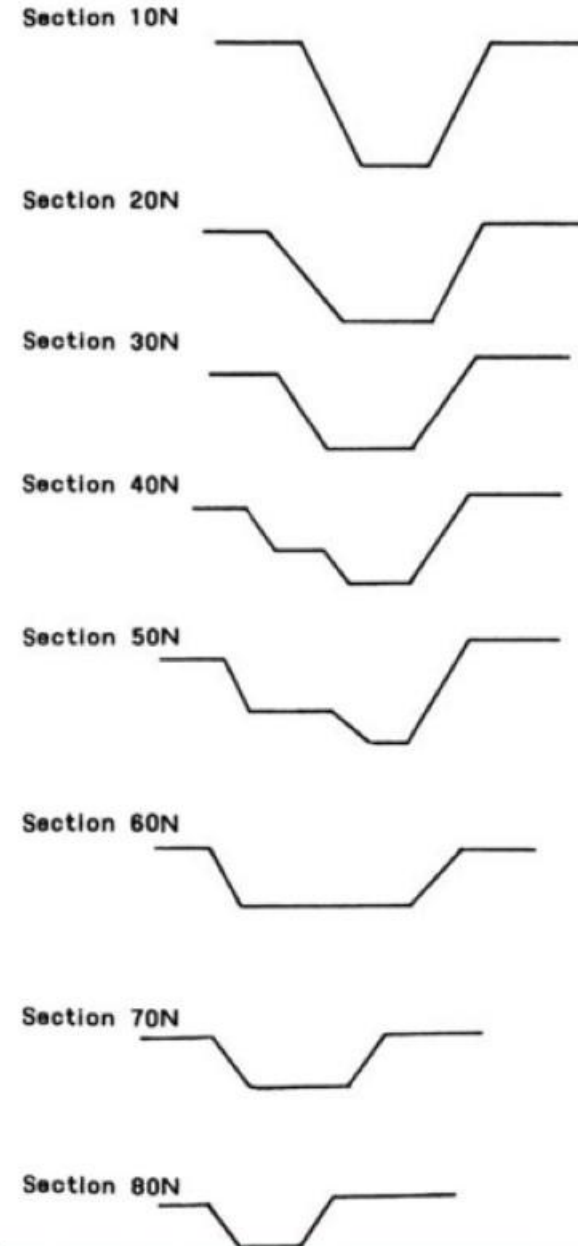


Figure 5.23. A series of initial pit outline sections for an orebody.

5.2.7 Generating a final pit outline

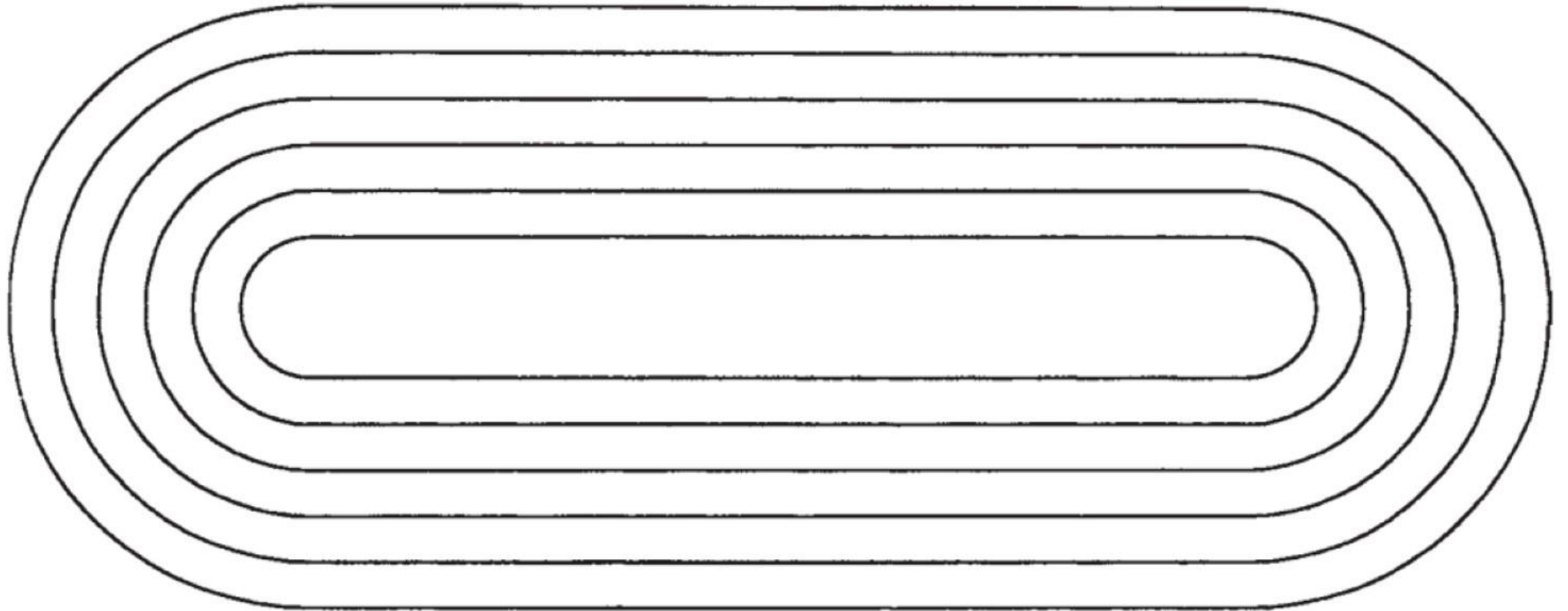


Figure 5.24. Plan view of a pit with constant slope angle.

5.2.7 Generating a final pit outline

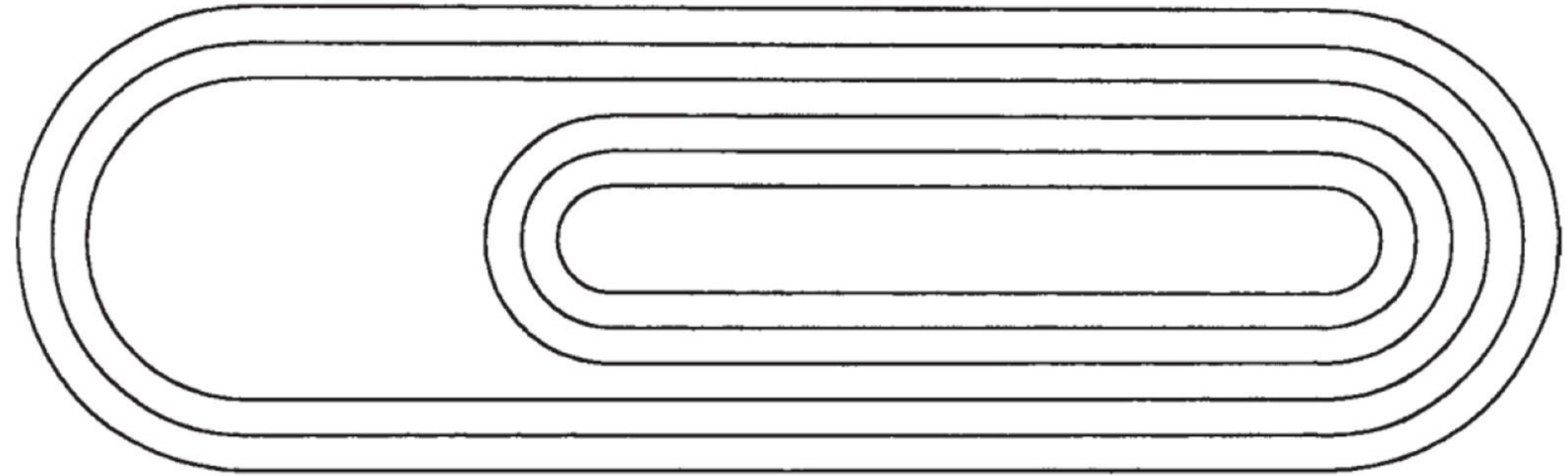


Figure 5.25. Plan view of a pit containing a wide bench.

5.2.7 Generating a final pit outline

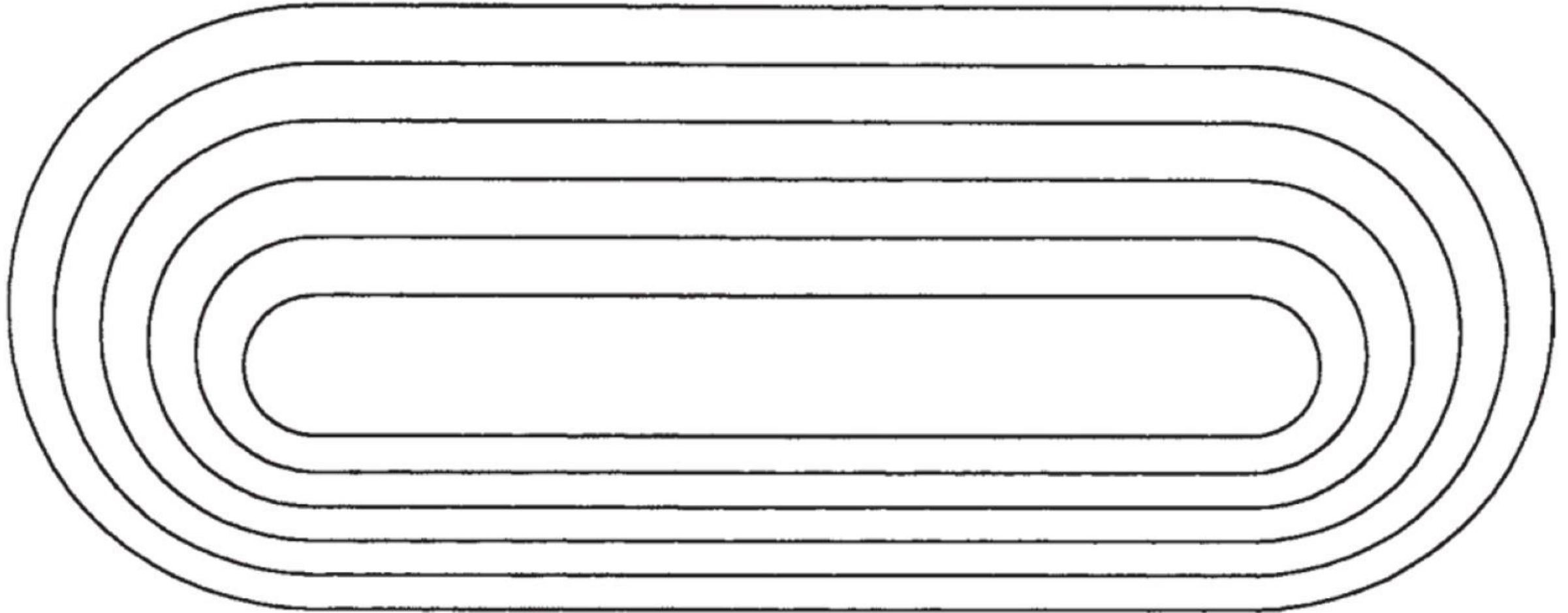


Figure 5.26. Plan view of a pit with different north and south wall slopes.

5.2.7 Generating a final pit outline

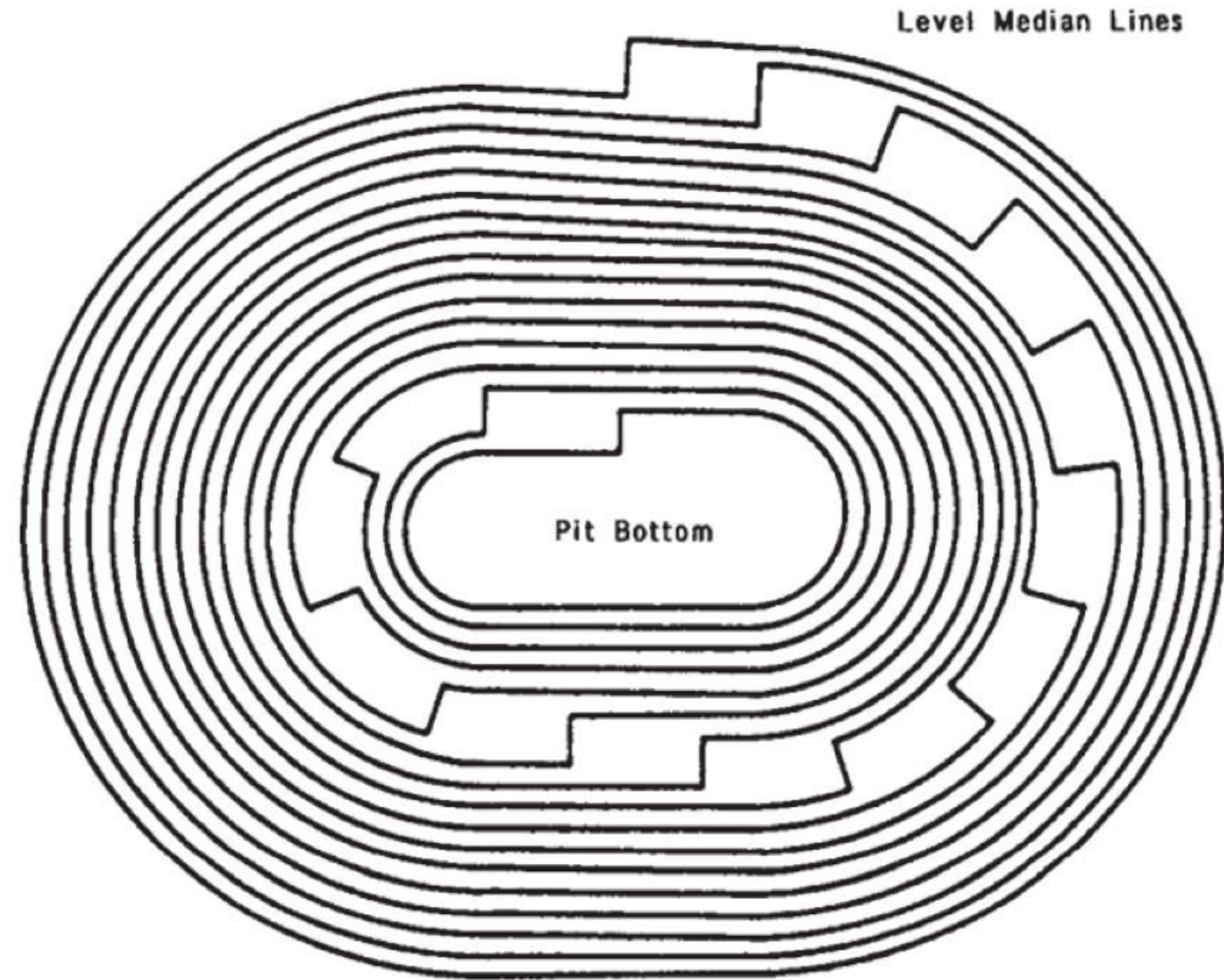


Figure 5.27. Composite ultimate pit plan (Koskiniemi, 1979).



PIT LIMITS

5.2.7 Generating a final pit outline

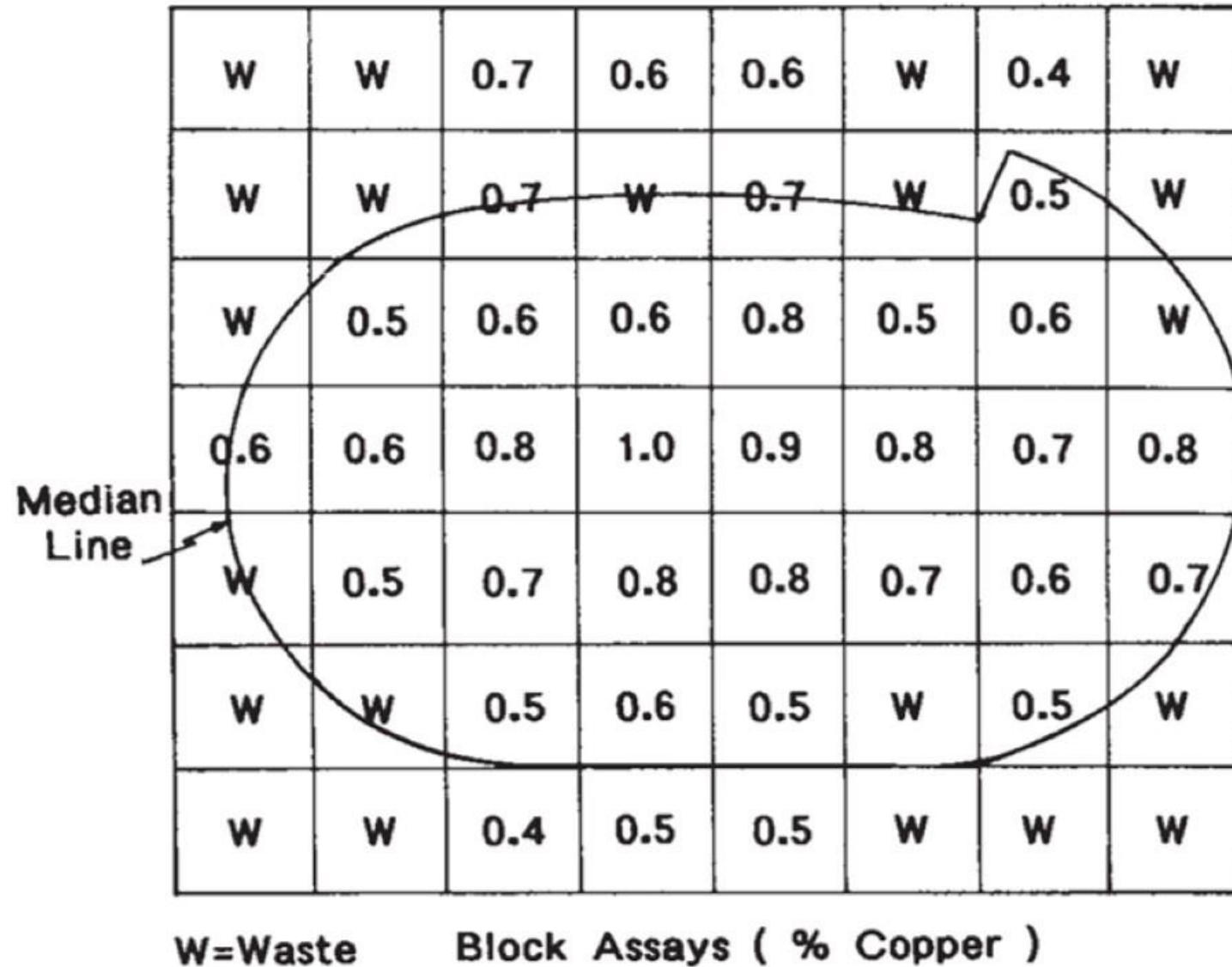


Figure 5.28. Level plan of bench a with the ultimate pit limit superimposed (Koskineemi, 1979).

5.4 THE FLOATING CONE TECHNIQUE

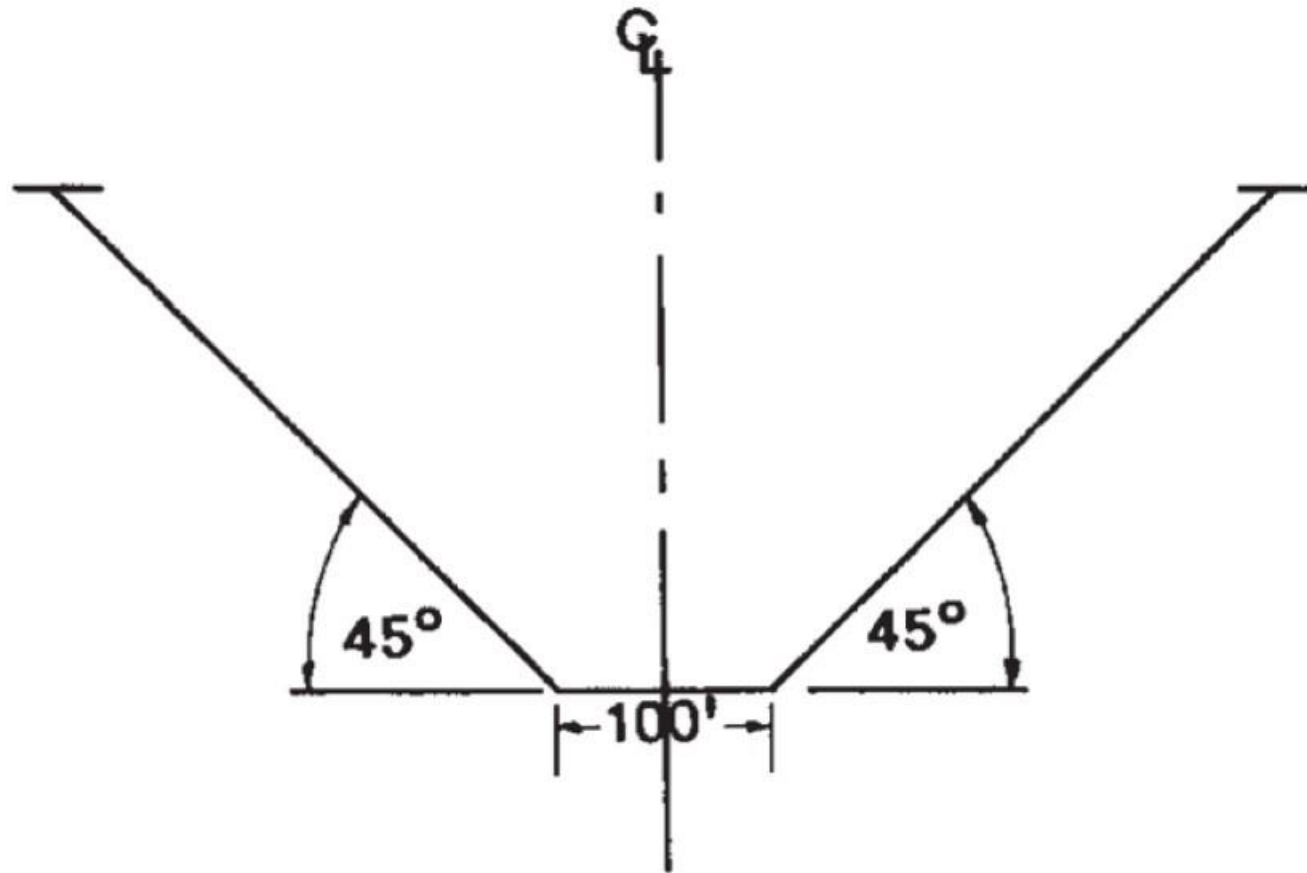


Figure 5.32. Pit profile to be 'floated'.

5.4 THE FLOATING CONE TECHNIQUE

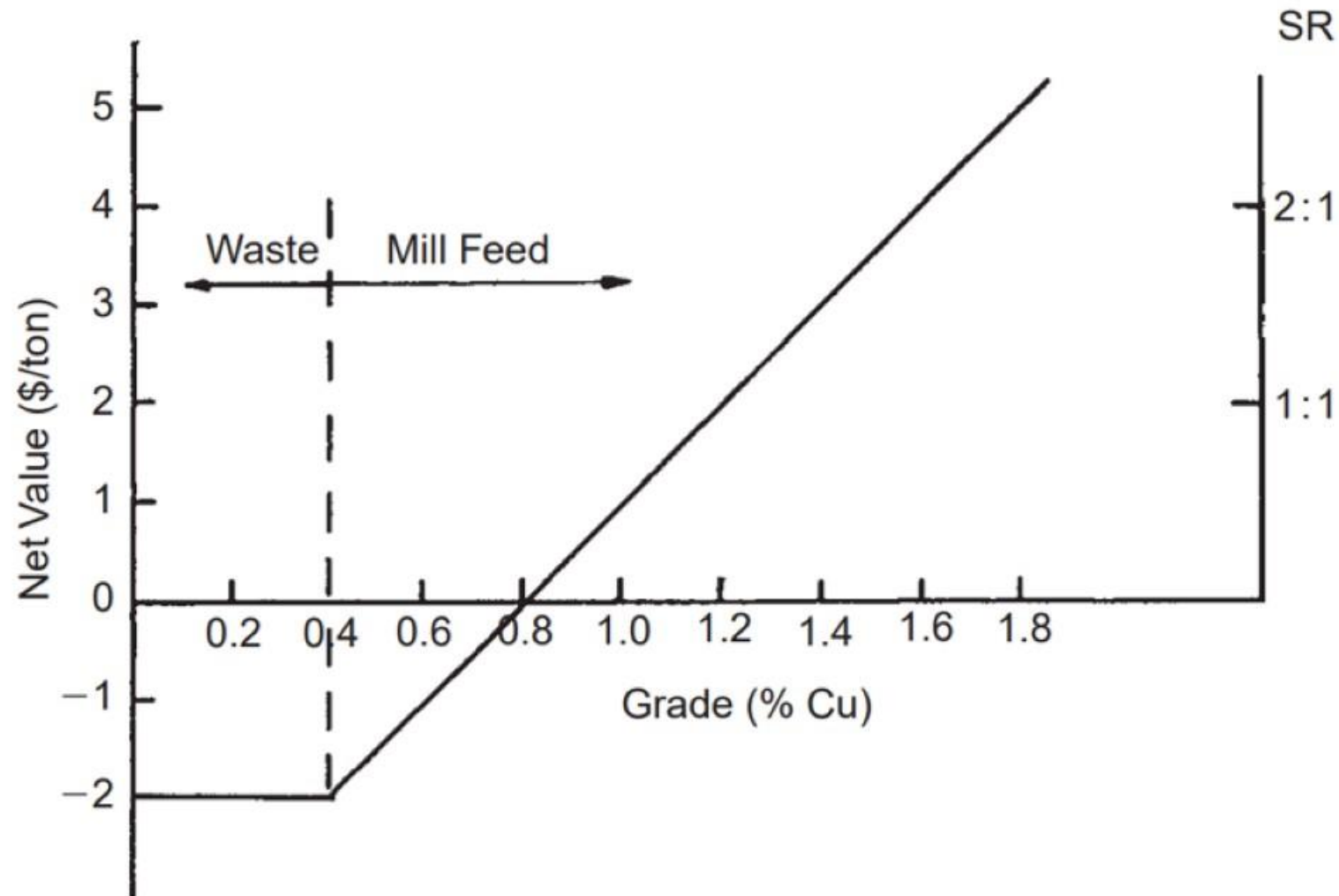


Figure 5.33. Net value – grade curve showing the waste/mill feed split.

5.4 THE FLOATING CONE TECHNIQUE

Section X

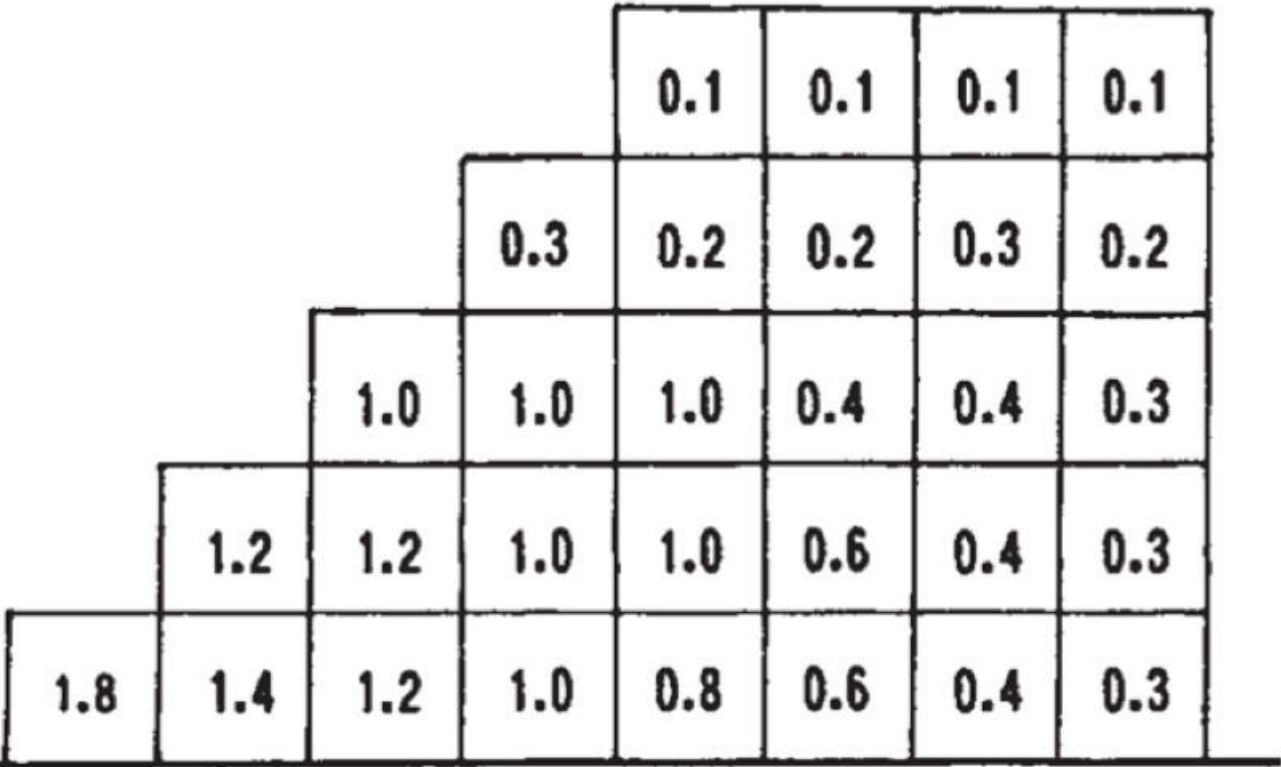


Figure 5.34. Grade block model for pit limit example.

PIT LIMITS

5.4 THE FLOATING CONE TECHNIQUE

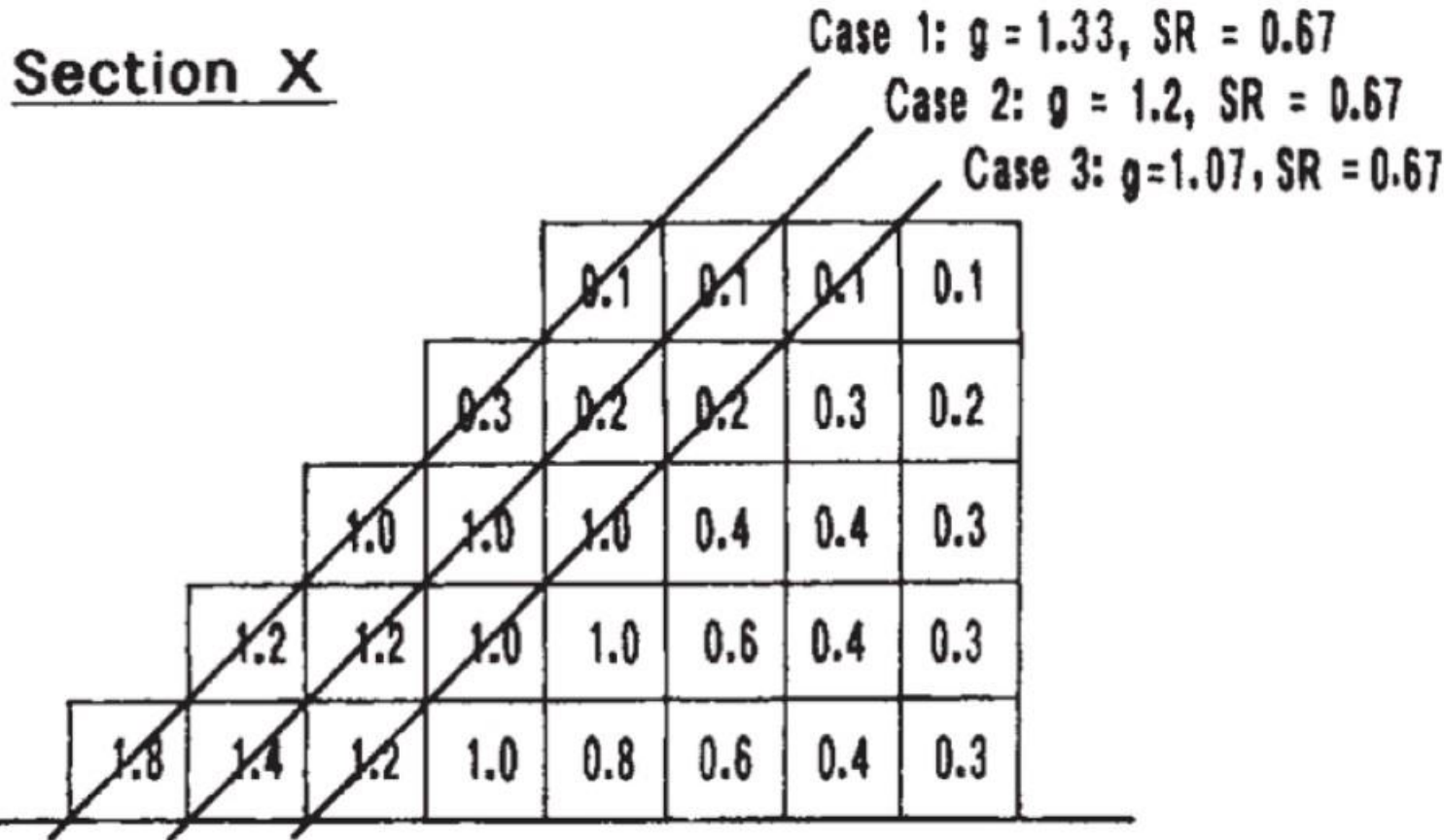


Figure 5.35. Trial final pit limits based on the manual procedure. Case 3 is the final limit.

5.4 THE FLOATING CONE TECHNIQUE

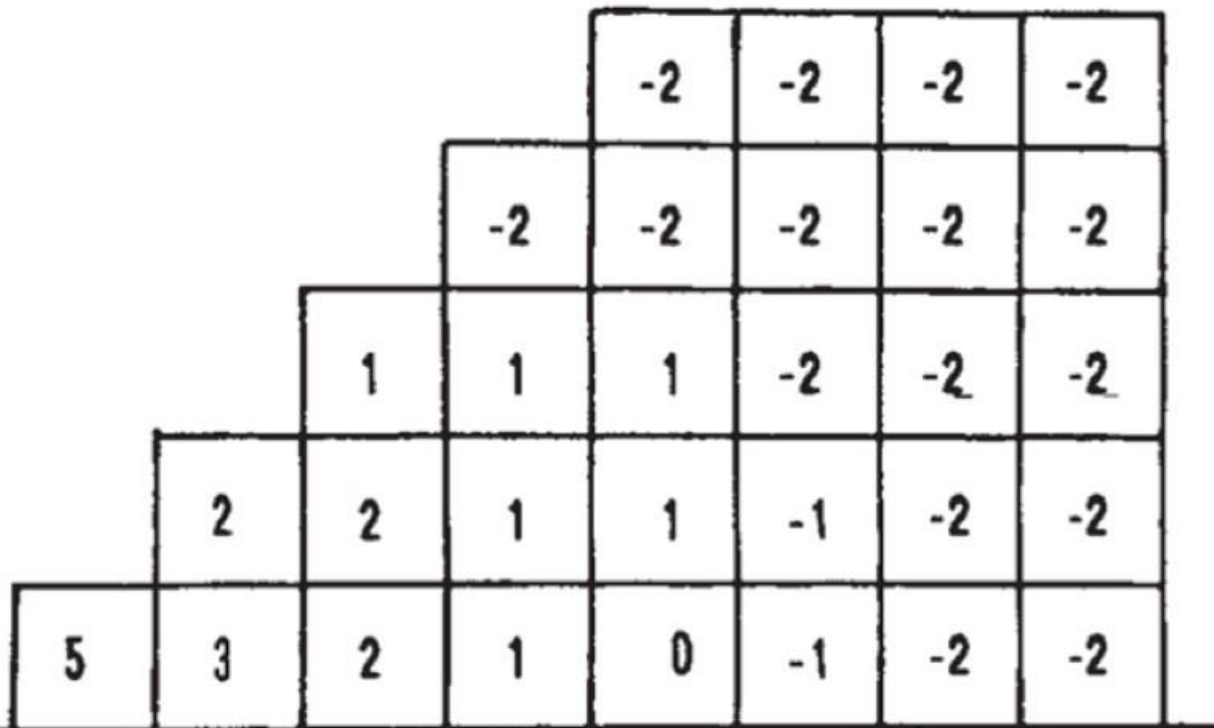


Figure 5.36. Corresponding economic block model.



PIT LIMITS

5.4 THE FLOATING CONE TECHNIQUE

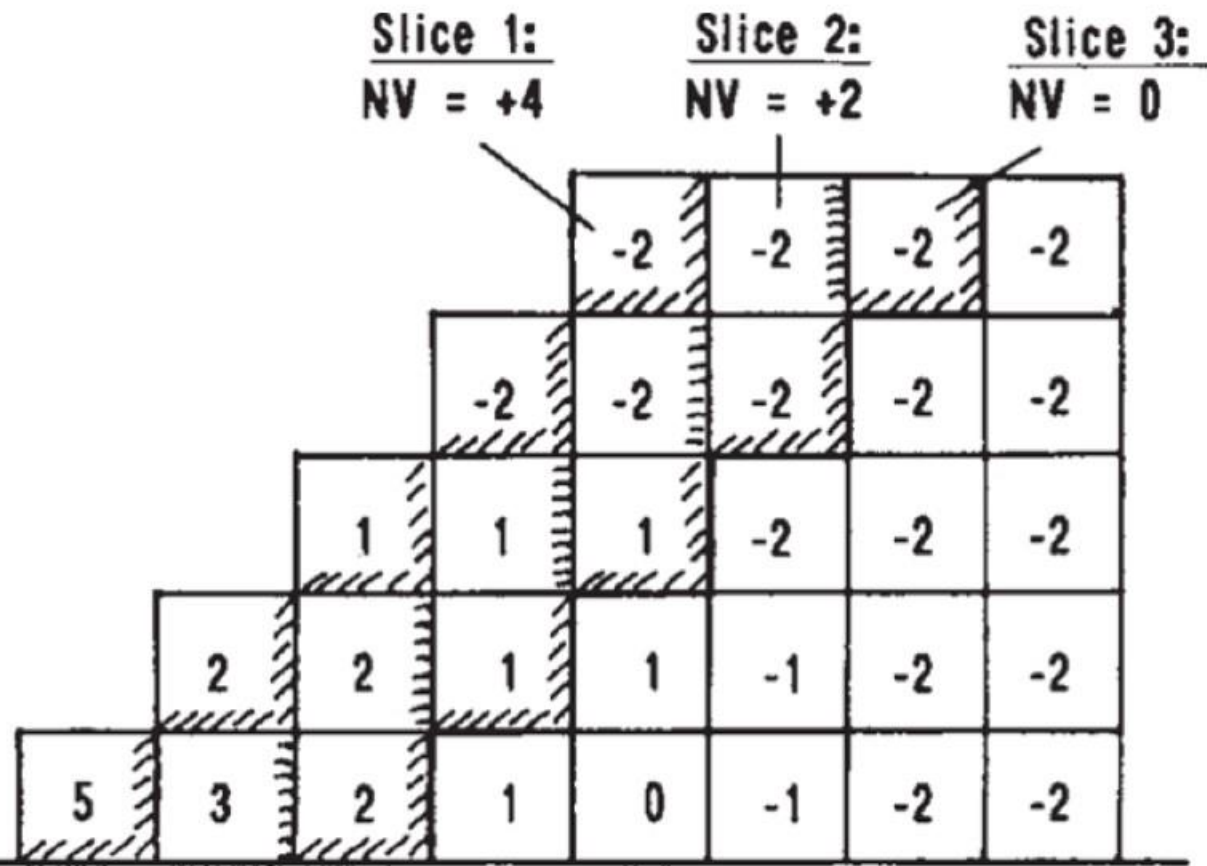


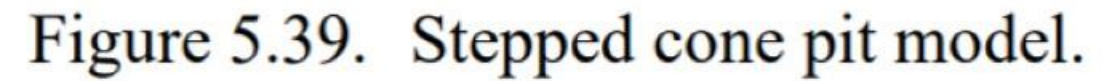
Figure 5.37. Trial final pit limits.

5.4 THE FLOATING CONE TECHNIQUE

	1	2	3	4	5	6	7
1	-1	-1	-1	-1	-1	+1	-1
2		-2	-2	+4	-2	-2	
3			+7	+1	-3		

Block
Value

Figure 5.38. Block model for Example 1 (Barnes, 1982).



5.4 THE FLOATING CONE TECHNIQUE

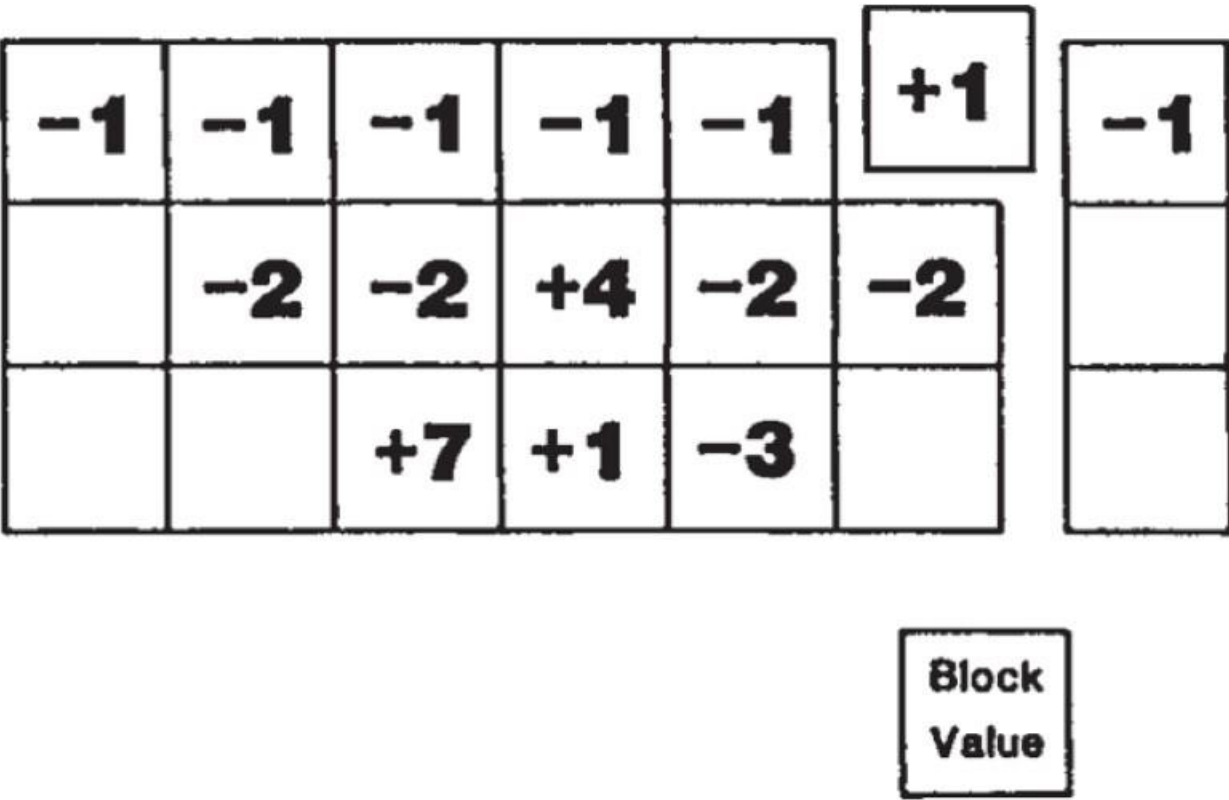


Figure 5.40. First incremental cone – Example 1 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

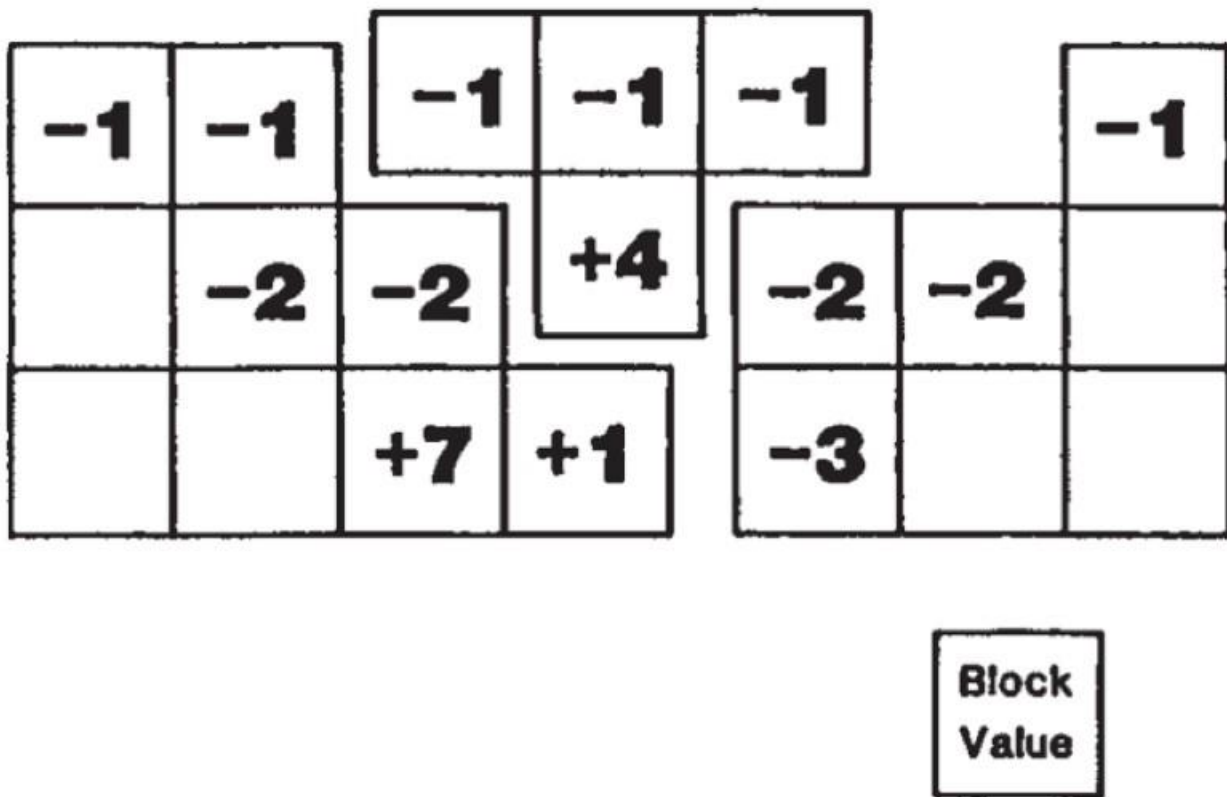


Figure 5.41. Second incremental cone – Example 1 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

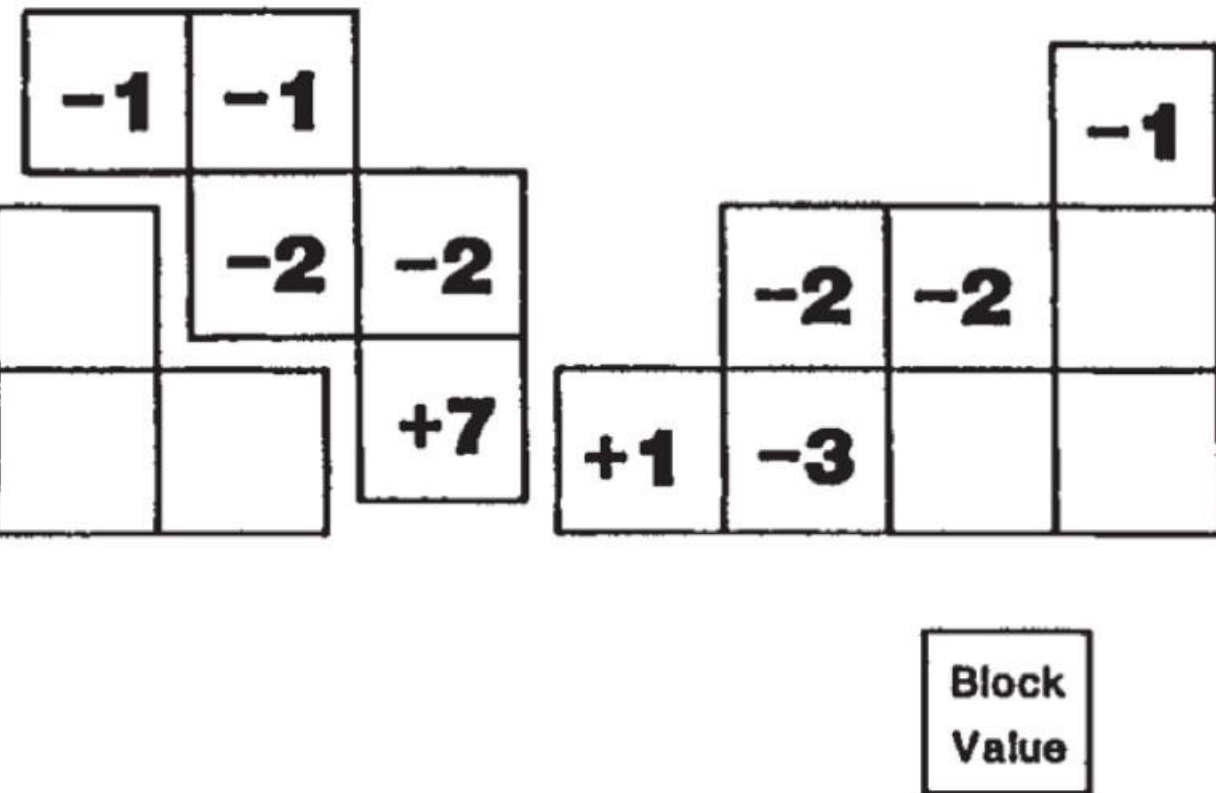


Figure 5.42. Third incremental cone – Example 1 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

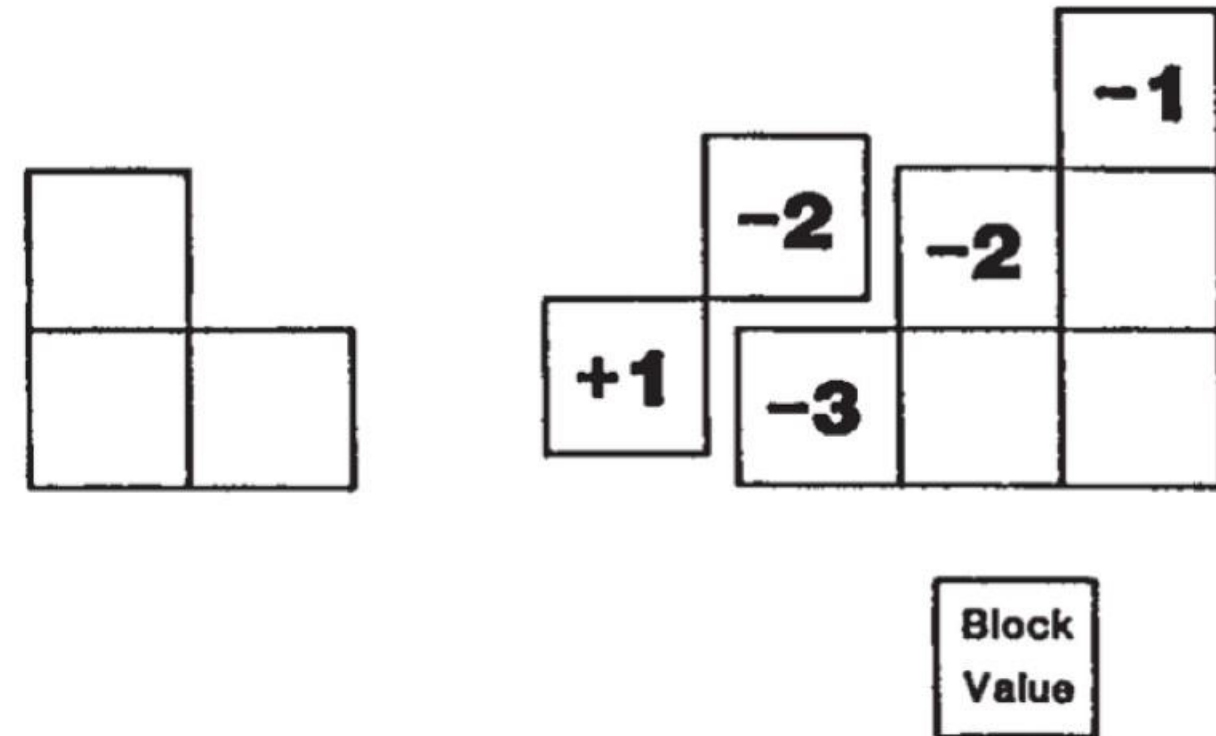


Figure 5.43. Fourth incremental cone – Example 1 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

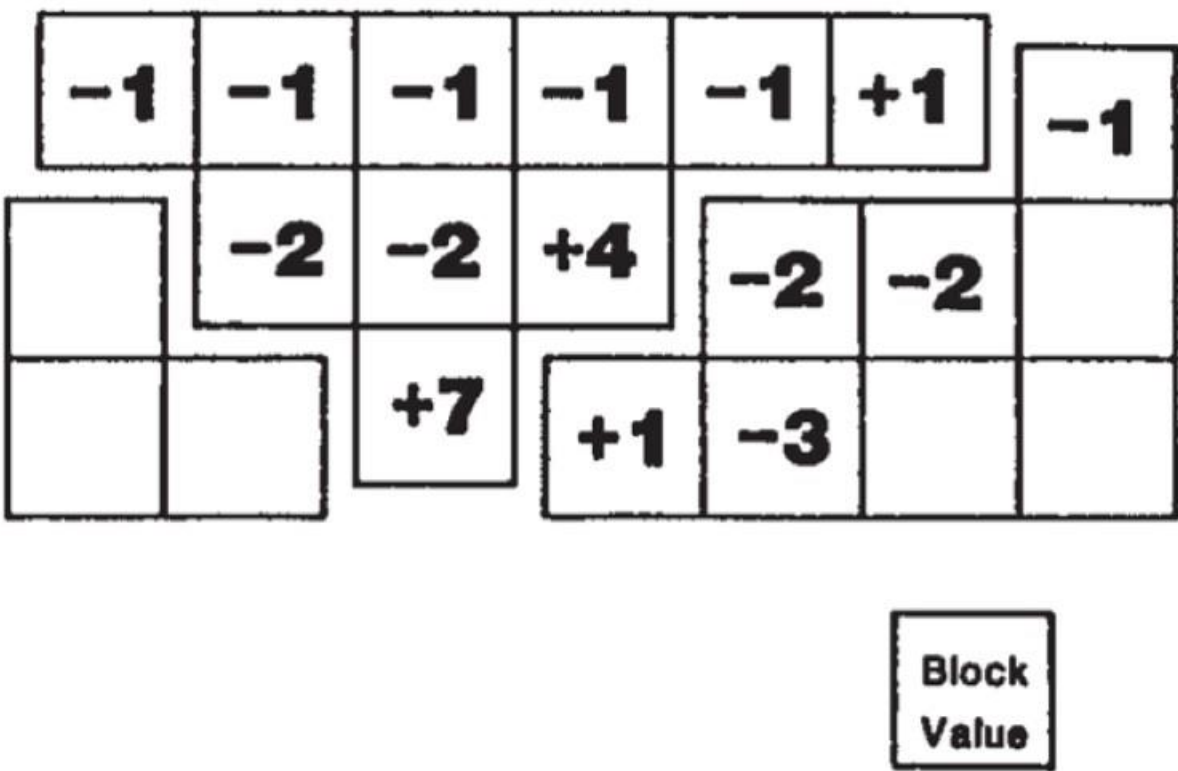


Figure 5.44. Ultimate pit – Example 2 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

Problem 1. Missing combinations of profitable blocks

	1	2	3	4	5	6	7
1	-1	-1	-1	-1	-1	-1	-1
2		-2	-2	-2	-2	-2	
3			+10	-3	+10		

Block
Value

Figure 5.45. Block model for Example 2 (Barnes, 1982).

-1	-1	-1	-1	-1	-1	-1
	-2	-2	-2	-2	-2	
		+10	-3	+10		

Block
Value

Figure 5.46. Second incremental cone – Example 2 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

Problem 1. Missing combinations of profitable blocks

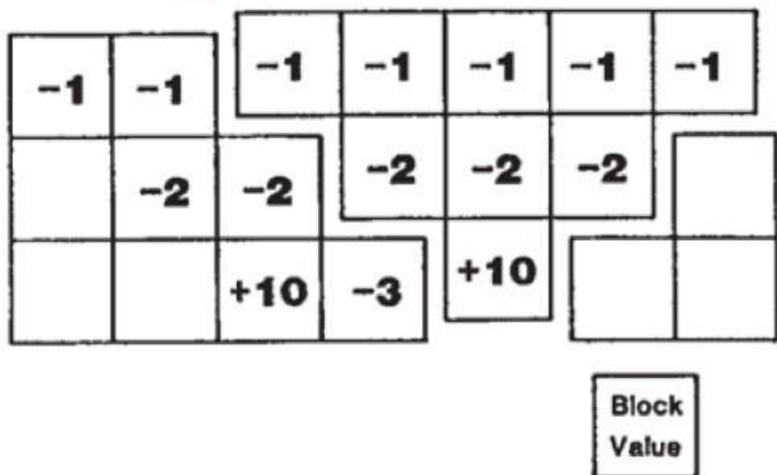


Figure 5.47. Second incremental cone – Example 2 (Barnes, 1982).

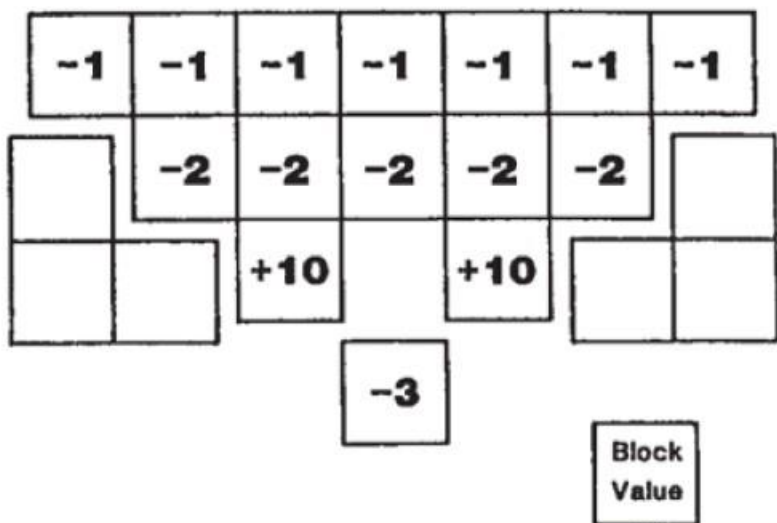


Figure 5.48. True optimal pit – Example 2 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

Problem 2. Extending the ultimate pit beyond the optimal pit limits

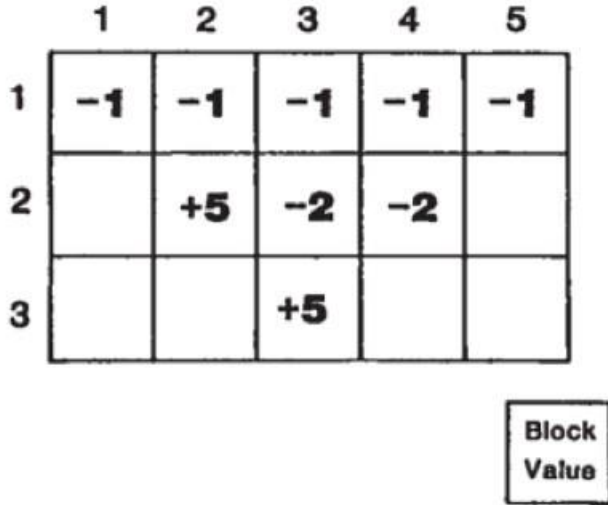


Figure 5.49. Block model for Example 3 (Barnes, 1982).

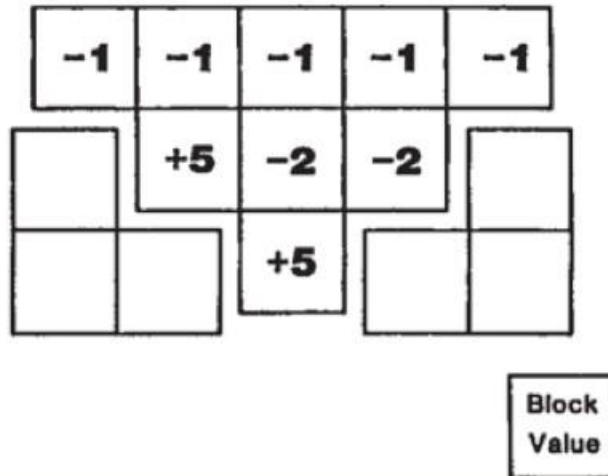


Figure 5.50. Large incremental cone for Example 3 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

Problem 2. Extending the ultimate pit beyond the optimal pit limits

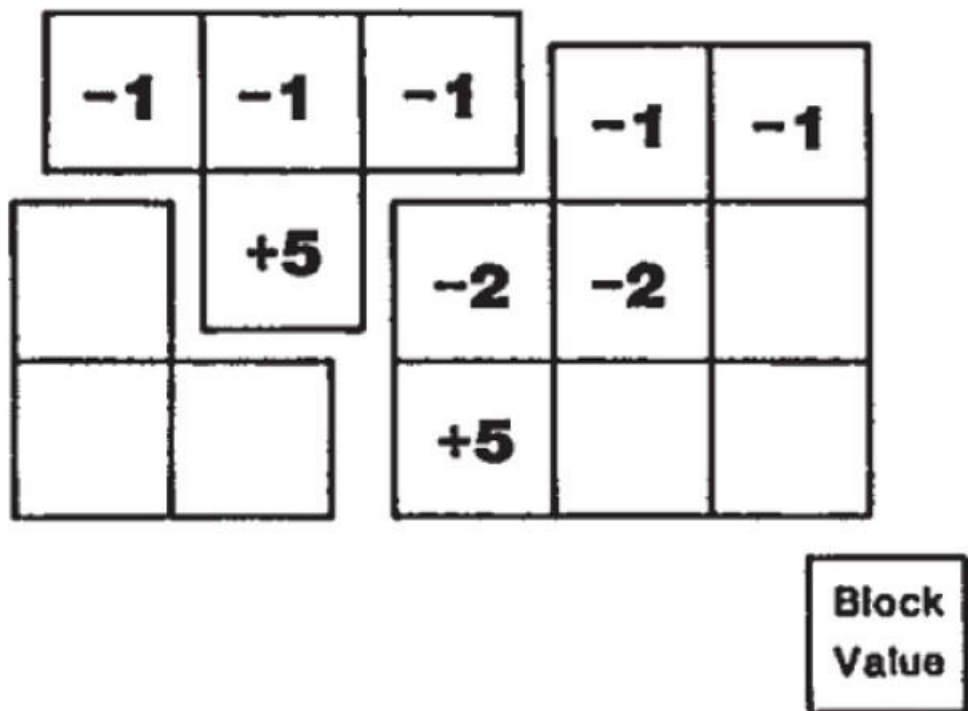


Figure 5.51. Small incremental cone for Example 4 (Barnes, 1982).

5.4 THE FLOATING CONE TECHNIQUE

Problem 3. Combination of problems 1 and 2

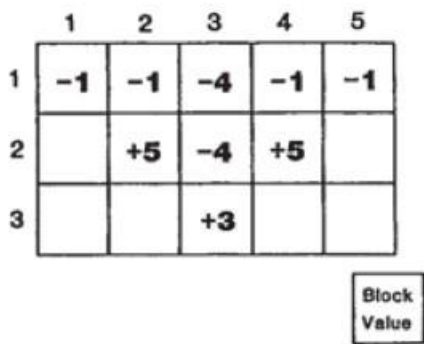


Figure 5.52. Block model for Example 4 (Barnes, 1982).

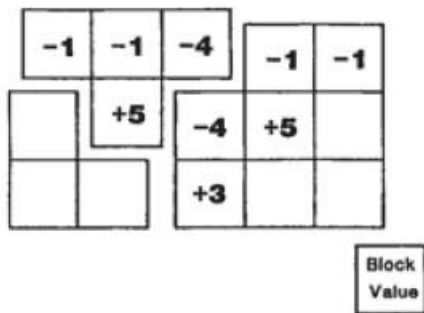


Figure 5.53. First incremental cone for Example 4 (Barnes, 1982).

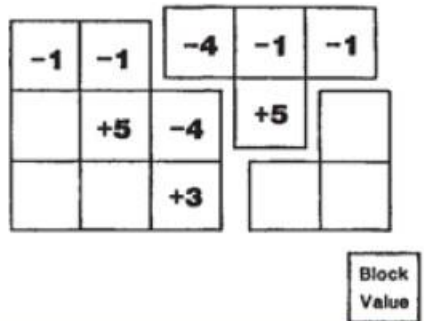
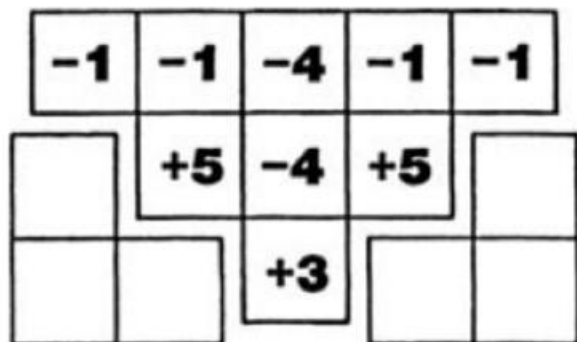


Figure 5.54. Second incremental cone for Example 4 (Barnes, 1982).

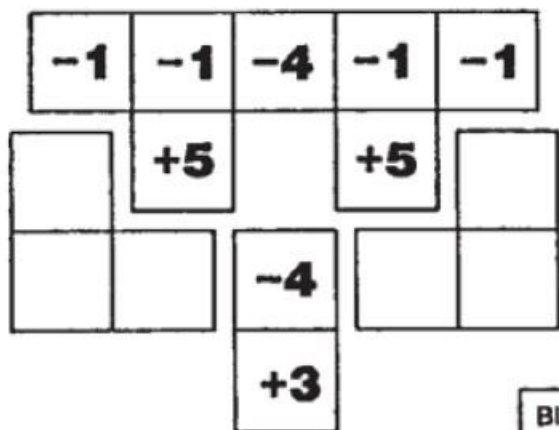
5.4 THE FLOATING CONE TECHNIQUE

Problem 3. Combination of problems 1 and 2



Block
Value

Figure 5.55. Third incremental cone for Example 4 (Barnes, 1982).



Block
Value

Figure 5.56. Optimal pit design for Example 4 (Barnes, 1982).

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

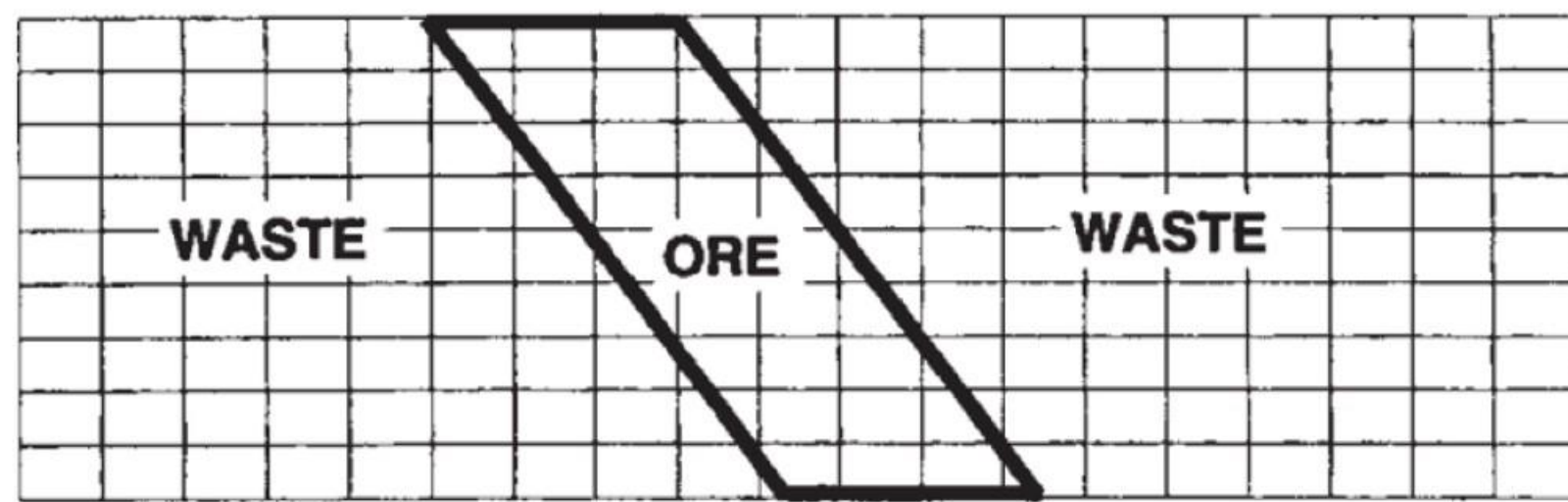


Figure 5.57. Orebody geometry for the Lerchs-Grossmann 2-D example (Sainsbury, 1970; Lerchs & Grossmann, 1965).

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

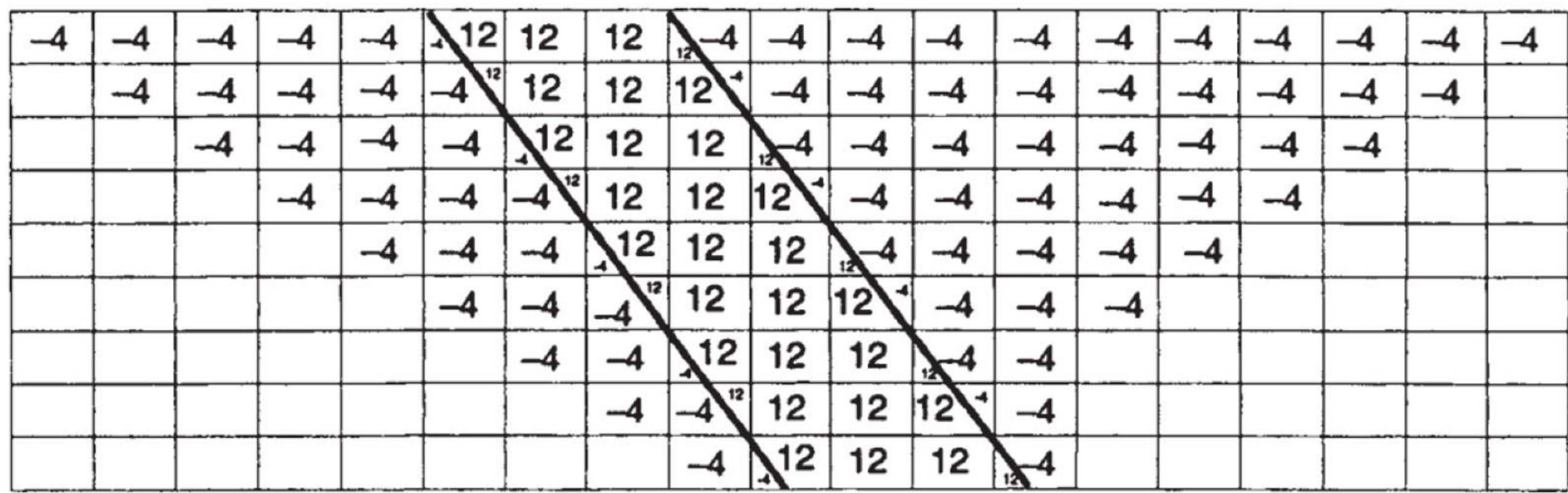


Figure 5.58. Initial economic block model.

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

$i \backslash j$	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
①	-4	-4	-4	-4	-4	8	12	12	0	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
②		-4	-4	-4	-4	0	12	12	8	-4	-4	-4	-4	-4	-4	-4	-4	-4	
③			-4	-4	-4	-4	8	12	12	0	-4	-4	-4	-4	-4	-4	-4		
④				-4	-4	-4	0	12	12	8	-4	-4	-4	-4	-4	-4			
⑤					-4	-4	-4	8	12	12	0	-4	-4	-4	-4				
⑥						-4	-4	0	12	12	8	-4	-4	-4					
⑦							-4	-4	8	12	12	0	-4						
⑧								-4	0	12	12	8	-4						
⑨									-4	12	12	12	0						

Figure 5.59. Final economic block model (Lerchs & Grossmann, 1965).



5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

Row	Current Value	Revised Value
$i = 1$	12	12
$i = 2$	12	$24 = 12 + 12$
$i = 3$	8	$32 = 12 + 12 + 8$
$i = 4$	0	$32 = 12 + 12 + 8 + 0$
$i = 5$	-4	$28 = 12 + 12 + 8 + 0 - 4$
$i = 6$	-4	$24 = 12 + 12 + 8 + 0 - 4 - 4$
$i = 7$	-4	$20 = 12 + 12 + 8 + 0 - 4 - 4 - 4$
$i = 8$	-4	$16 = 12 + 12 + 8 + 0 - 4 - 4 - 4 - 4$

Figure 5.60. Calculation of the cumulative sums for column 6.



PIT LIMITS

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

Columns

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
1	-4	-4	-4	-4	-4	-8	12	12	0	-4	-4	-4	-4	-4	-4	-4	-4	-4	-4
2	-8	-8	-8	-8	-8	8	24	24	8	-8	-8	-8	-8	-8	-8	-8	-8	-8	
3	-12		-12	-12	-12	4	32	36	20	-8	-12	-12	-12	-12	-12	-12	-12		
4				-16	-16	0	32	48	32	0	-16	-16	-16	-16	-16	-16			
5					-20	-4	28	56	44	12	16	-20	-20	-20	-20				
6						-8	24	56	56	24	-8	-24	-24	-24					
7							20	52	64	36	4	-24	-28						
8							16	48	64	48	16	-16	-32						
9									60	56	28	-4	32						

Figure 5.61. Completed cumulative sums (Lerchs & Grossmann, 1965).

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

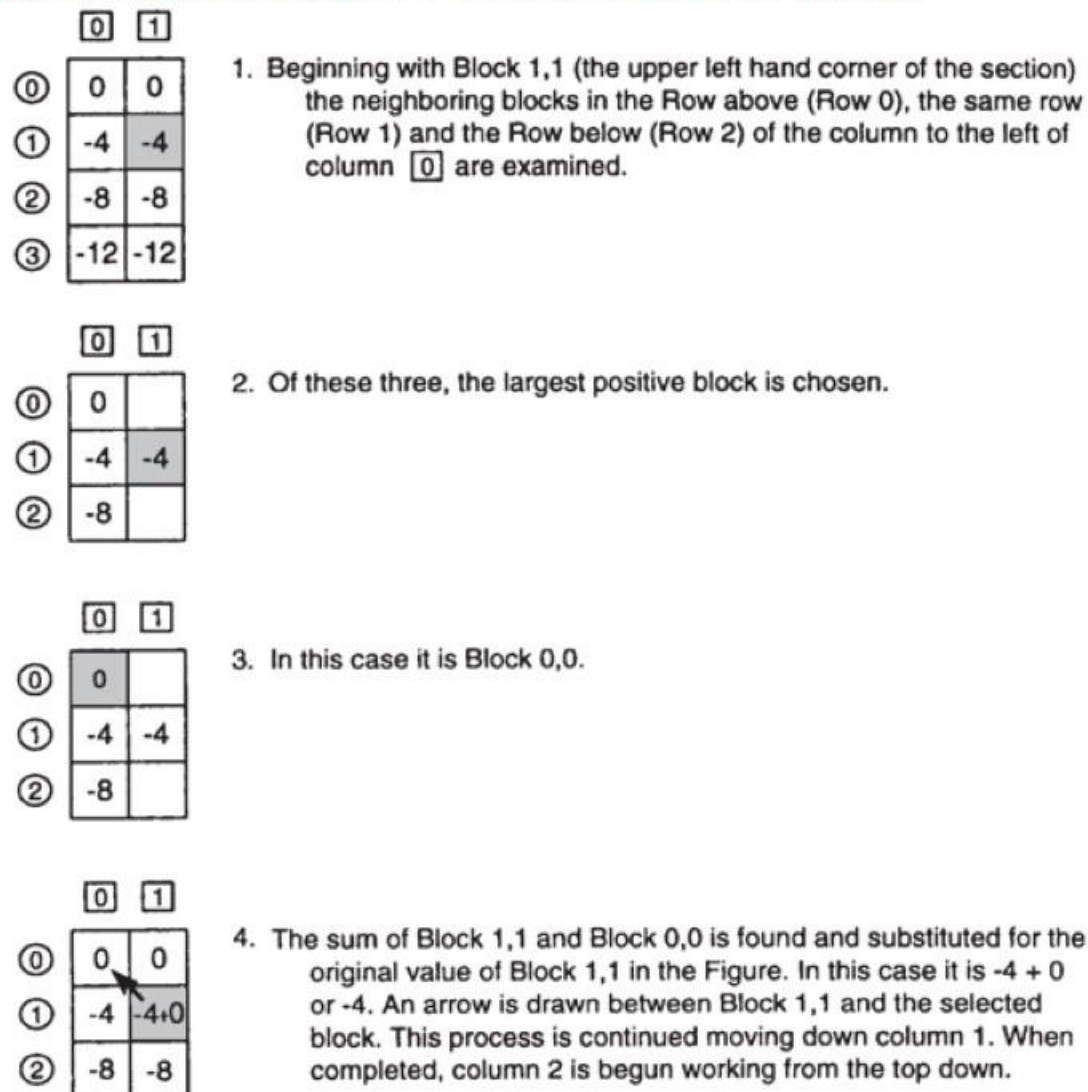


Figure 5.62. Procedure for determining the cumulative maximum value and maximizing direction.



PIT LIMITS

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

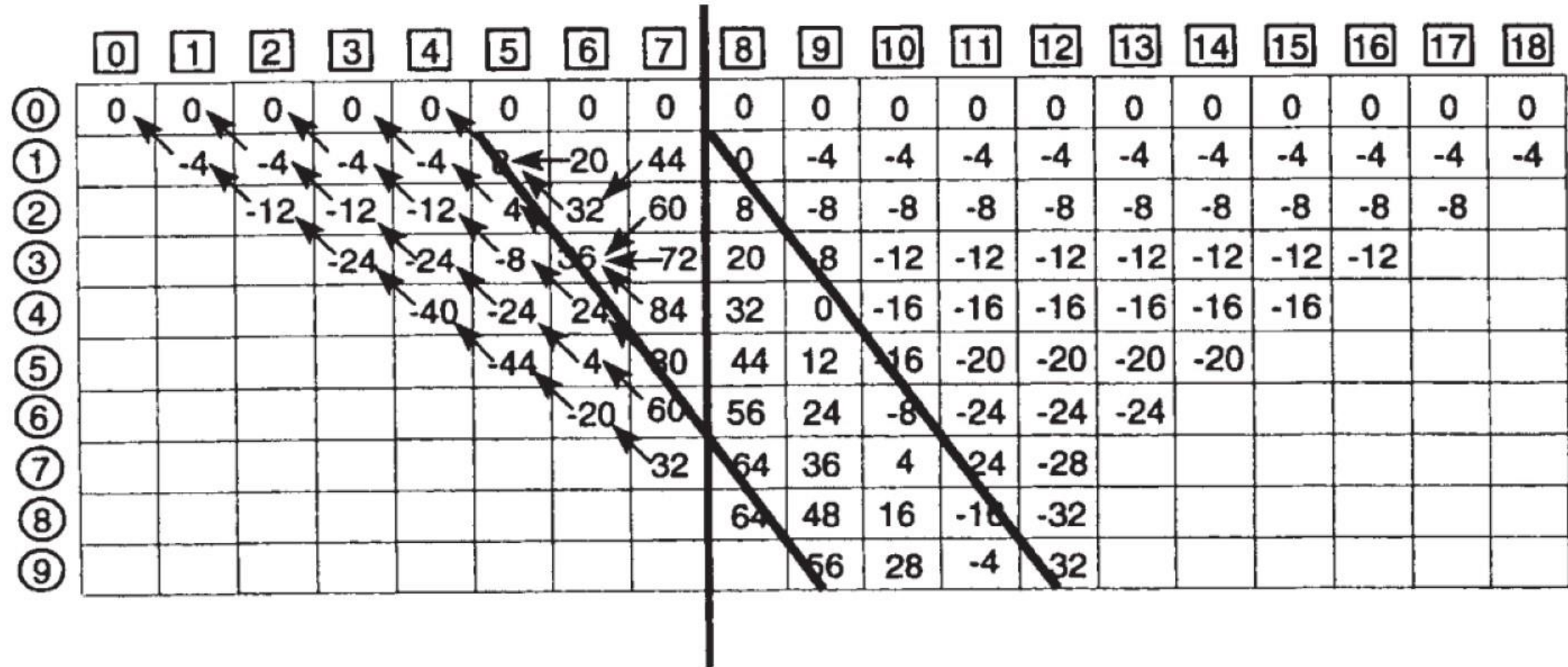


Figure 5.63. Progression of summing process through column 7.

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

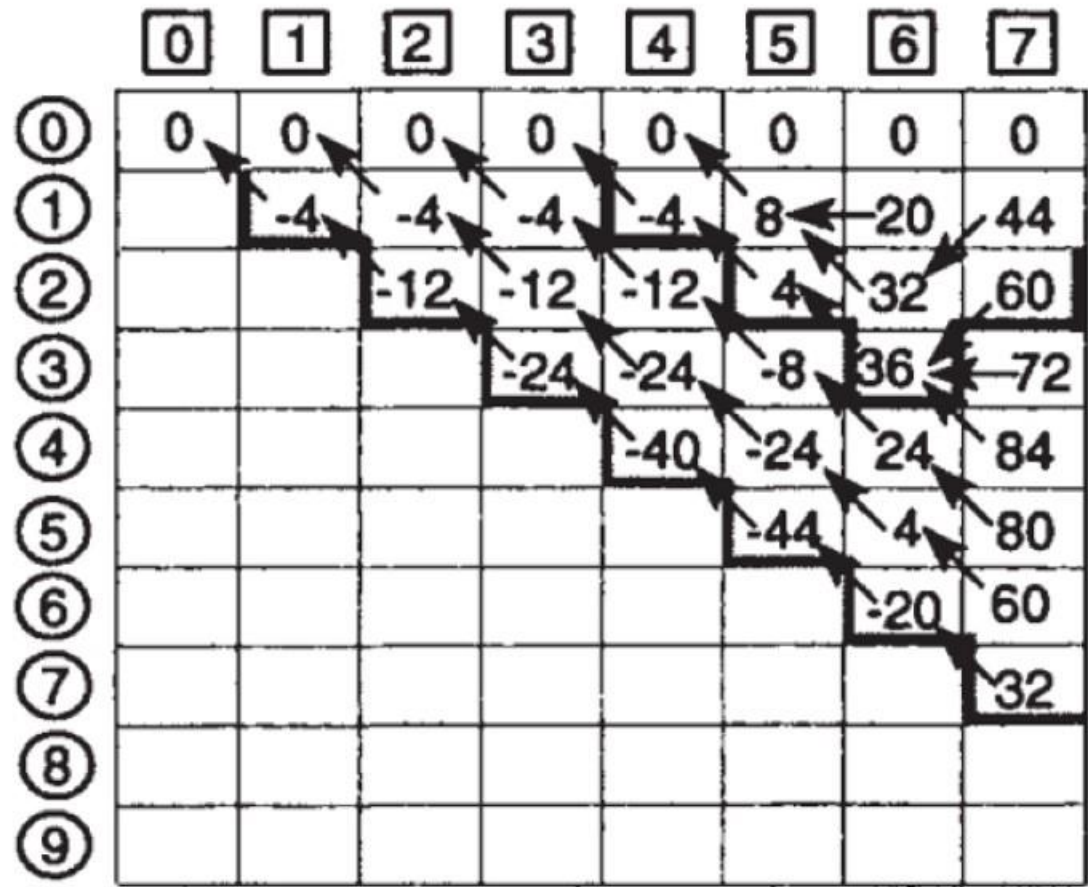


Figure 5.64. Pit determination and total value by following the arrows.

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

	0	1	2	3	4	5	6	7
0	0	0	0	0	0	0	0	0
1		-4	-4	-4	-4	8	12	12
2			-4	-4	-4	0	12	12
3				-4	-4	-4	8	12
4					-4	-4	0	12
5						-4	-4	8
6							-4	0
7								-4
8								
9								

Figure 5.65. Individual block values for the two partial pits.



5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

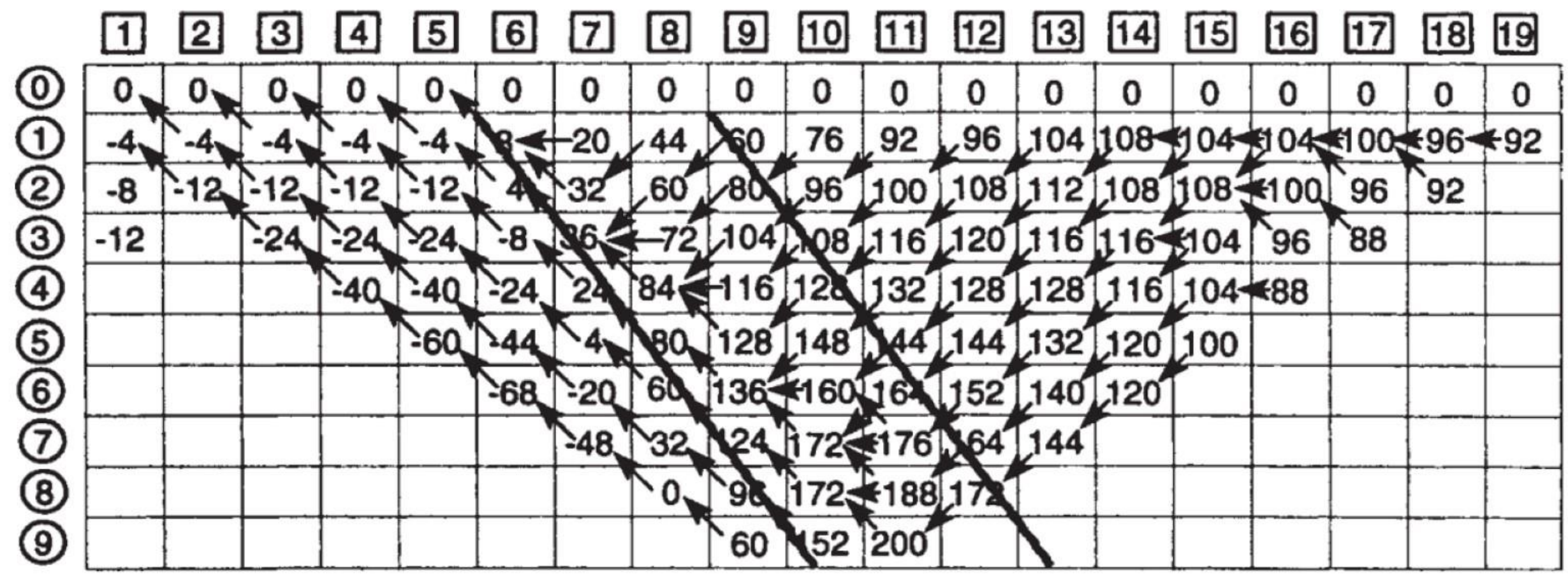


Figure 5.66. The summing process has progressed through the entire section (Lerchs & Grossmann, 1965).

PIT LIMITS

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

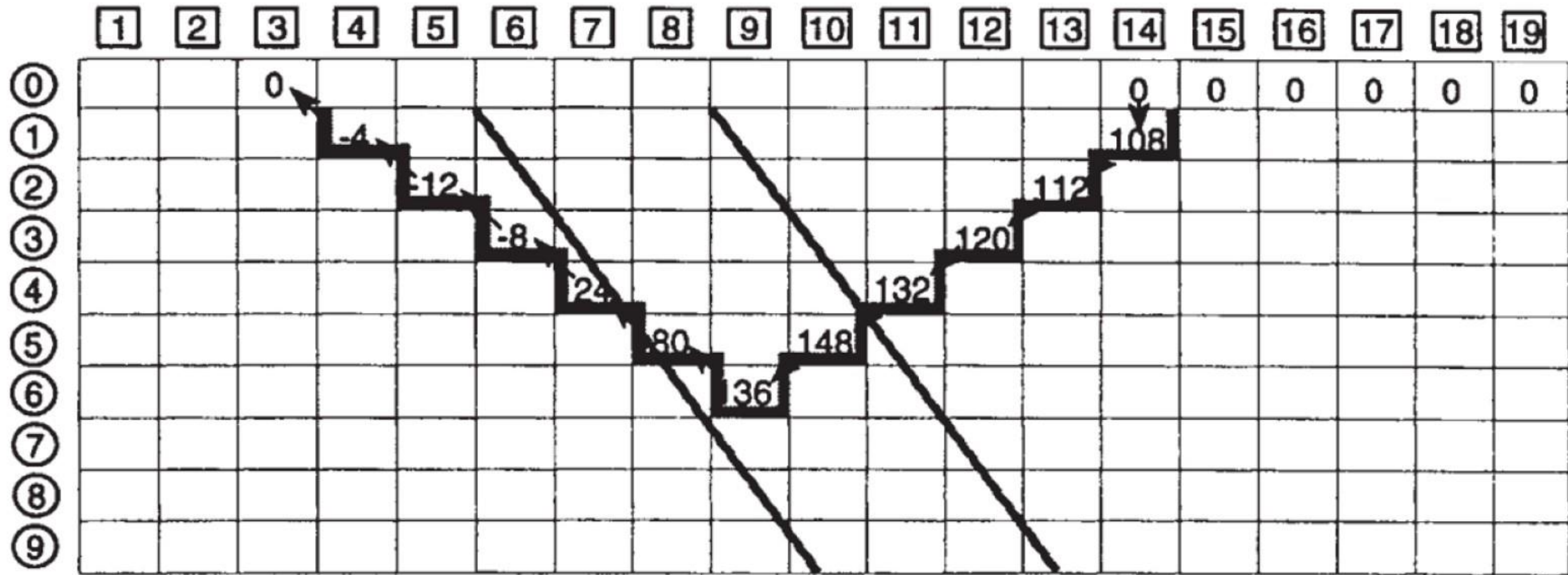


Figure 5.67. Optimum pit determination.



PIT LIMITS

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

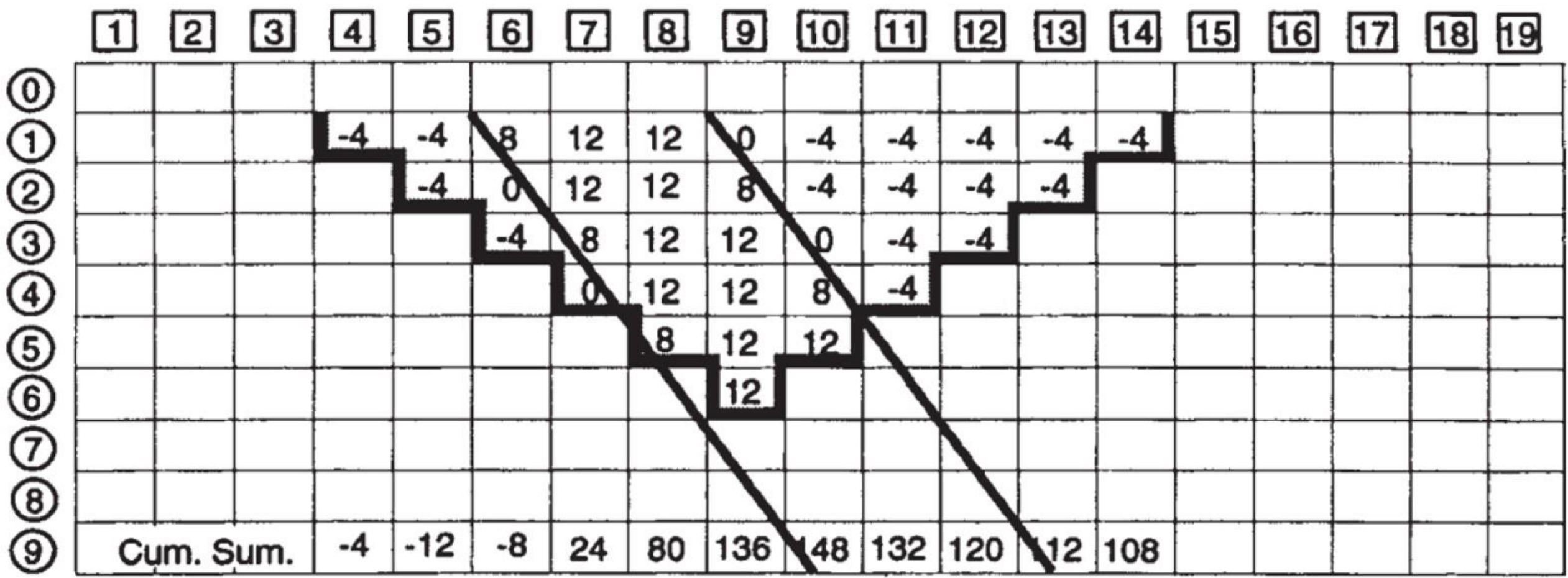


Figure 5.68. Optimum pit limits superimposed on the block model.

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

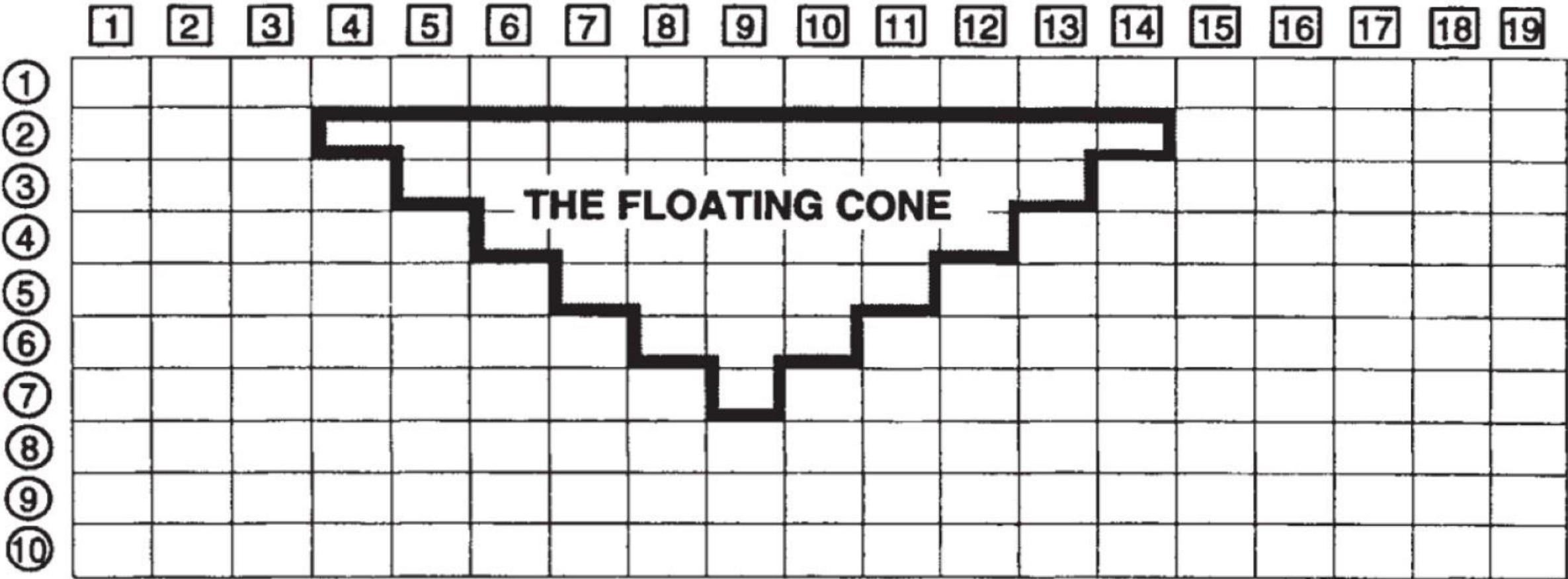


Figure 5.69. The floating cone used to evaluate the final pit limits.

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM



Figure 5.70. Situation after floating down two rows.



Figure 5.71. Situation after floating down three rows.

5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

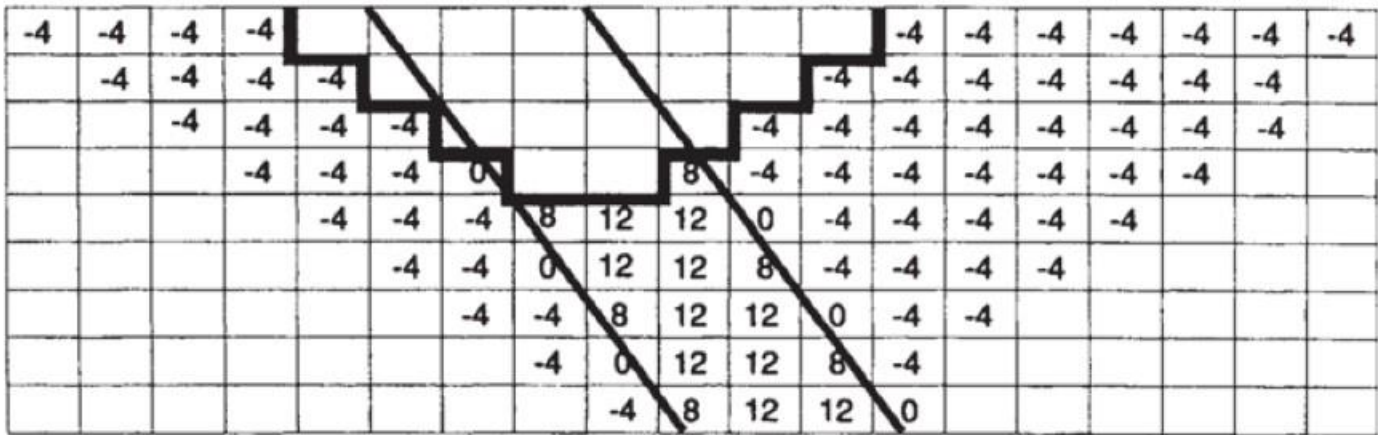


Figure 5.72. Situation after floating down four rows.

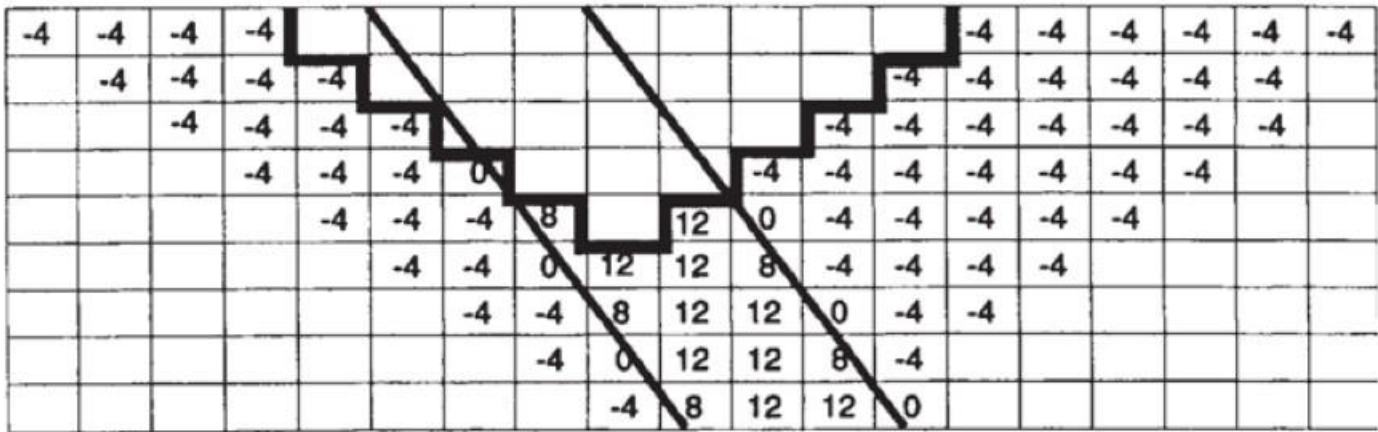


Figure 5.73. Situation after floating down five rows.



5.5 THE LERCHS-GROSSMANN 2-D ALGORITHM

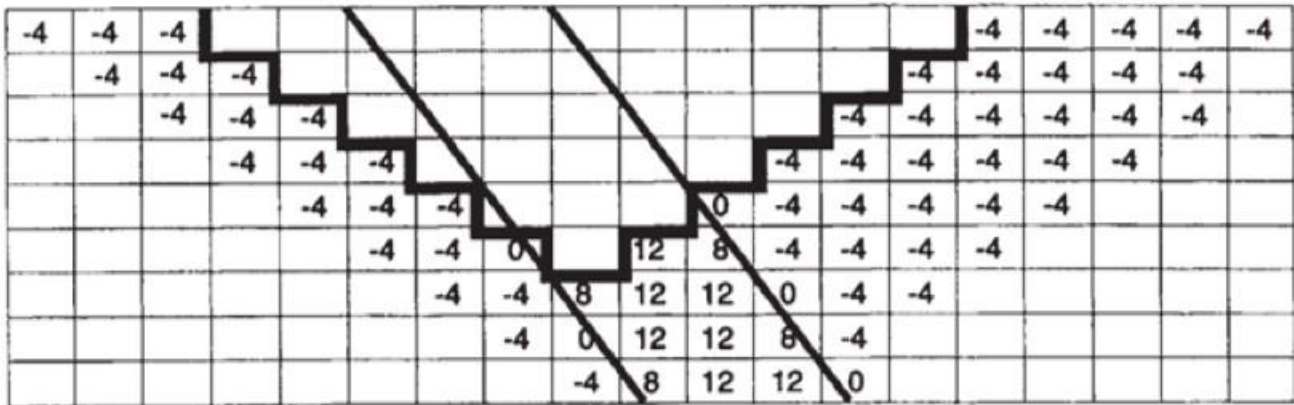


Figure 5.74. Situation after floating down six rows.

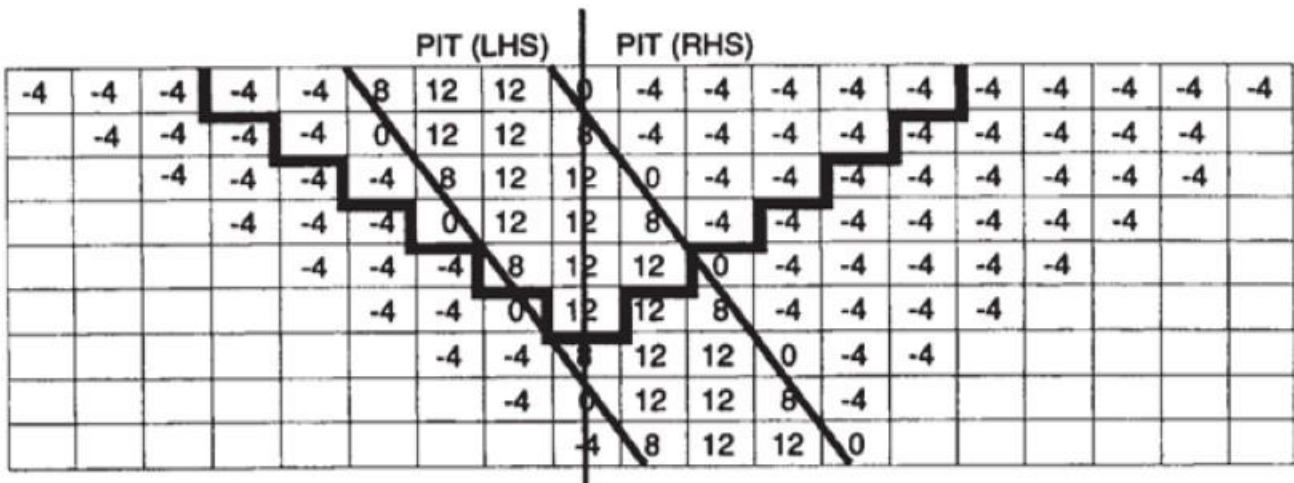
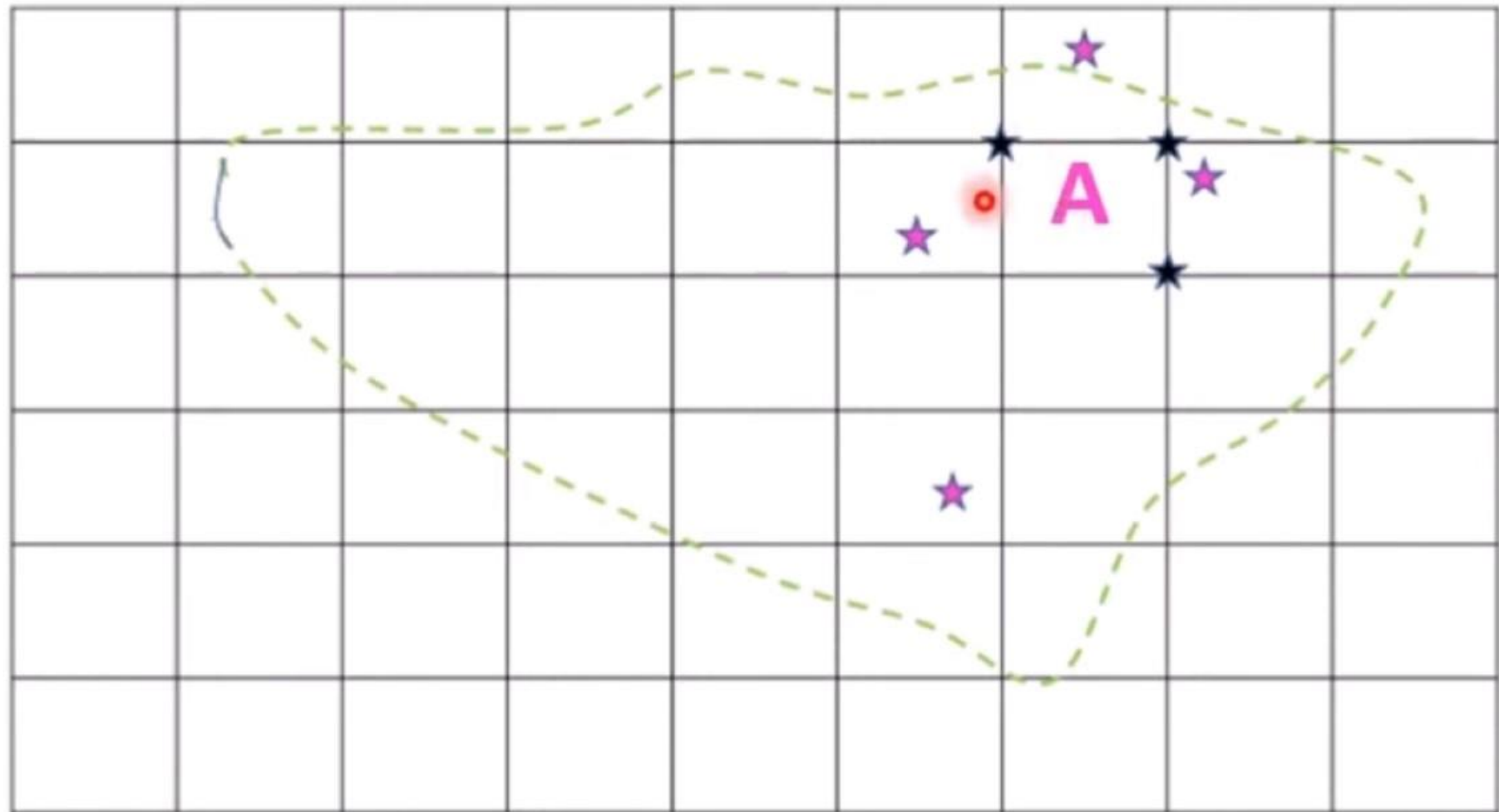


Figure 5.75. Pit resulting from the manual approach using the stripping curves.

Slice plan at 60 MRL

★ Grid point

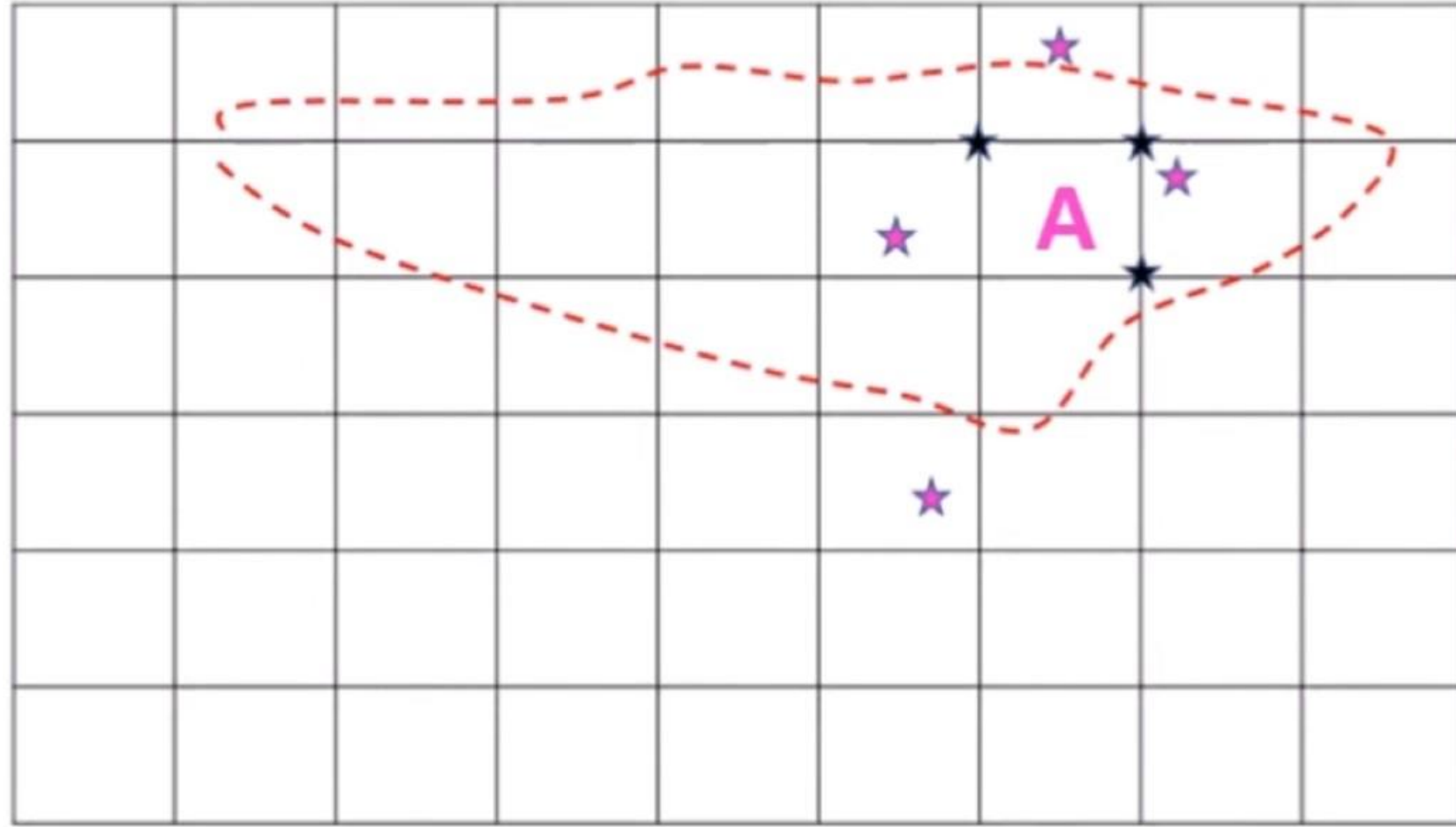
★ Bore hole



Slice plan at 70 MRL

★ Grid point

★ Bore hole



The following apply:

$$NV_{st} = -\$4 \times 10^3/\text{block}$$

$$NV_{mf} = \$12 \times 10^3/\text{block}$$

$$\text{Slope angle} = 35.5^\circ$$

$$\text{Bench height} = 40 \text{ ft}$$

$$\text{Tonnage factor} = 12.5 \text{ ft}^3/\text{st}$$

For ease in calculation the values \$12 and -\$4 will be used. At the conclusion, the factor of 1000 will be reintroduced. To apply this technique the grid (block geometry) is selected based upon the slope angle. The slope is formed by moving up one block and over one block. Hence for a bench height of 40 ft and a slope angle of 35.5° one finds using Equation (5.14)

$$\alpha = \frac{H}{B} = \tan \theta \quad (5.14)$$

where α is the ratio of block height/block width, H is the block height, B is the block width, and θ is the slope angle, that

$$B = \frac{H}{\tan \theta} = \frac{40}{\tan 35.5^\circ} = 56 \text{ ft}$$

Hence

$$\alpha = \frac{5}{7}$$

This grid system has been superimposed on the section in Figure 5.57. In Figure 5.58 the net values have been added. As can be seen, the boundary blocks contain both ore and waste elements. A weighted averaging procedure has been used to obtain the block model of Figure 5.59. The block positions will be denoted using an i, j numeration system. In keeping with the nomenclature used by Lerchs and Grossmann, i refers to the rows and j to the columns. The first step in this procedure is to calculate cumulative profits for each column of blocks starting from the top and moving downward. Each vertical column of blocks is independent of the others. This process is shown in Figure 5.60 for column $j = 6$.

The equation which describes this process is

$$M_{ij} = \sum_{k=1}^i m_{kj} \quad (5.15)$$

where M_{ij} is the profit realized in extracting a single column with block (i, j) at its base and m_{kj} is the net value of block (k, j) . Applying the equation to find the value of the column for $j = 6, i = 3$

$$\begin{aligned} M_{36} &= \sum_{k=1}^3 m_{k6} = m_{16} + m_{26} + m_{36} \\ &= 12 + 12 + 8 = 32 \end{aligned}$$

The new table of values obtained by applying this process to all columns is shown in Figure 5.61. The next step in the process is to add row $i = 0$ containing zero's. A zero is also added at position $(i = 0, j = 0)$. This revised table is shown in Figure 5.61.

It is now desired to develop an overall cumulative sum as one moves laterally from left to right across the section. Beginning with the extreme top left hand real block, the values of three blocks are examined:

1. One directly above and to the left.
2. One on the left.
3. One directly below and to the left.

Of the three, that block which when its value is added to the block in question yields the most positive sum is selected. An arrow is drawn from the original block to that block. This sum is substituted for that originally assigned and becomes the value used for subsequent calculations.

Figure 5.62 shows the process for block (1, 1). This process is continued, working down the first column, then down the next column to the right, until all blocks have been treated. It should be pointed out that the reason some of the blocks on the section have not been filled in is that they fall outside of the bounds of the ultimate pit. Figure 5.63 shows the results when the process has been completed through column 7.

One can examine the relationship between the values in the current table to the initial blocks (Fig. 5.59). If one follows the arrows beginning at the value 32, the pit which results is indicated by the shaded line (Fig. 5.64).

Superimposing the same pit on the block model one finds that the cumulative value of the blocks is 32 (Fig. 5.65). Moving up to the block containing the value 60 in Figure 5.64 one follows the arrows to outline the pit. The value obtained by summing the blocks is also 60 (Fig. 5.65). Therefore the technique provides a running total of the value of the pits defined by following the arrows. At this point in the calculation, the optimum pit has a value of 84. Figure 5.66 shows the results when the summing process is completed. The optimum pit is that which has the maximum cumulative value. To determine this, one moves from right to left along row 1 until the largest value is encountered. The arrows are then followed around to give the optimum pit outline on the section.

$$\text{Tons waste} = 16 \text{ blocks} \times 10,000 = 160,000 \text{ tons}$$

$$\text{Stripping ratio} = \frac{16}{20} = 0.8$$

$$\text{Average profitability/ton} = \frac{\$108,000}{360,000} = \$0.30/\text{ton}$$

The expressions used (Lerchs & Grossmann, 1965) to calculate the 'derived' profit P_{ij} (as given in Fig. 5.66) are

$$P_{ij} = \begin{cases} 0, & i = 0 \\ M_{ij} + \max_{k=-1,0,1} P_{i+k,j-1} & i \neq 0 \end{cases} \quad (5.16)$$

The maximum is indicated by an arrow going from (i, j) to $(i + k, j - 1)$. P_{ij} is the maximum possible contribution of columns 1 to j to any feasible pit that contains (i, j) on its contour. If the element (i, j) is part of the optimum contour, then this contour to the left of element (i, j) can be traced by following the arrows starting from element (i, j) . Any feasible pit must contain at least one element of the first row. If the maximum value of P in the first row is positive then the optimum contour is obtained by following the arrows from and to the left of this element. If all elements of the first row are negative, then no contour with positive profit exists.

This same section (Fig. 5.59) can be evaluated using the floating cone technique. The ‘stepped’ cone of Figure 5.69 has been used to float over the section. Figures 5.70 through 5.73 show the intermediate pits with the final pit given in Figure 5.74. As can be seen the result is the same as when using the Lerchs-Grossmann approach. An advantage of the floating cone procedure is that a variety of slope angles can be modelled.

As shown in Figure 5.75, one can also examine the pit limits by considering the lengths along the perimeter. Consider the left-hand side (LHS) of the pit. The horizontal segments

are considered first to determine the vertical position. When the pit is slightly above the known true position, the sum of the horizontal segments is

$$-4 - 4 - 4 + 0 + 8 + 6 = +2$$

When it is slightly below than the sum is

$$-4 - 4 - 4 - 4 - 4 + 0 + 4 = -16$$

Examining the vertical segments to find the correct horizontal position is done next. When the contour is slightly inside the correct position the sum is

$$-4 - 4 - 4 + 0 + 8 + 12 = +8$$

When it is slightly outside, the sum is

$$-4 - 4 - 4 - 4 - 4 + 0 = -20$$

Therefore the correct position is as given by the floating cone and Lerchs-Grossmann procedures. The same procedure could be followed on the right hand side.

5.6 MODIFICATION OF THE LERCHS-GROSSMANN 2-D ALGORITHM TO A 2½-D ALGORITHM

The two-dimensional dynamic programming algorithm for determining the optimal configuration of blocks to be mined in cross-section was presented in the previous section. The technique is elegant yet simple. Like all other two-dimensional techniques it has the drawback that extensive effort is usually required to smooth out the pit bottom and the pit ends as well as to make sure that the sections fit with one another. As indicated by Johnson and Sharp (1971), this smoothing seldom results in an optimal three-dimensional pit. The modification which they proposed to the basic algorithm, as described in this section has been referred to as the 2½-D algorithm (Barnes, 1982). An overall view of the block model to be used is shown in Figure 5.76.

The column of blocks representing in the longitudinal section ($i = 4, j = 1$)

- 2 Level 1
- 3 Level 2
- 1 Level 3
- 7 Level 4

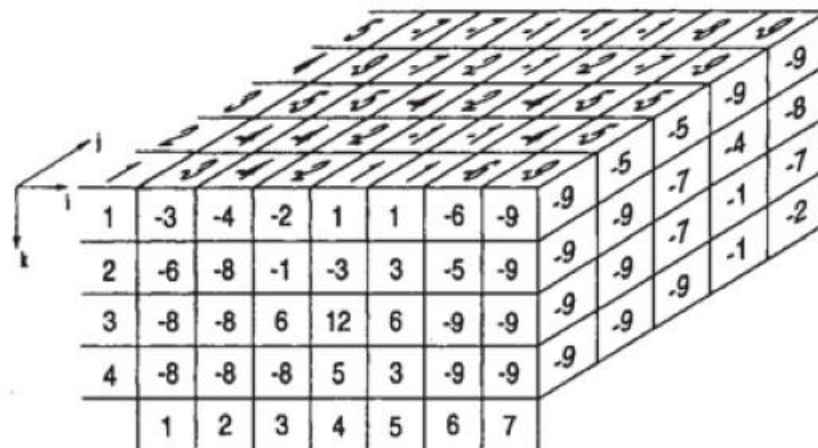


Figure 5.76. Block model used for the 2½-D projection bound (Johnson & Sharp, 1971).

This process is repeated for each of the other sections. The resulting longitudinal section is shown in Figure 5.79. The 2-D algorithm is now applied and the optimal contour becomes that given in Figure 5.80.

Figure 5.77 shows the 5 sections of a block model. Beginning with section 1, the optimum pit outlines are determined, given that one must mine to a given level k . In this case there are four levels to be evaluated. The results are shown in Figure 5.78. Each outline

-3	-4	-2	1	1	-6	-9
-6	-8	-1	-3	3	-5	-9
-8	-8	6	12	6	-9	-9
-8	-8	-8	5	3	-9	-9

a. Section 1 ($j=1$)

-4	-4	-2	-1	-1	-4	-5
-9	-9	3	4	3	-9	-9
-8	-8	5	5	5	-9	-9
-9	-9	-9	4	4	-9	-9

b. Section 2 ($j=2$)

-5	-5	-4	-2	-4	-5	-5
-8	-2	4	5	4	-3	-7
-9	-9	3	12	7	-7	-7
-9	-9	-9	7	7	5	-9

c. Section 3 ($j=3$)

-9	-7	-2	-1	-2	-7	-9
-6	-5	1	5	1	-5	-4
-9	-9	4	5	9	-2	-1
-4	-1	-2	9	3	-2	-1

d. Section 4 ($j=4$)

-7	-7	-1	-1	-1	-8	-9
-9	-9	-1	-2	-1	-9	-8
-9	-5	1	3	1	-5	-7
-2	2	1	1	2	1	-2

e. Section 5 ($j=5$)

Figure 5.77. Block sections for the 2½-D model (Johnson & Sharp, 1971).

Level $k=1$ S_{ik} value=2

1	-3	-4	-2	1	1	-6	-9
2							
3							
4							
k/i	1	2	3	4	5	6	7

Level $k=3$ S_{ik} value=1

1	-3	-4	-2	1	1	-6	-9
2	-6	-8	-1	-3	3	-5	-9
3	-8	-8	6	12	6	-9	-9
4							
k/i	1	2	3	4	5	6	7

Level $k=2$ S_{ik} value=-3

1	-3	-4	-2	1	1	-6	-9
2	-6	-8	-1	-3	3	-5	-9
3							
4							
k/i	1	2	3	4	5	6	7

Level $k=4$ S_{ik} value=-7

1	-3	-4	-2	1	1	-6	-9
2	-6	-8	-1	-3	3	-5	-9
3	-8	-8	6	12	6	-9	-9
4	-8	-8	-8	5	3	-9	-9
k/i	1	2	3	4	5	6	7

Figure 5.78. Analysis of section 1.

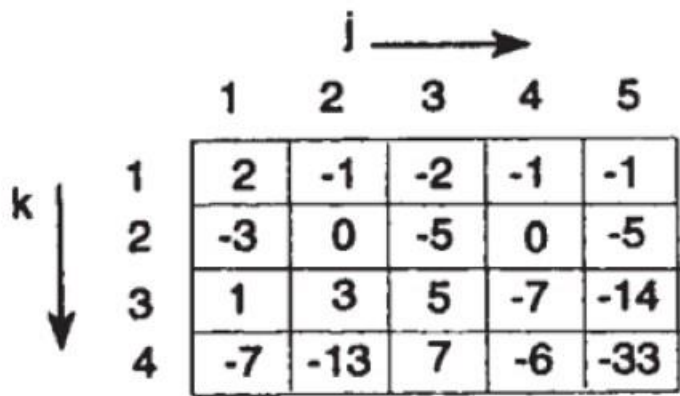


Figure 5.79. Longitudinal section (Johnson & Sharp, 1971).

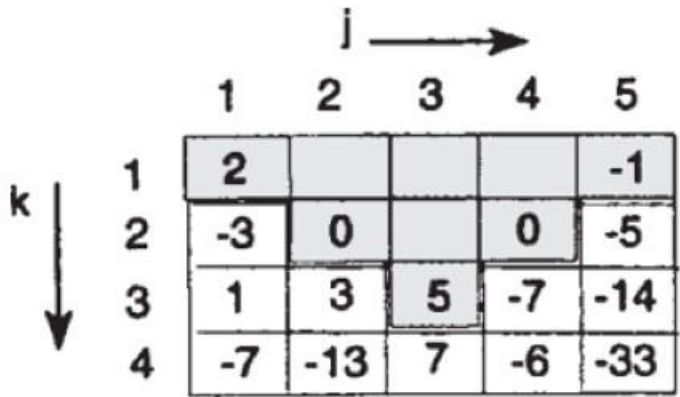


Figure 5.80. Optimal longitudinal contour (Johnson & Sharp, 1971).

is found using a slightly modified version of the Lerchs-Grossmann 2-D algorithm. The net values for each of the four pits on this section are then determined. As can be seen these are:

is found using a slightly modified version of the Lerchs-Grossmann 2-D algorithm. The net values for each of the four pits on this section are then determined. As can be seen these are:

$$\text{Level 1: } S_{11} = 2$$

$$\text{Level 2: } S_{12} = -3$$

$$\text{Level 3: } S_{13} = 1$$

$$\text{Level 4: } S_{14} = -7$$

To combine all of the transverse (ik) sections, a longitudinal section will now be taken. In viewing Figure 5.78, it is seen that the deepest mining on section 1 ($j = 1$) occurs at column $i = 4$. A column [$i = 4, j = 1$ (section 1)] of blocks will now be formed representing the net value of the cross-section as mined down to the various levels.

The net value for the material contained within the pit is found by summing the block values along the final contour. Here it becomes

$$\text{Pit value} = 2 + 0 + 5 + 0 - 1 = 6$$

Now that the bottom levels for each of the transverse sections have been determined, one goes back to these sections and selects the appropriate one. These are summarized in Figure 5.81.

The data and solution for a somewhat more complicated problem are given in Figures 5.82 and 5.83, respectively. The block height to length ratio along the longitudinal section (the axis of mineralization) is 1 to 2 while for the cross-sections the block height and length are equal (1 to 1). Hence to maintain a maximum pit slope of 45 degrees, the pit slope can change by two blocks per section along the length but by only block per column across the width.

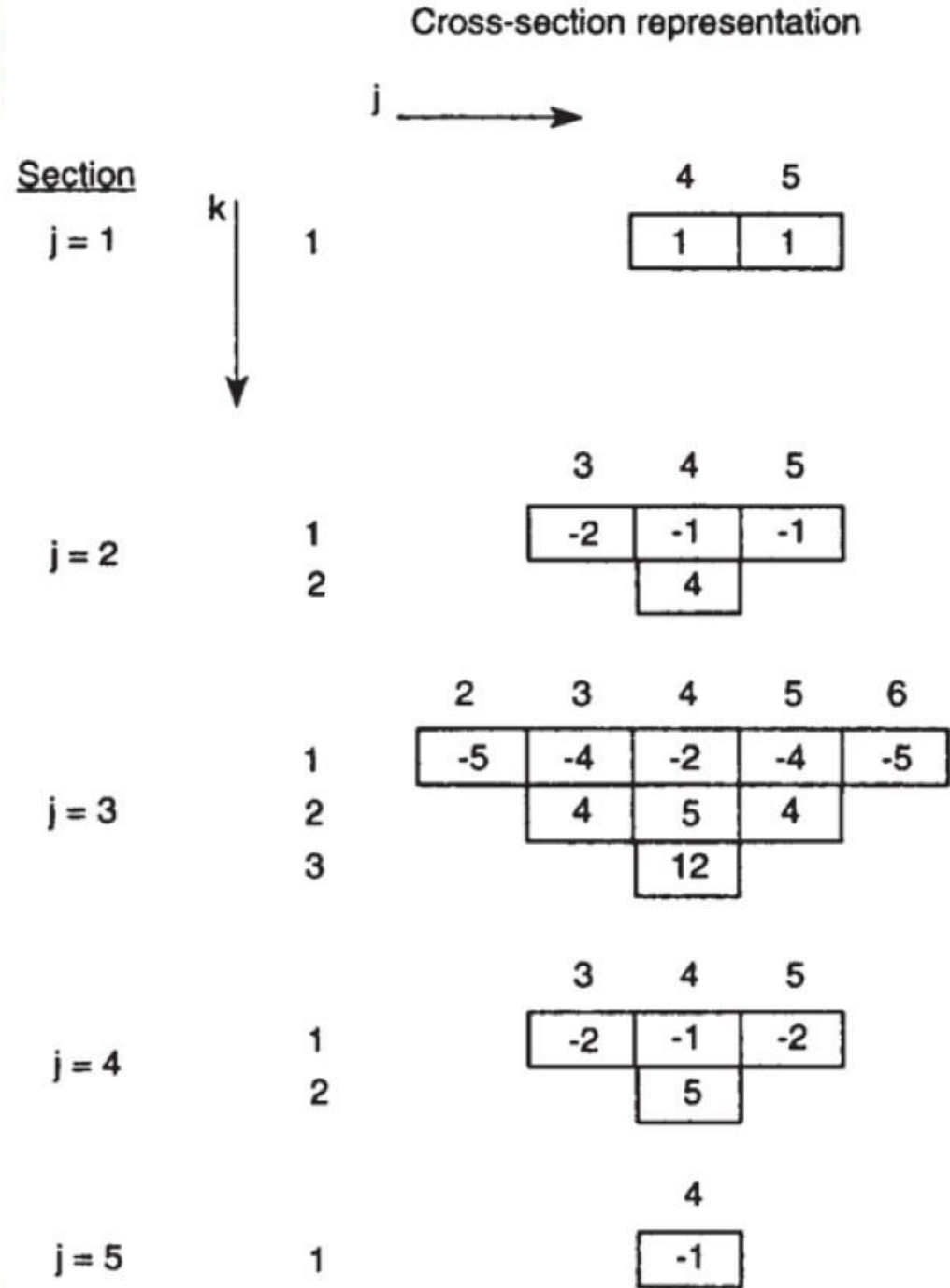
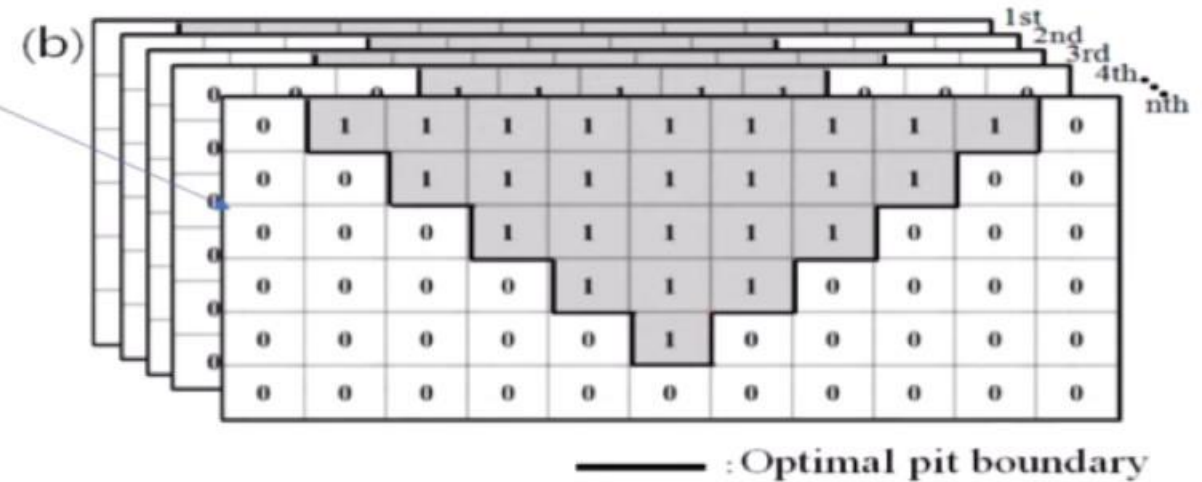
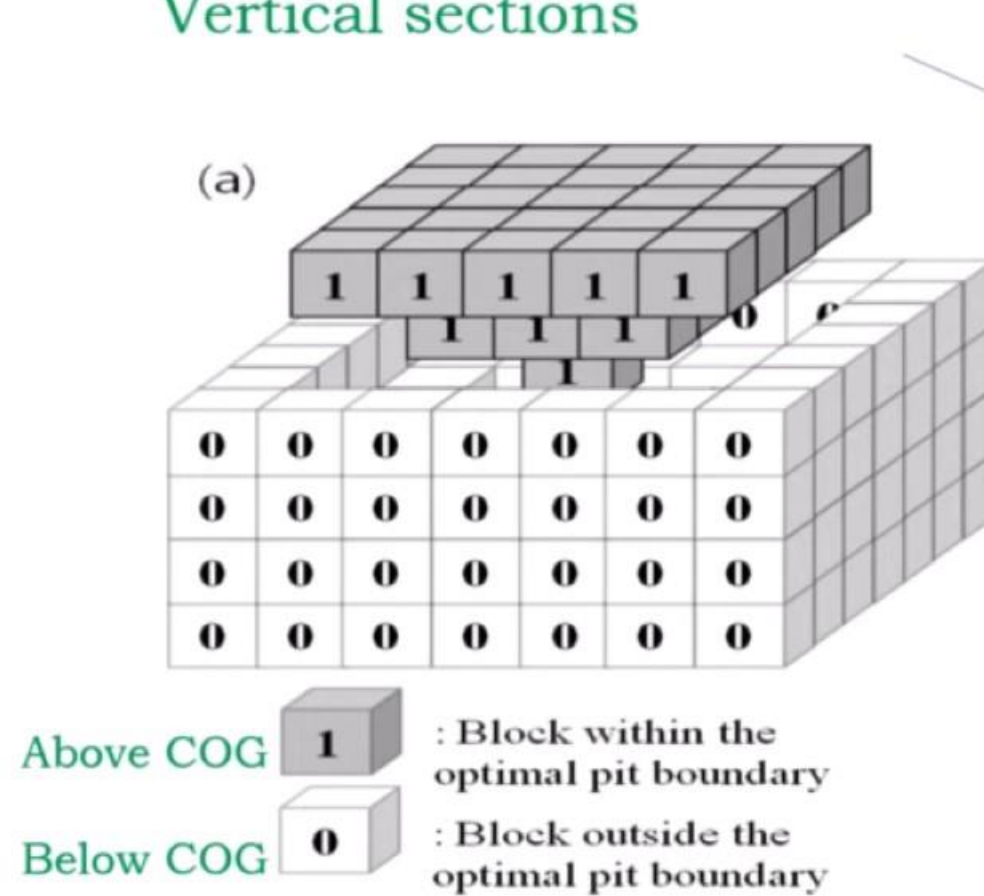


Figure 5.81. Optimal longitudinal pit sections (Johnson & Sharp, 1971).

Block diagram

Vertical sections



Pit design based on stripping ratio

1. Having found the reserves roughly by at least two methods & prepared the block diagram based on cut off limits, next step is to finalize pit profile.

On the basis of rock mechanics study ultimate pit slopes on hang wall and foot wall sides are determined. Pit opening place is decided & keeping minimum floor width of 35 m , ultimate pit slope is marked on all transverse, longitudinal & horizontal sections for different depths.

2. Lateral ultimate pit limit at surface as well as at all **bench levels** is plotted on the surface and horizontal sections / slice plans .

3. Stripping ratio at different ultimate depths is determined at all the sections at these slopes.

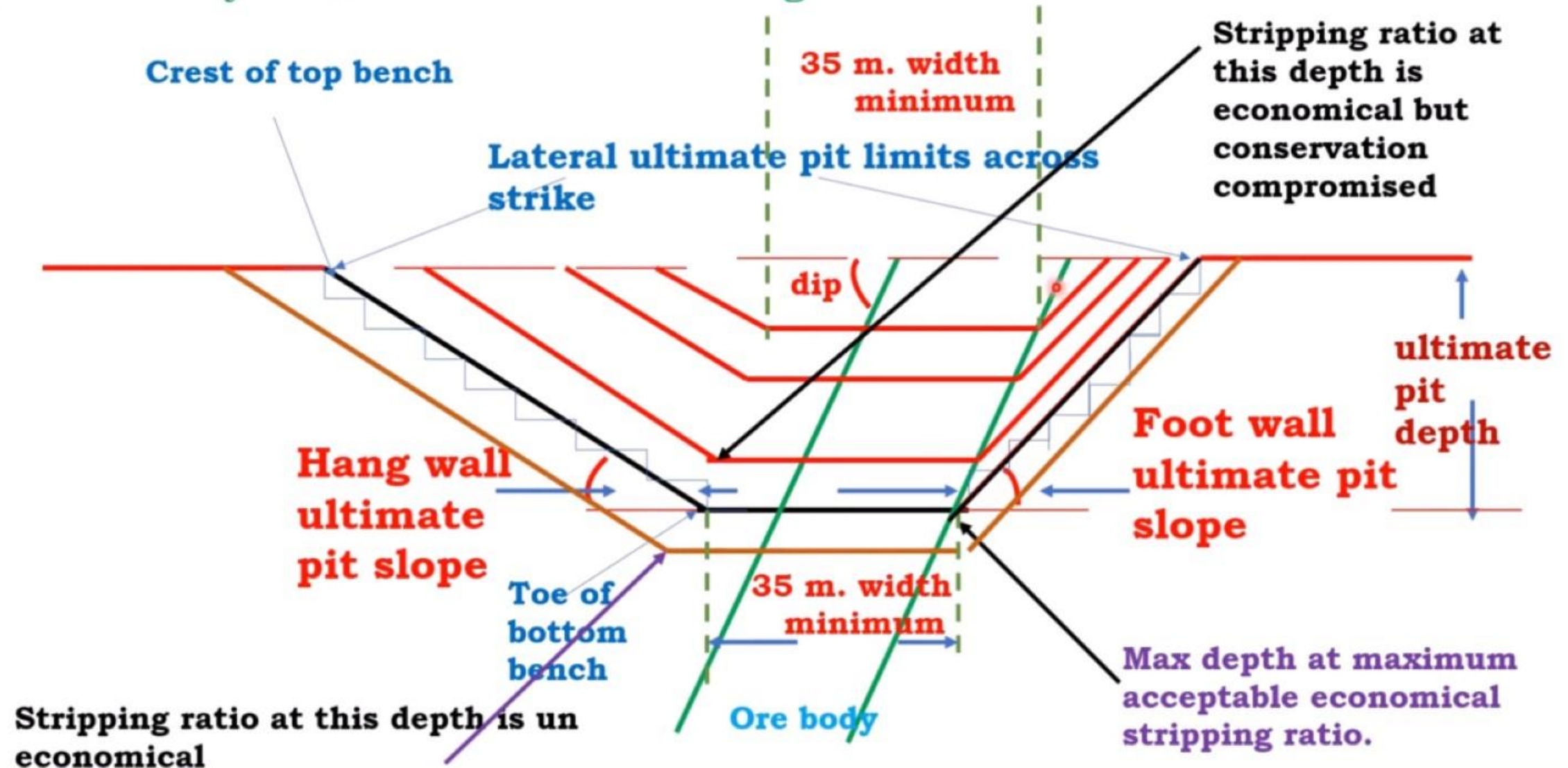
4. Ultimate pit bottom level for OC mining is considered at **maximum acceptable economical stripping ratio** to balance conservation and economics both.

5. Corresponding to arrived ultimate pit bottom pit profile is marked on all the transverse, longitudinal & horizontal sections. This will give final bench wise position of reserves available in the pit with grade and quantity depending on grade and quantity expressed by blocks falling in the bench.
6. Pit opening and progress is so planned that maximum profit is targeted in earlier periods for greater internal rate of return.
7. Based on year wise production/ development targets, tentative year wise working limits are marked on all the vertical sections and surface plans as well as on all slice plans, that will show Year wise bench progress & production/ development.
8. All benches are designed confirming to ultimate pit slope on all sides to ensure safety and no where the pit slope should exceed ultimate pit slope.
9. Rest of the mine planning includes deciding pit parameters, bench parameters dump and machine requirements etc discussed later.

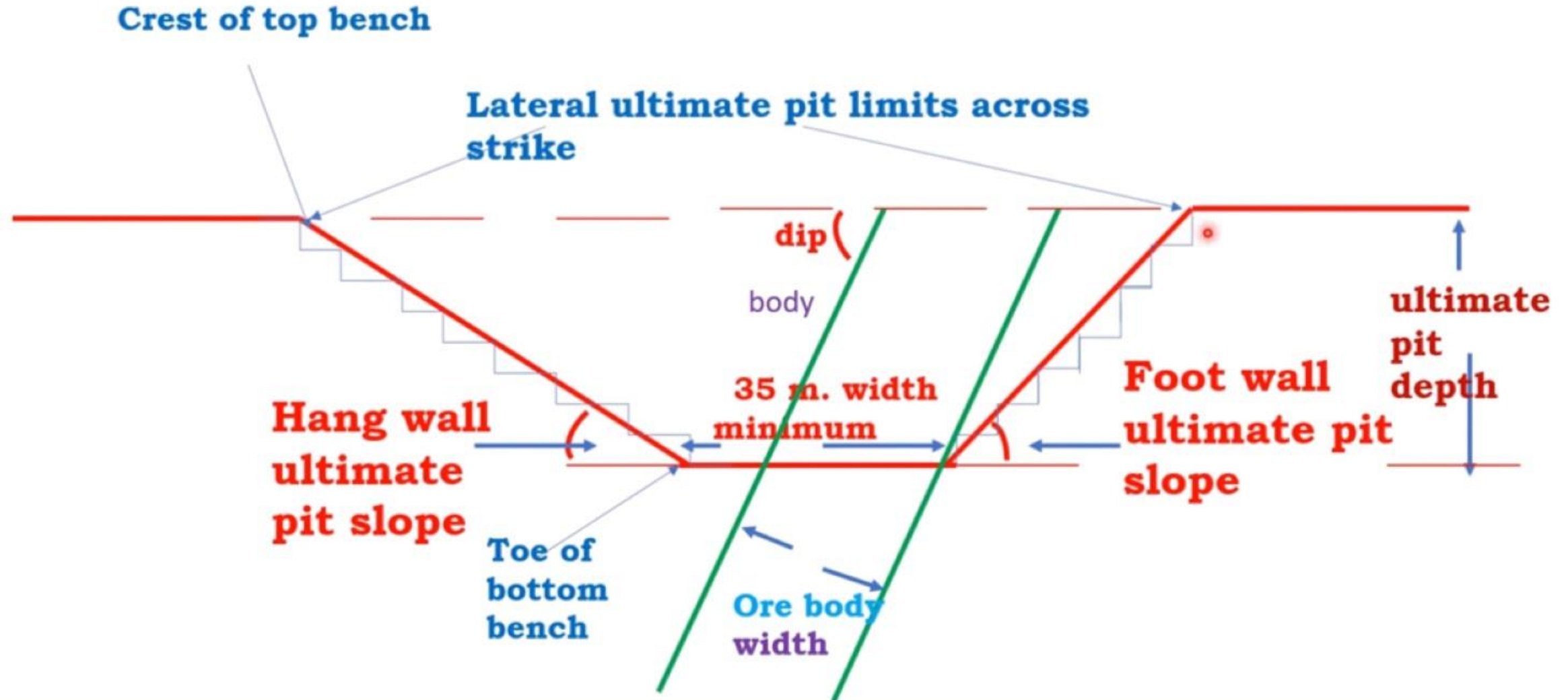


PIT LIMITS

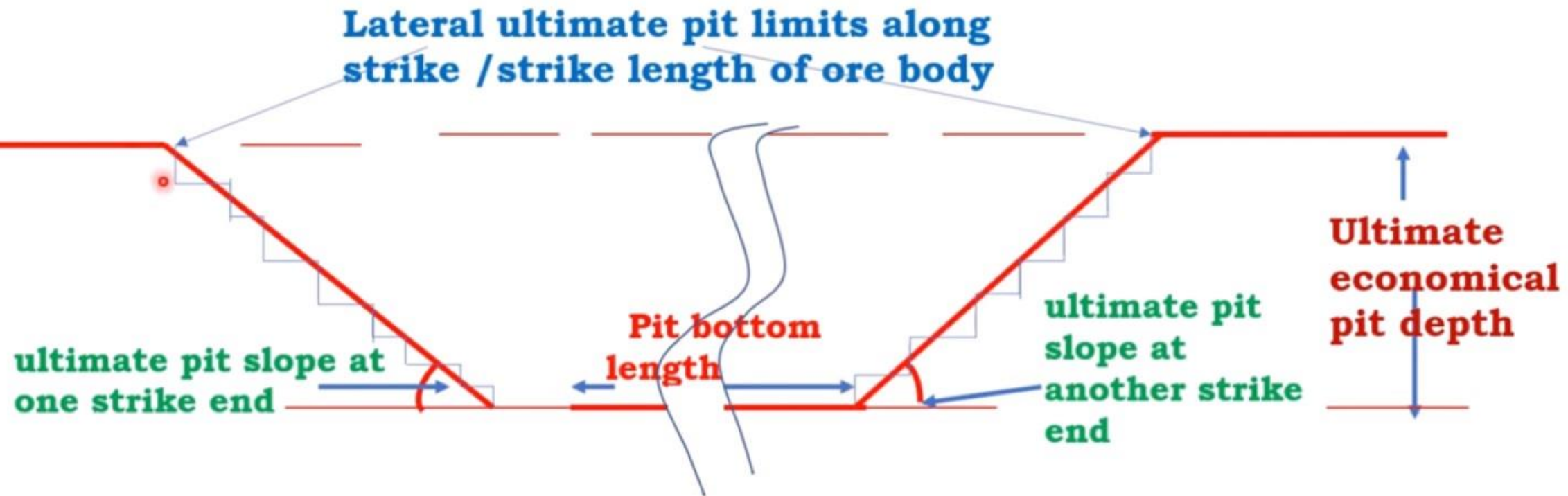
Floor width of minimum 35 m. is kept or mechanized mine and 10 m. or as permitted by DGMS in manual workings.



Typical Ultimate Transverse section

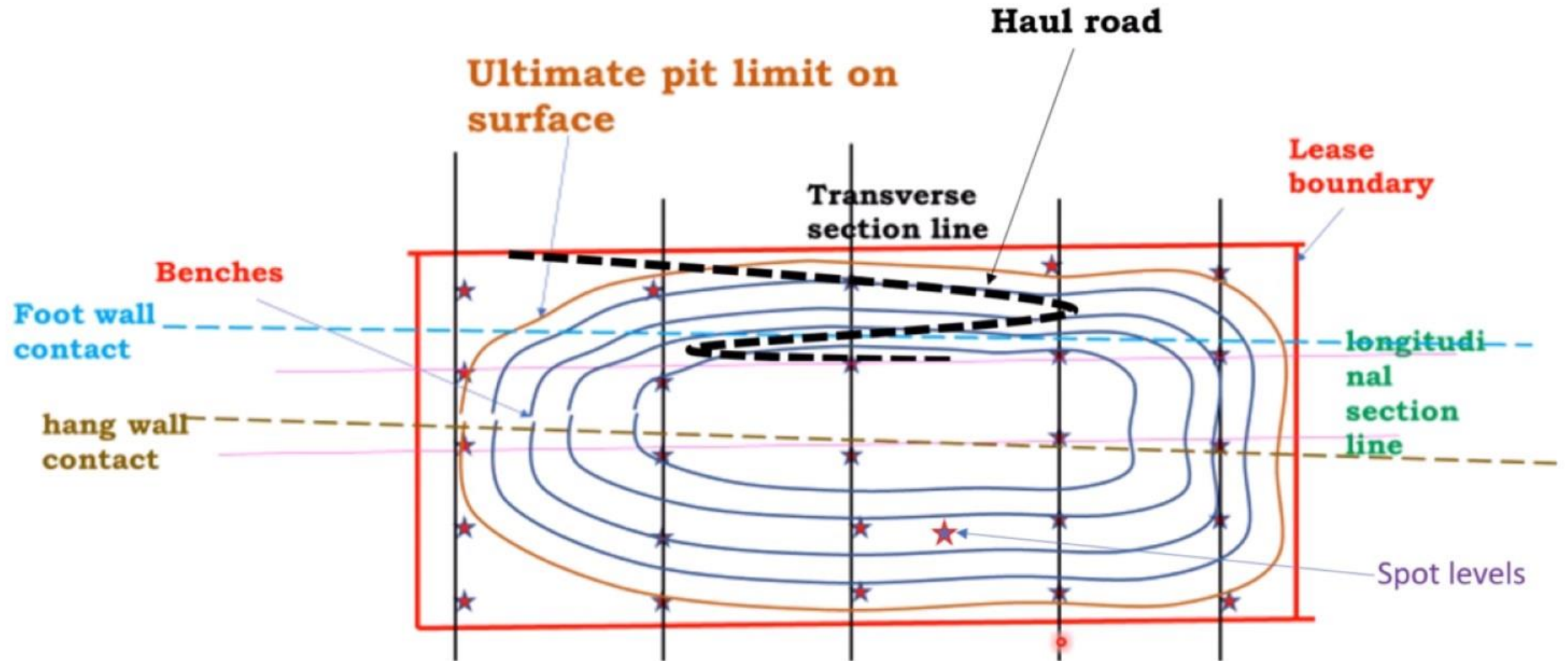


Typical Ultimate longitudinal section



Vertical ore body

Surface plan of a working mine



PIT LIMITS



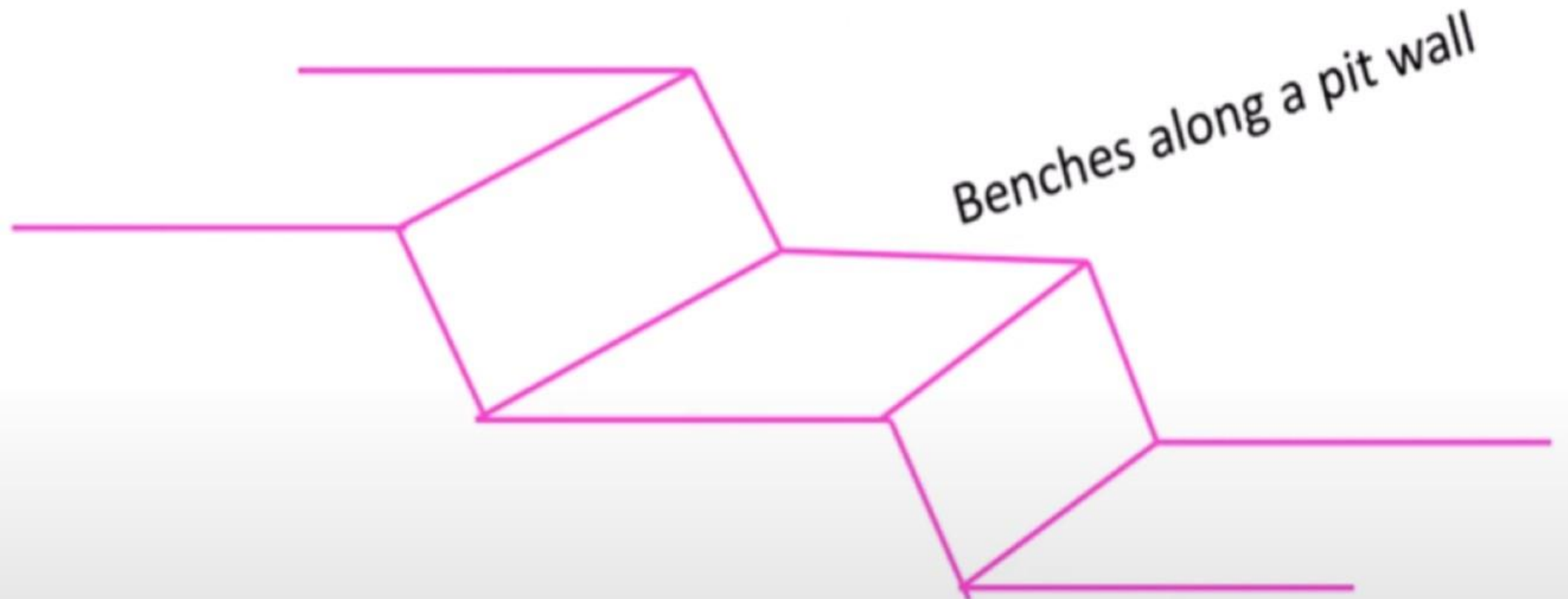
Ore body is mined from top to bottom in series of uniformly thick layers called benches



Benches along
pit wall

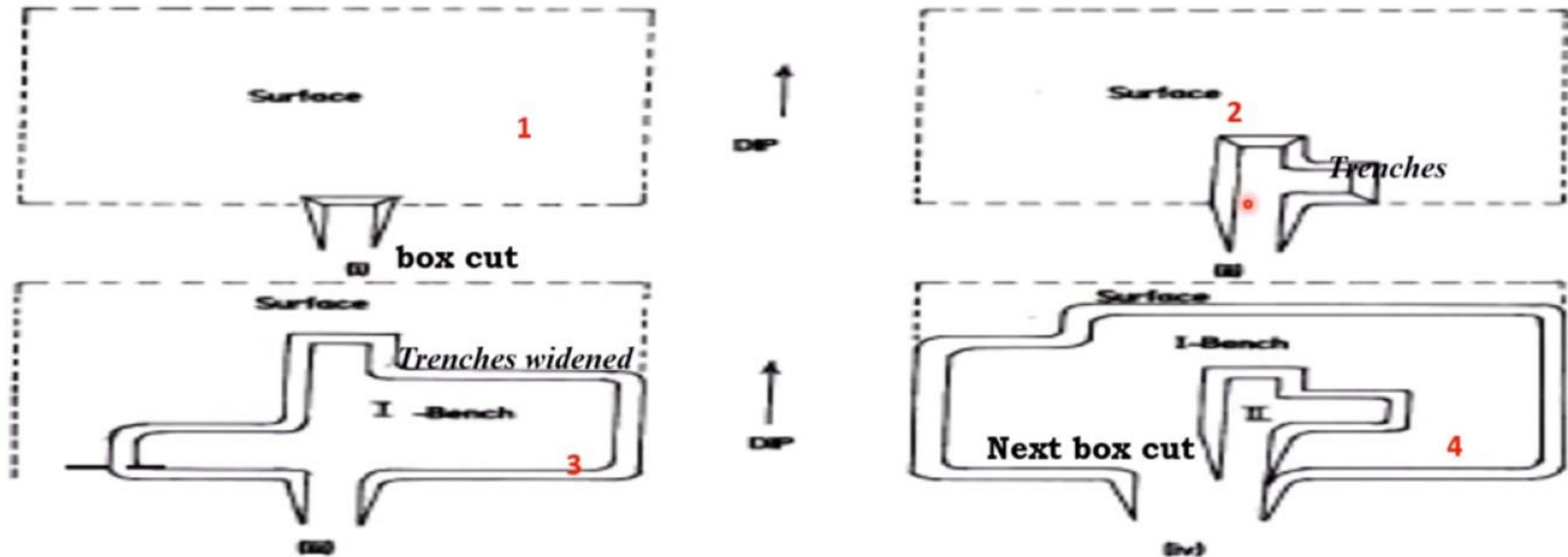
Isometric view

Bench :A ledge all along pit wall in step like fashion.



DEVELOPMENT & FORMATION OF BENCHES

Once the box cut is established up to the 1st bench floor level, 1st bench is developed by making trenches and widening them. Once the 1st bench is developed over a **considerable** area, **new box cut is made to reach the next lower bench floor level** for development of next bench. The process of development of benches continues till **UPL** and **UPD** is reached.



All trenches extended from box cut **convert** into benches after their widening. Formation of benches is for maintaining pit slope, **arresting falling objects**, **ensuring safety against slope failure**, facilitating safe & productive working and convenient handling of material by mechanised or manual means.

The floor of each bench is the **bench working level**, thus the crest of 1st bench is at surface level and its toe is at 1st bench floor level. **Similarly toe of the bottom most bench corresponds to ultimate pit floor level.** Sometimes benches are named after their reduced level related with an arbitrary bench mark.

Benches are almost horizontal except with little gradient towards toe as well as along length to direct drainage water to sump and keep the work places dry.

High wall /face of a bench always remains inclined. All benches are connected to haul road of adequate size and gradient through inbye roads for transport.

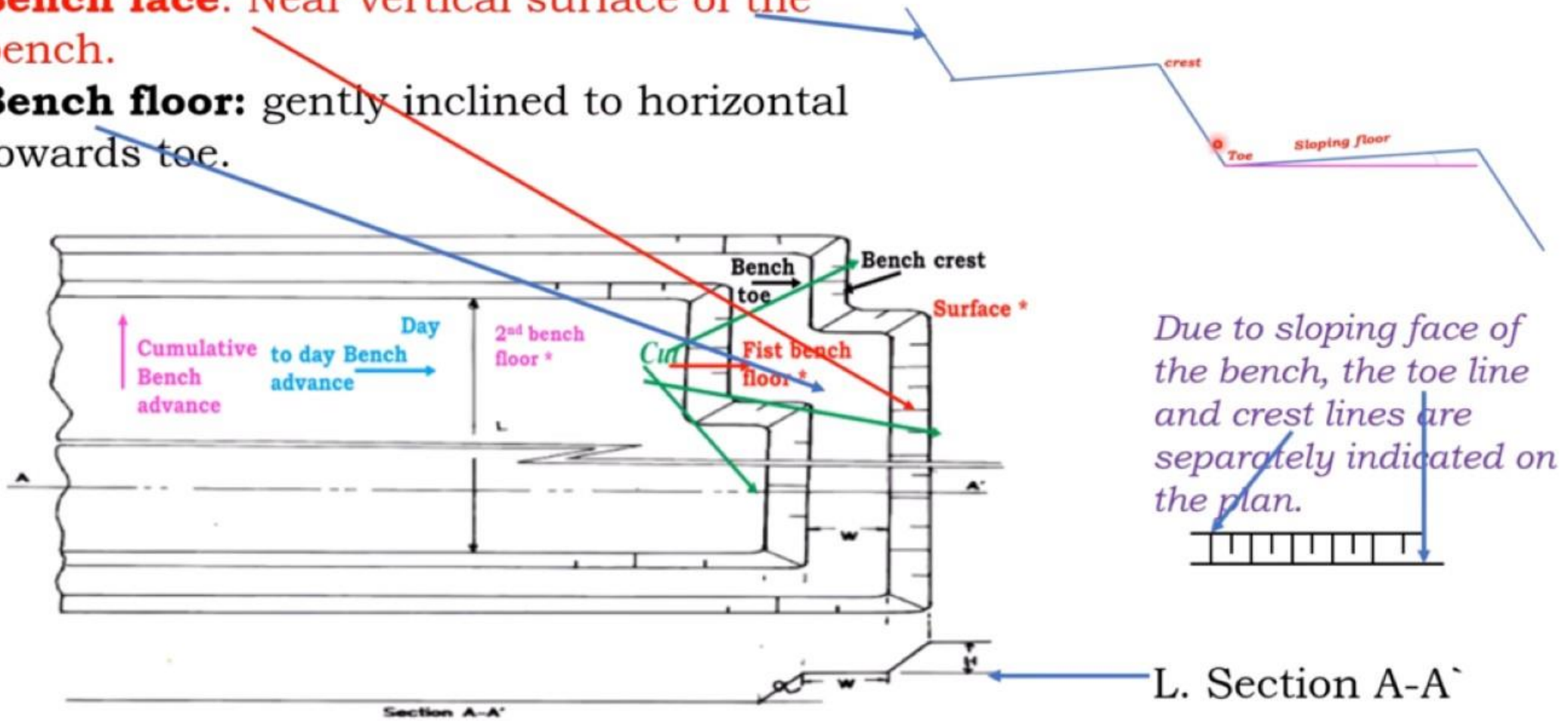
*Unlike closing benches, working benches are kept **wide** enough to accommodate blasted muck pile and equipment plying on them, **for one or two way traffic as required.** Benches are guarded by safety berms at the edge against the falling of operating equipment off working level. Excavation from a bench is mostly done by drilling blast holes in single/multiple rows from level above and blasting them. Blasted muck is loaded & transported from bench level to surface **via haul roads.***

PIT LIMITS

Bench Terminology

Bench face: Near vertical surface of the bench.

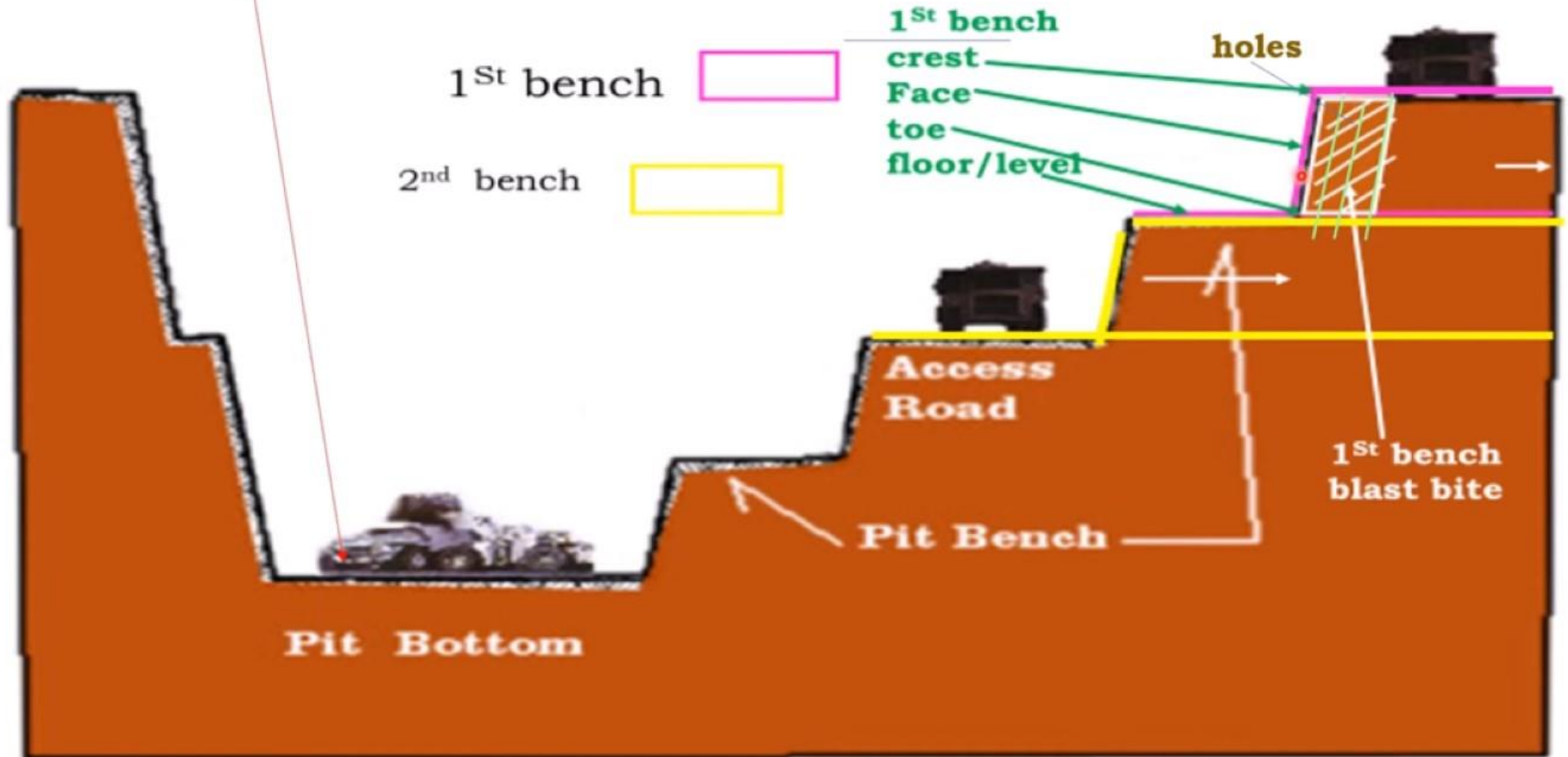
Bench floor: gently inclined to horizontal towards toe.



Plan of bench

Bench Terminology

Open-pit floor: Horizontal surface of the quarry bottom.

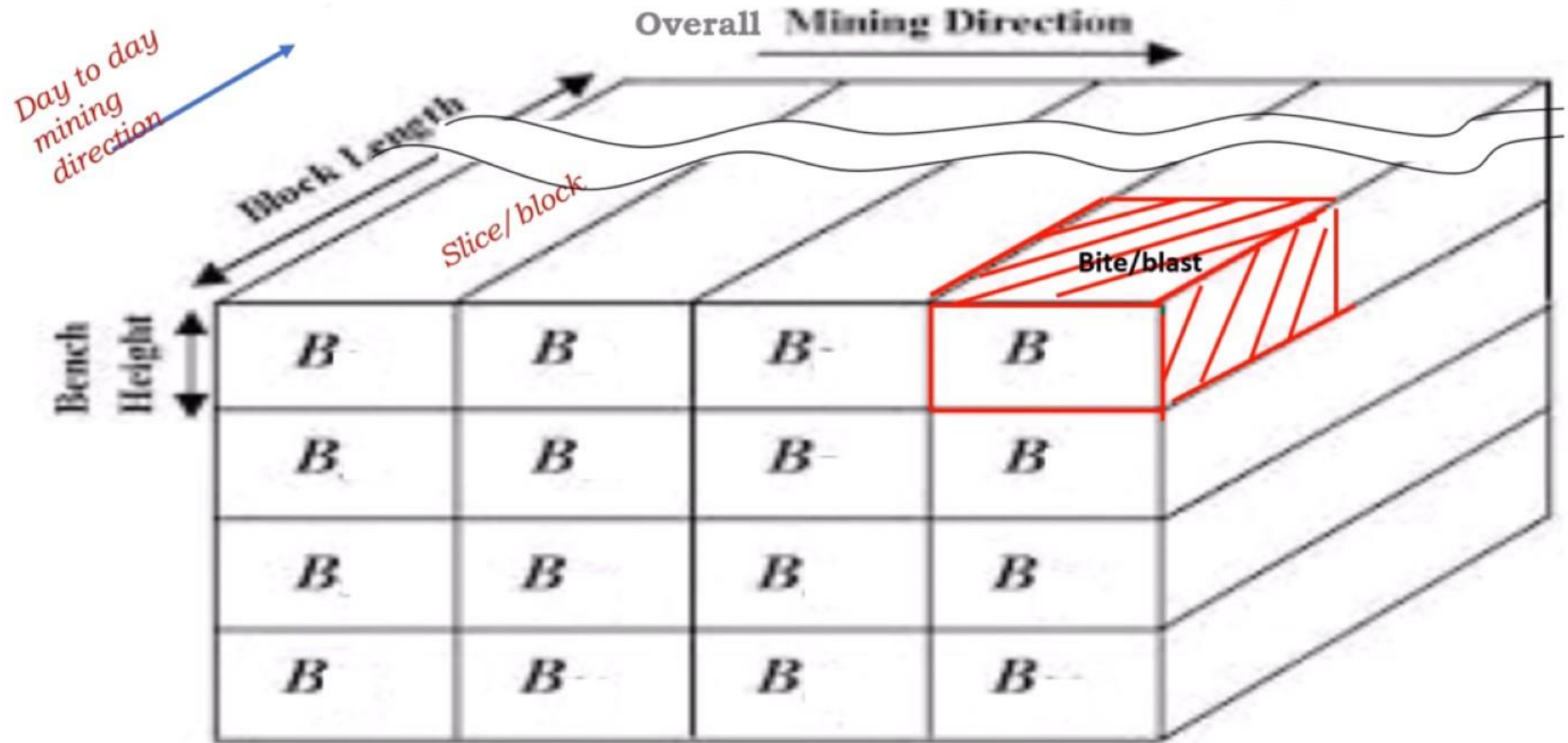




PIT LIMITS



Benches are worked whether by continuous miners or by cyclic mining in parallel strips called cuts

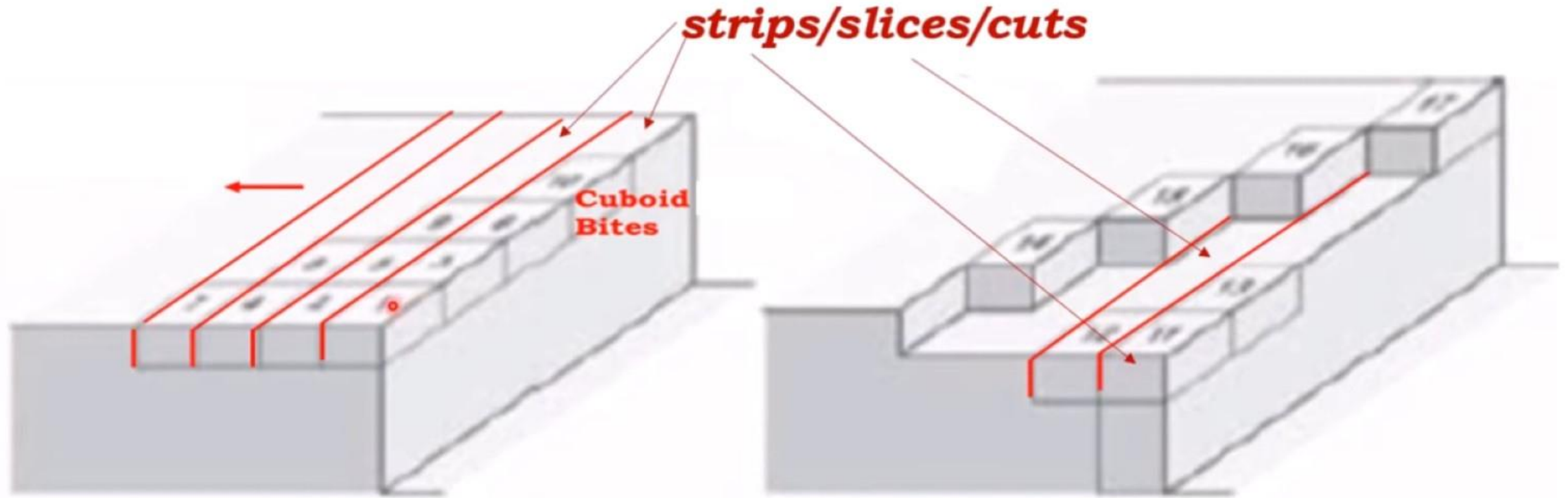


Bench is longitudinally divided into no. of parallel strips. Length of cut / slice / strip is equal to length of bench. Its width is equal to width of blast which in turn = burden $\times$ no. of rows and height of cut = height of bench

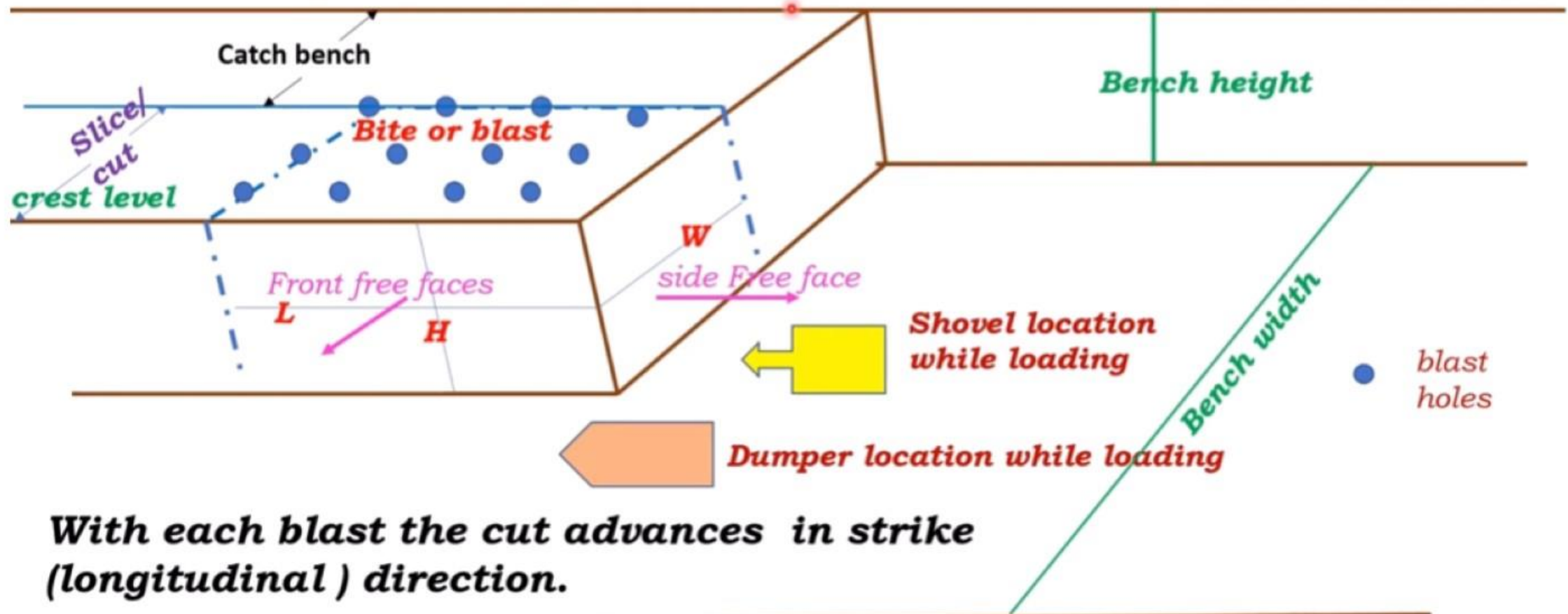
PIT LIMITS

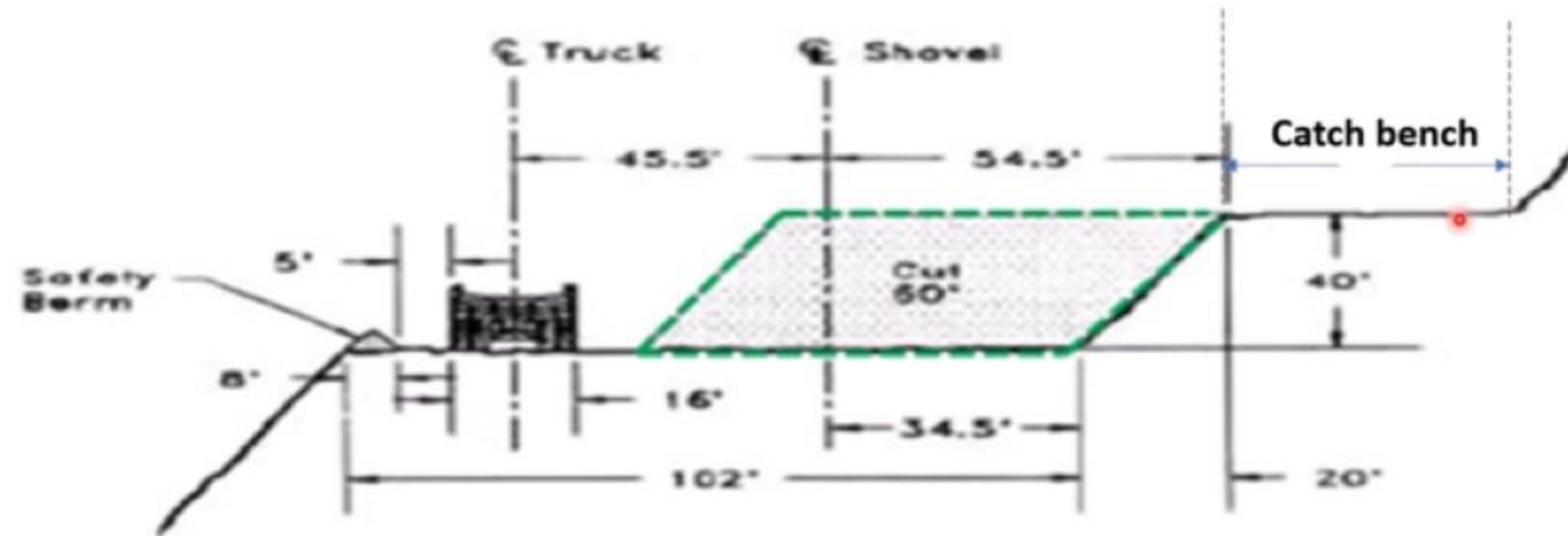
Cut: A parallel strip by which a bench is worked so that two free faces are available for blast.

Push back: Advance/progress of bench across the strike is called push back.

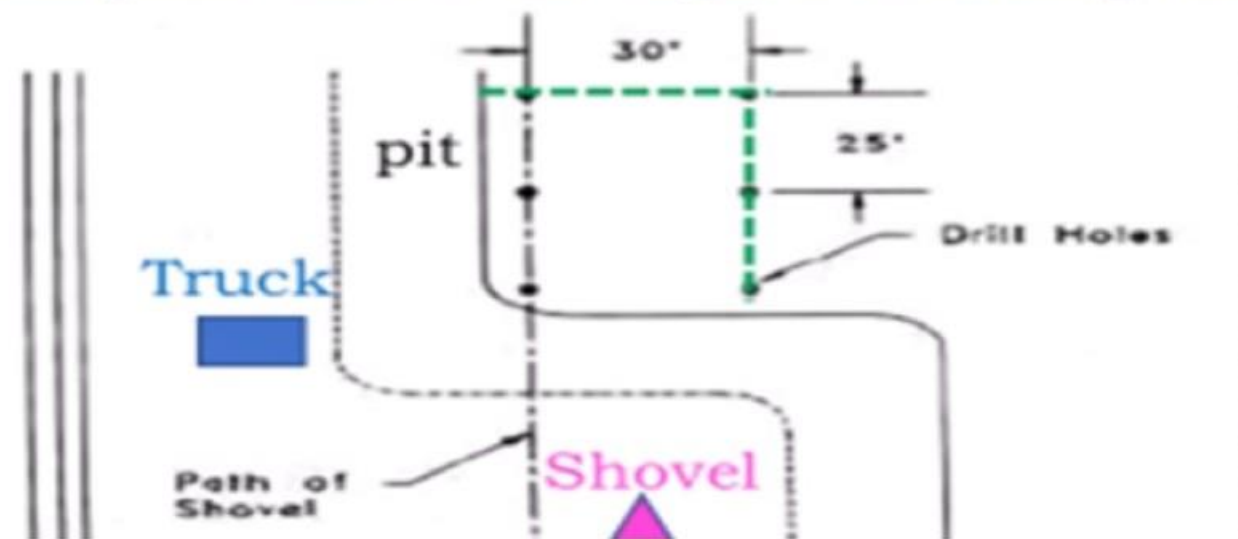


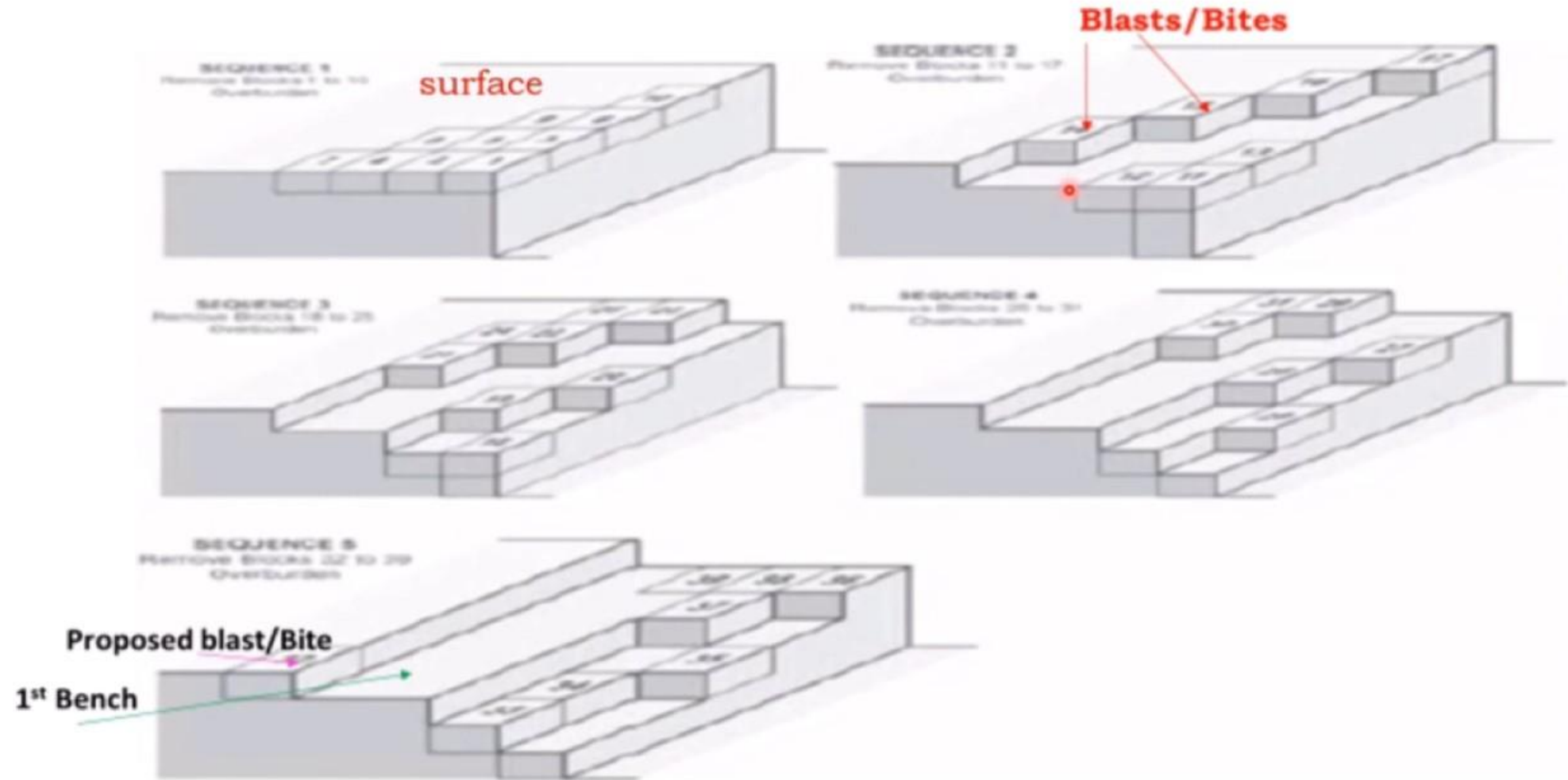
Bench Progression : Excavation from a bench is done in single or multiple cuts/slices. Each cut of bench height progresses bit by bit with every blast/bite. Holes are drilled & blasted in single or multiple rows to win a **bite**. The bite is part of bench from one end having 2 lateral free faces for blast. Blasted muck is excavated by shovel & loaded into dumper and transported to surface.





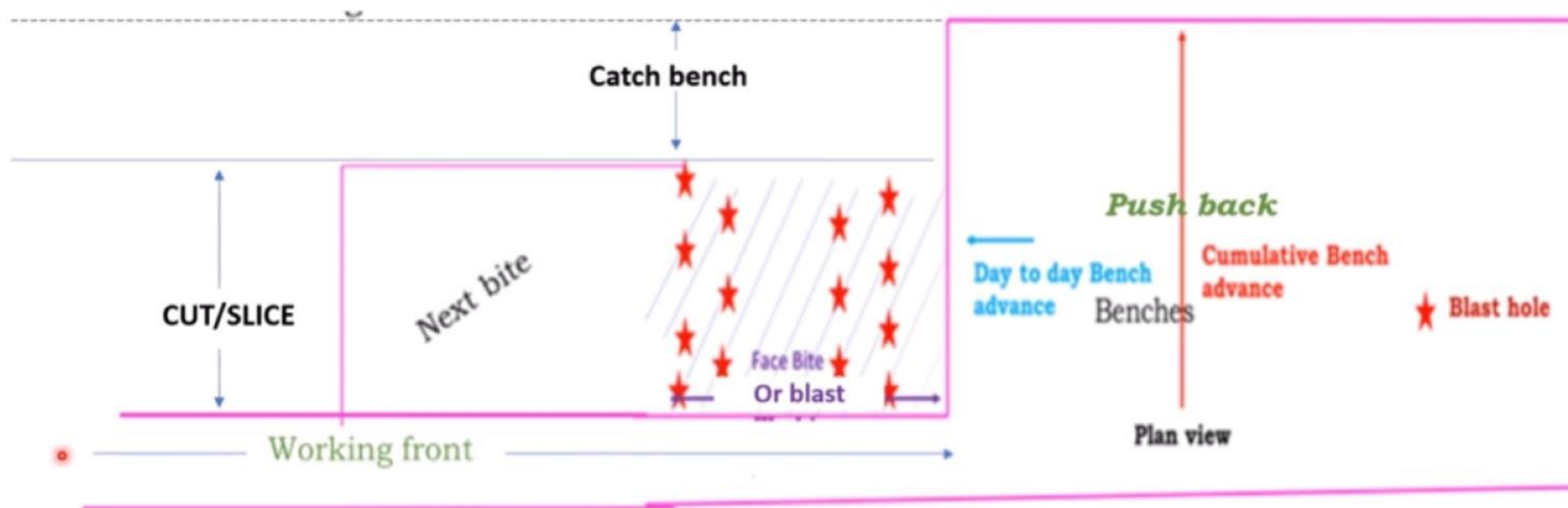
Section and plan views through a working bench.





Number of benches are worked simultaneously and number of cuts/strips on a bench are worked at a time so as to have multiple working faces.

Advance of bench /face : Benches are ideally kept parallel to strike. Blasting is so arranged that with every blast the bench progress in the strike & widens across strike :



The Day to day Bench advance is in the strike direction and Cumulative Bench advance is across the strike.

Plain topography

The figures 1, 2, 3, 4, etc. represent slices in coal or stone benches. Each slice has a height suitable for dipper shovel operation in and length upto the limit of the Section A. The width of the slice (in the dip rise direction) may be 18 to 20 m.

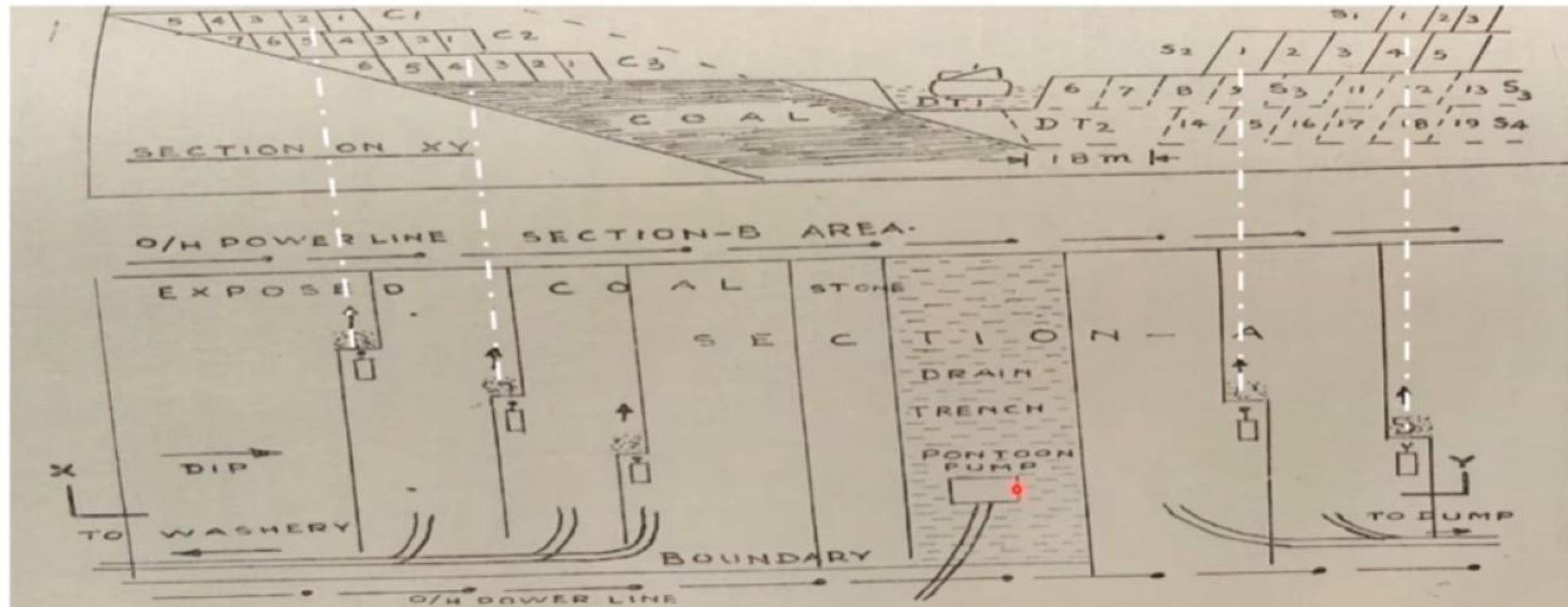
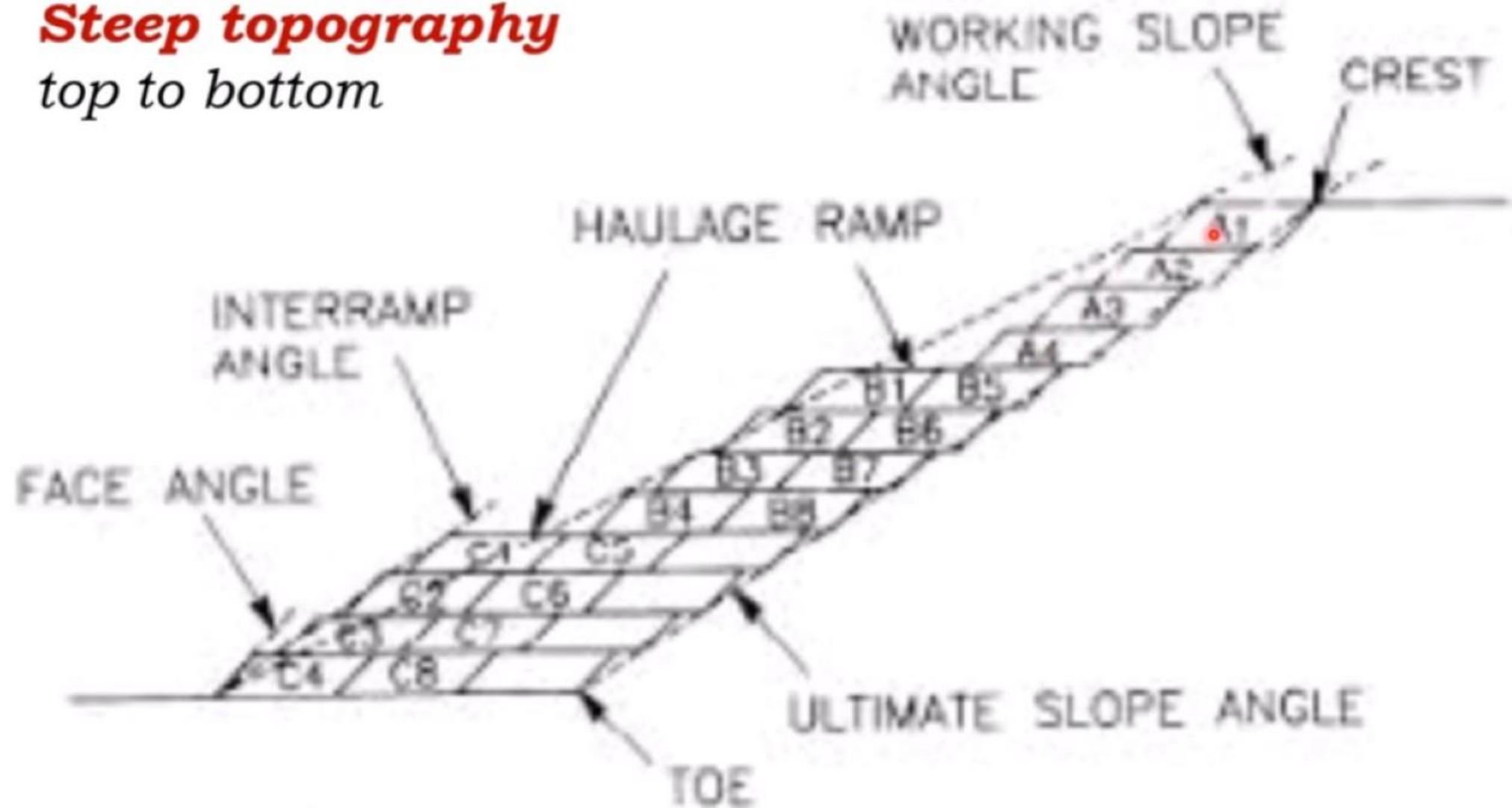


Fig. 9—Mechanised quarry layout with pontoon mounted pump (shovel dumper combination).

The day-to-day advance of the shovel is along the strike to remove a slice and the general progress of the bench extraction, considered over a quarter or longer period is in the direction of dip or rise.

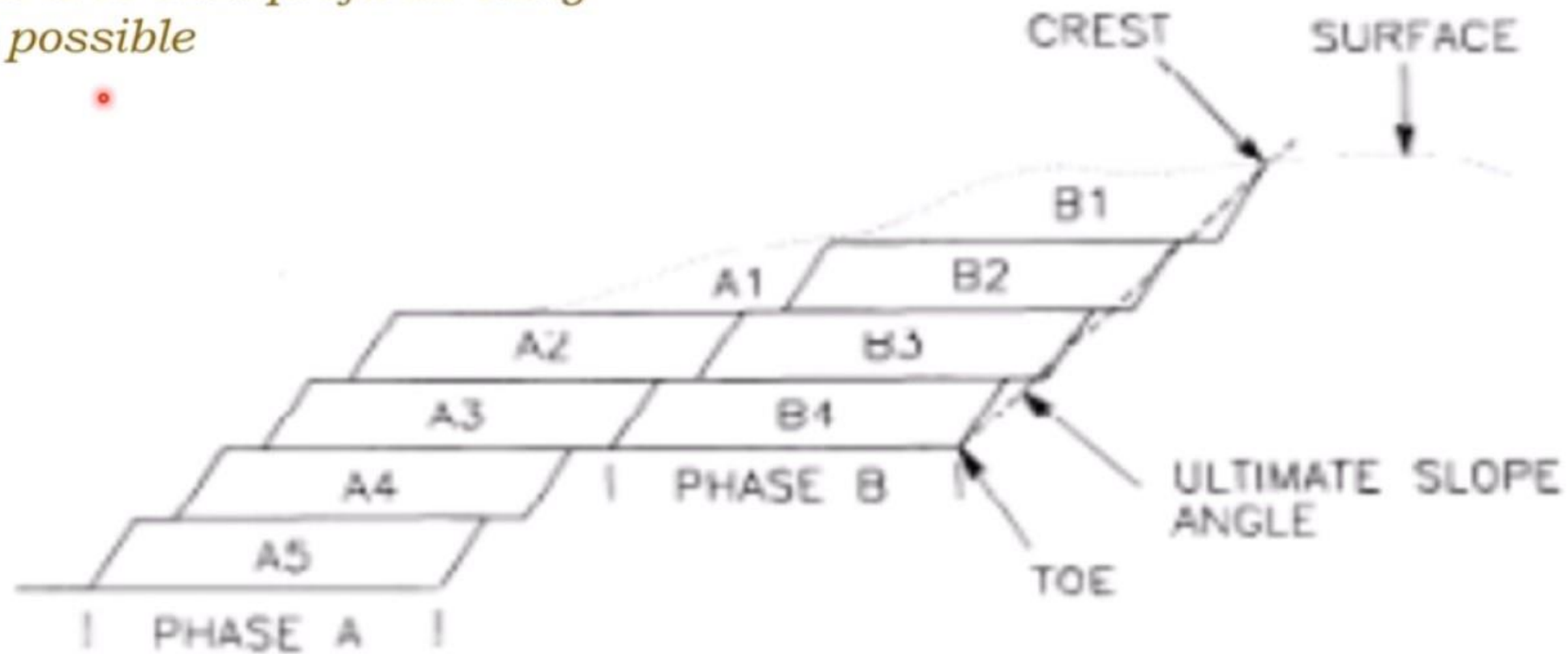
Steep topography
top to bottom



Cut sequence for sequential pushbacks.

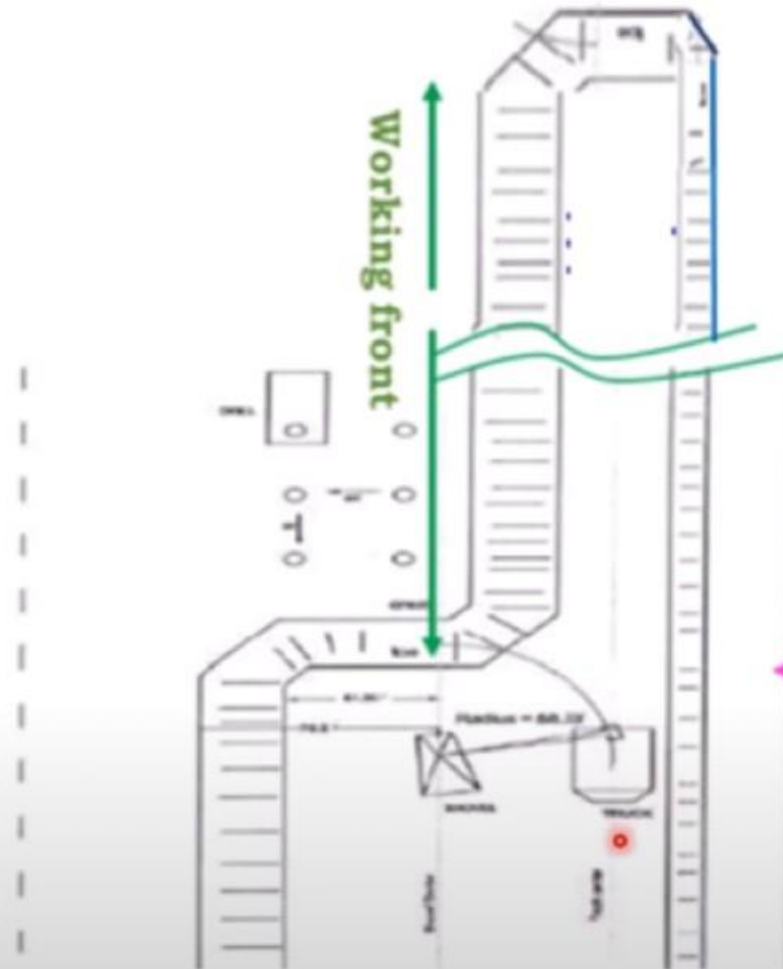
Gentle topography

Aim is to earn profit as early as possible



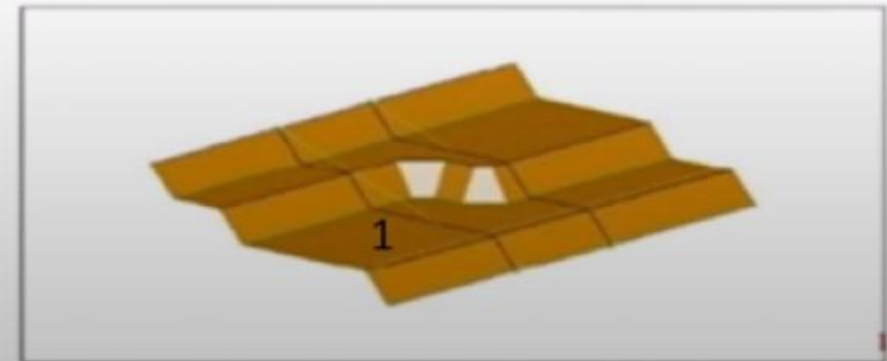
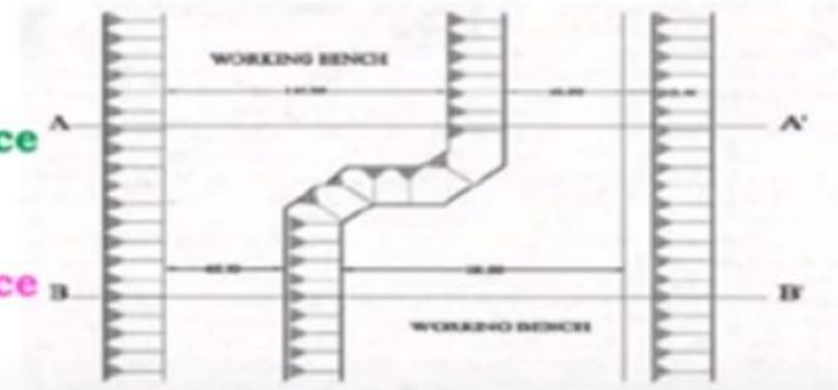
Cut sequence for conventional pushbacks.

Single working bench

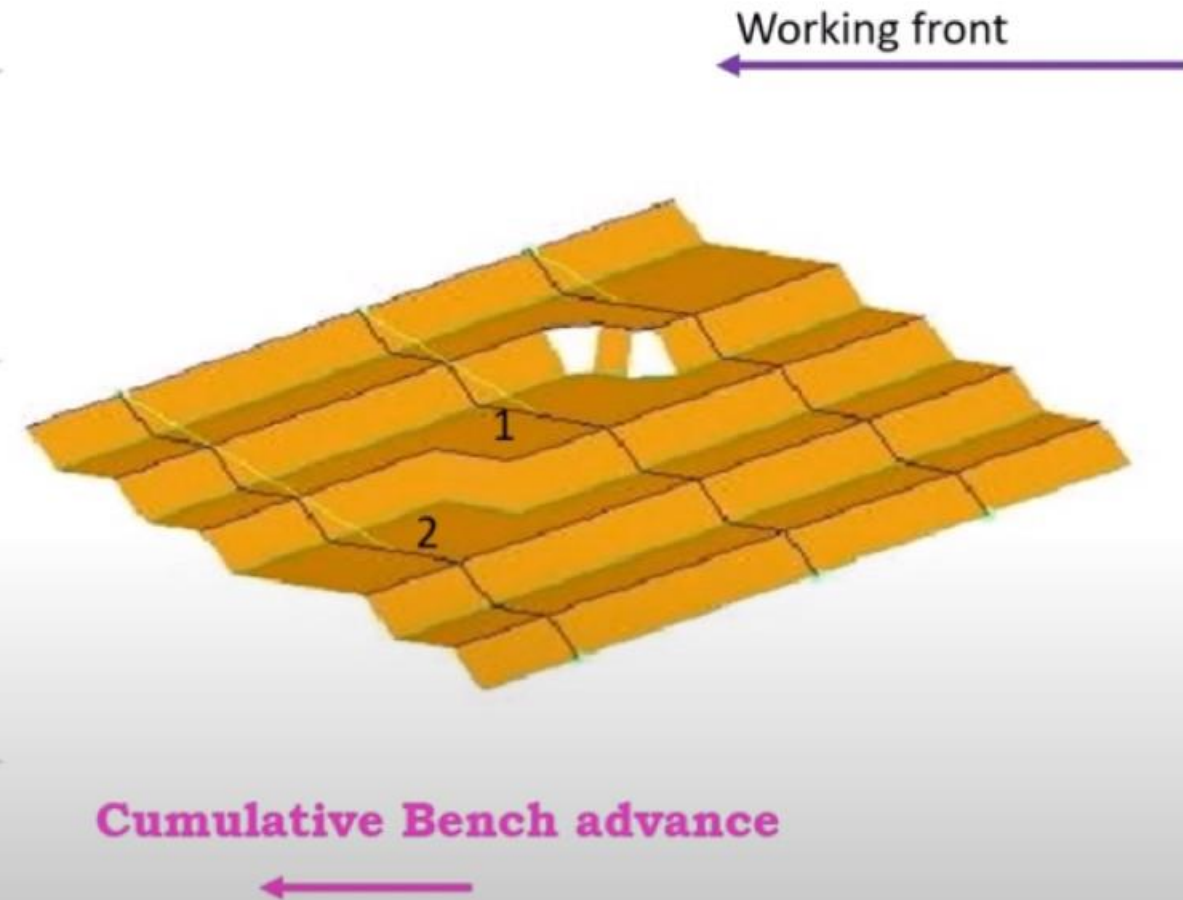
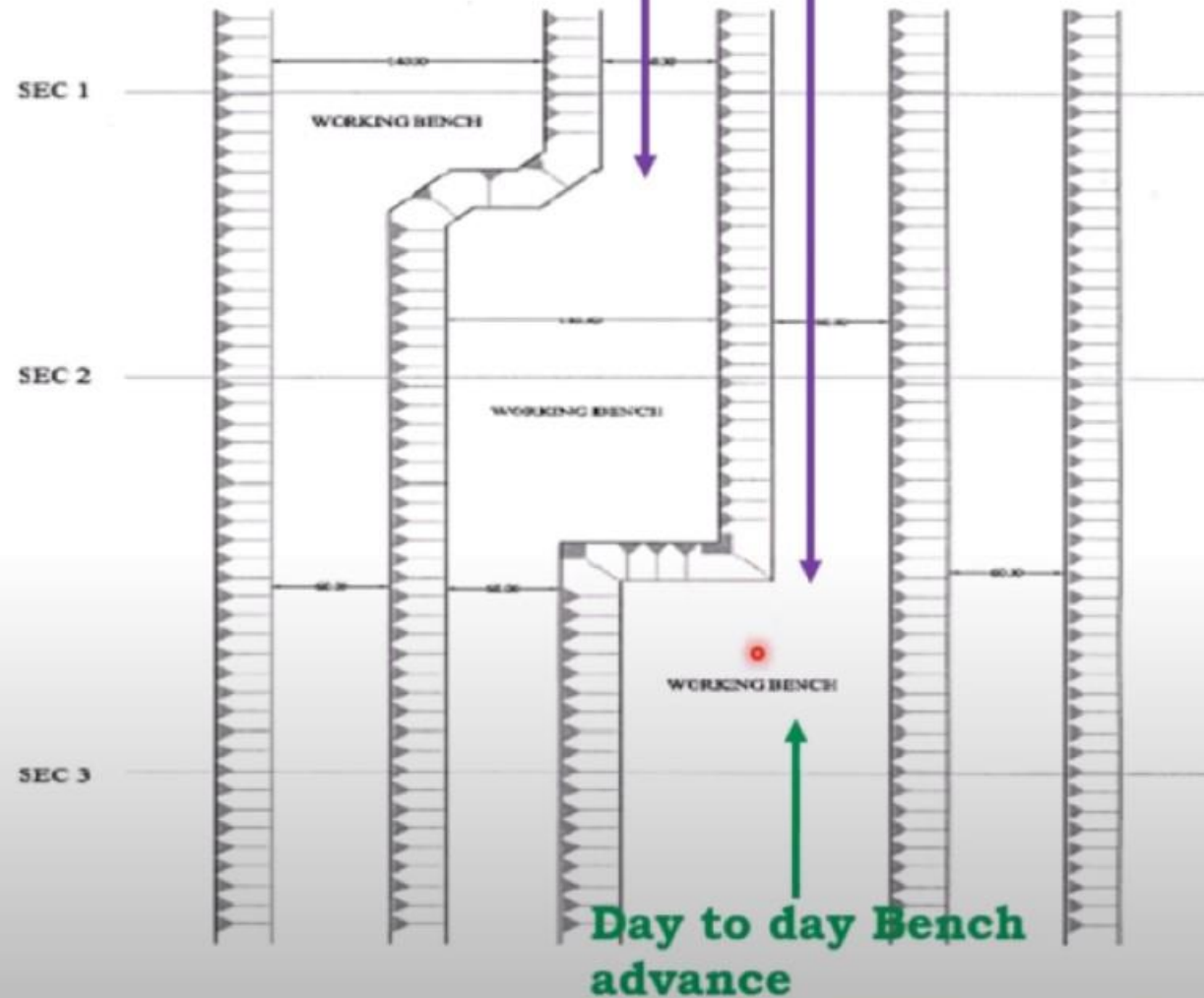


Day to day
Bench advance

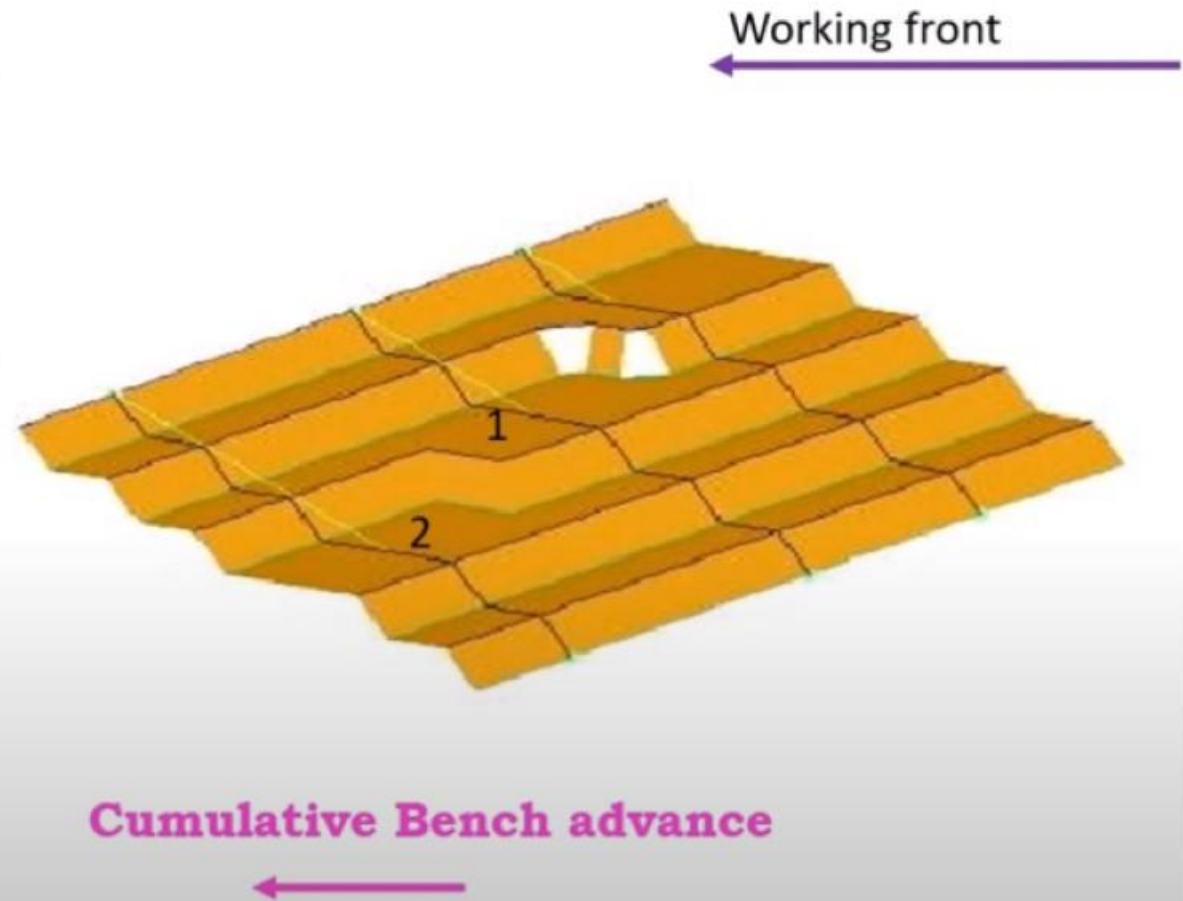
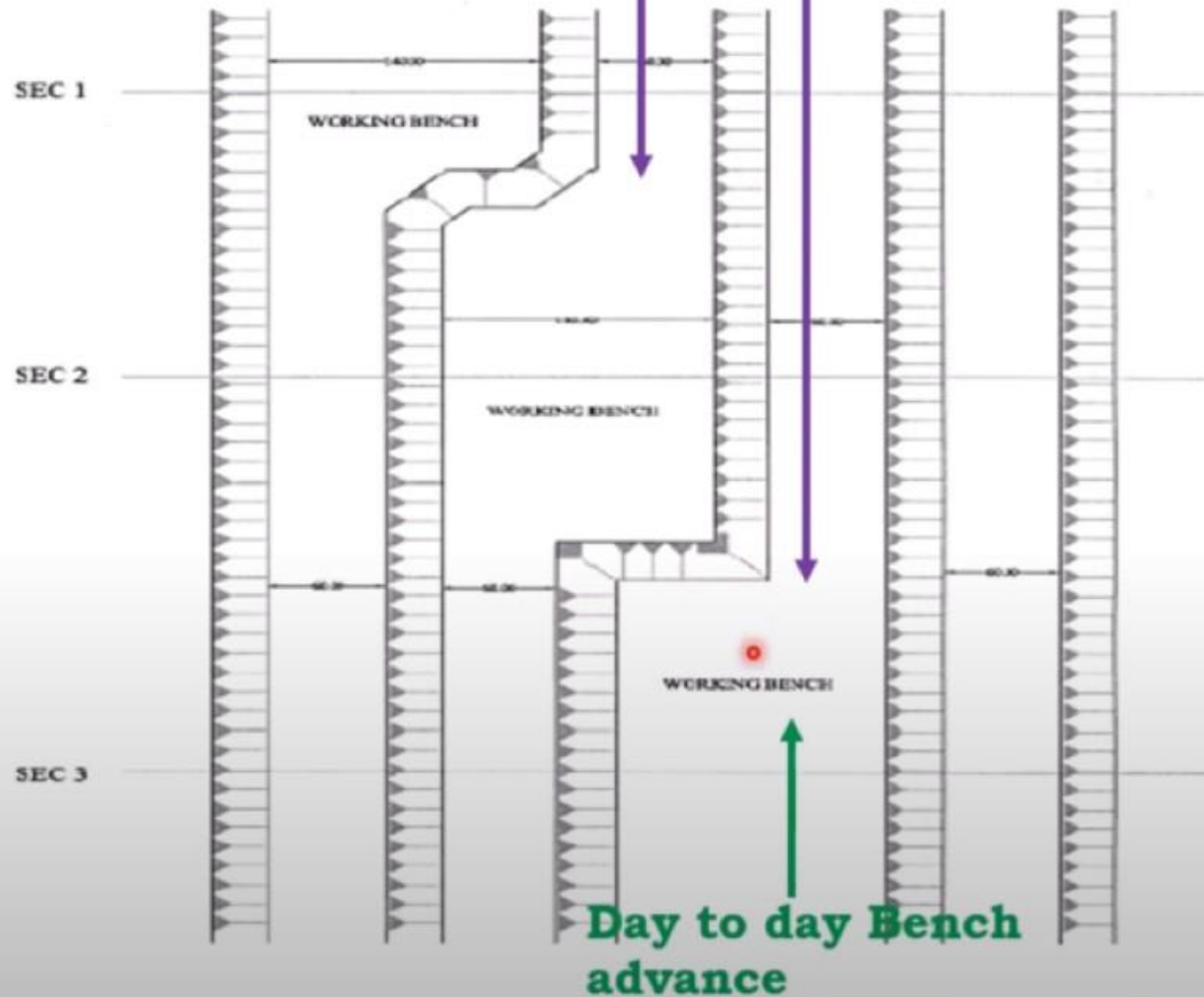
Cumulative
Bench advance



Two Working Benches



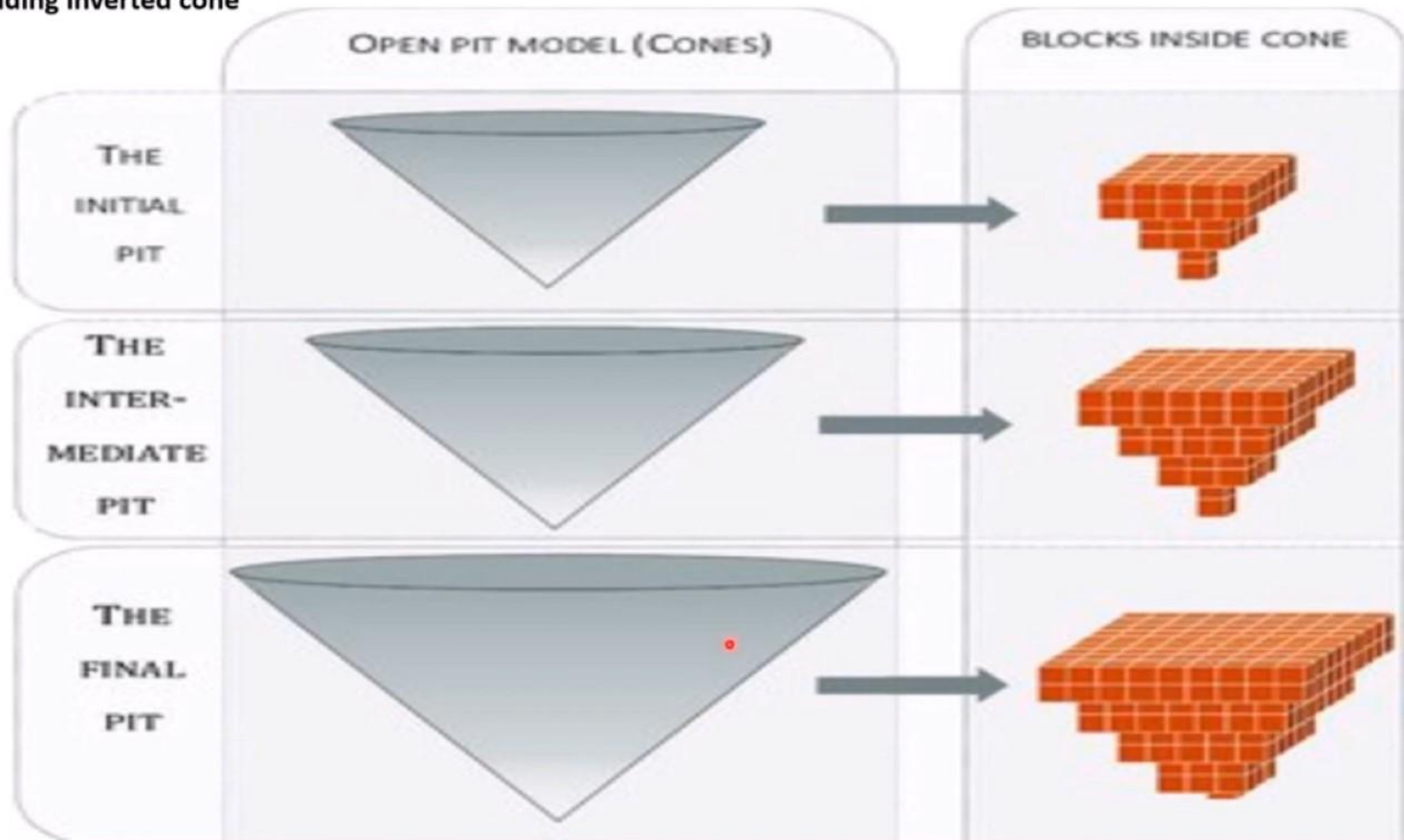
Two Working Benches



PIT LIMITS

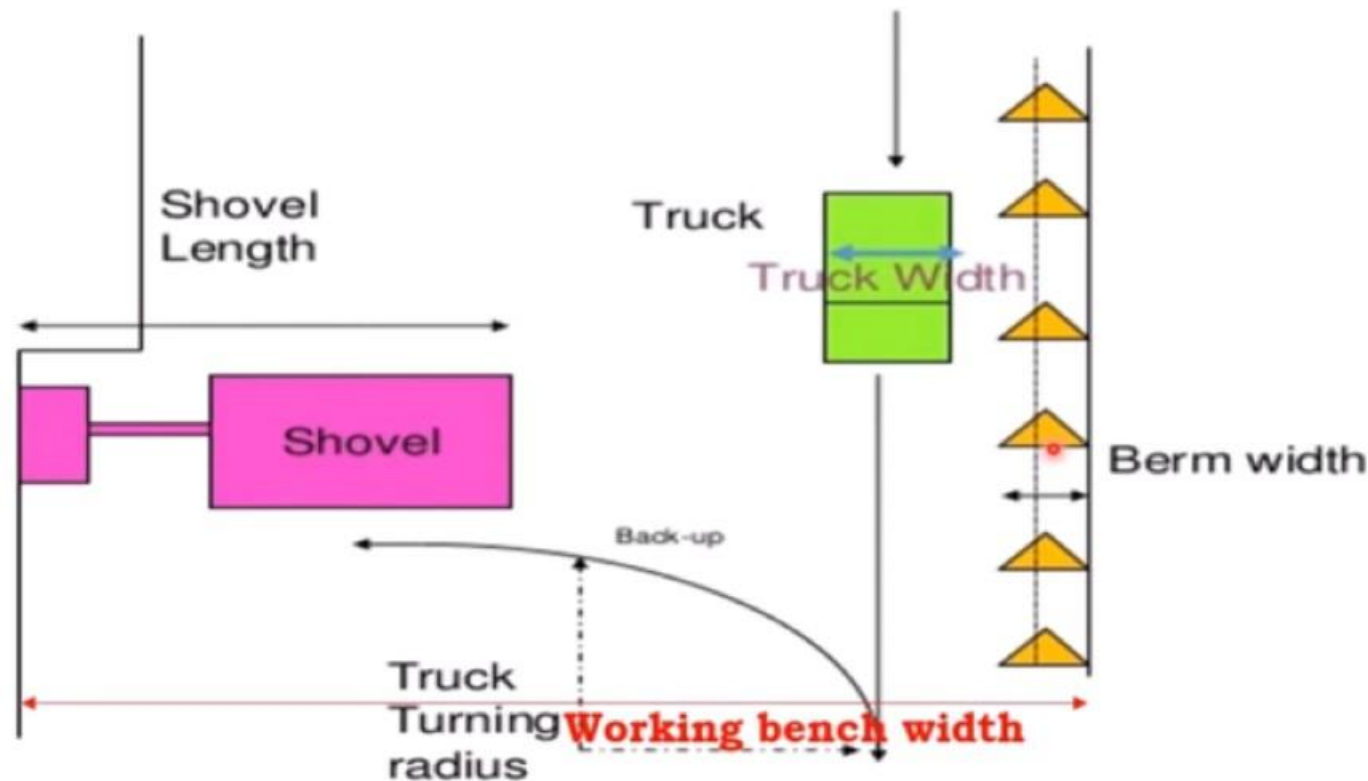


Expanding inverted cone



Working bench

Geometry of a Working Bench

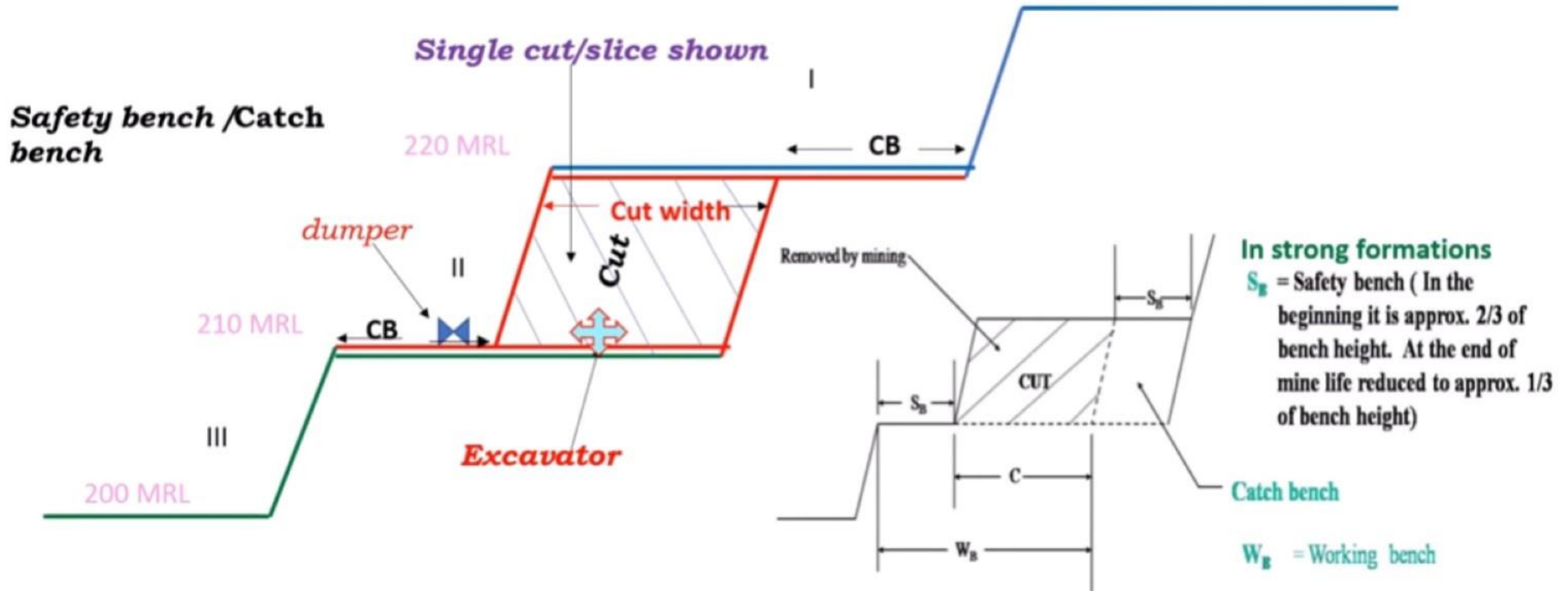


A bench big enough
To accommodate
Equipment working is
Much wider than one
Only intended to catch
Rolling rocks.

Working bench is the bench under operation in an active pit where dumpers and excavators ply & that has not reached UPL. Their width is always more than bench height. It is wider than Closing bench.

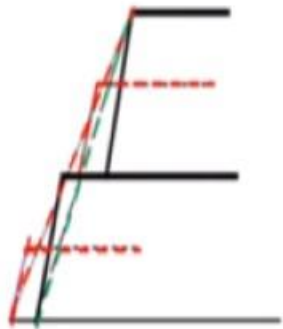
Safety bench /Catch bench/Berm

After the removal of cut / slice the remainder of the bench is called the safety bench or catch bench.



Safety bench/Berm/Catch bench /Closing bench

Horizontal ledges situated on the non-productive side of the open pit and serving to accommodate the transportation roads or to protect the operating equipment from falling lumps of ground.



Double benches at final pit limits.

Avoiding intermediate catch benches facilitates steepening the overall pit slope angle

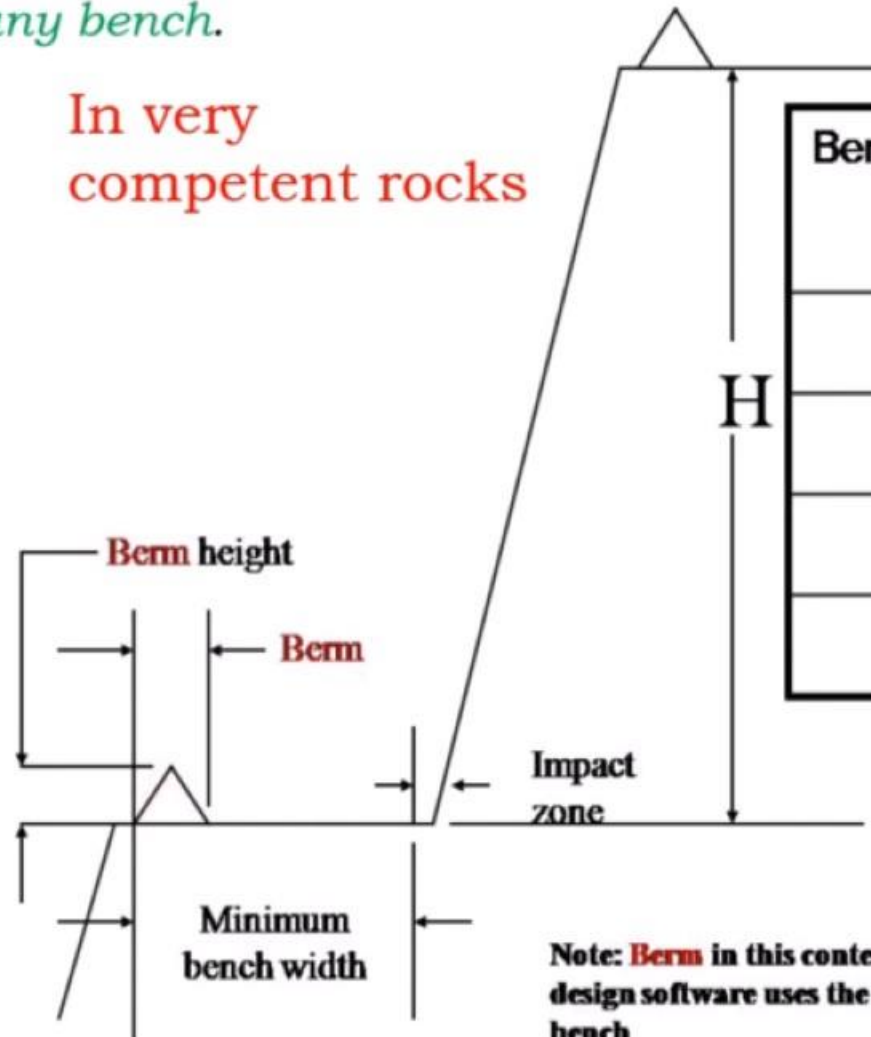
Closing bench :It is the bench on the quarry side that has reached the (UPL), Their purpose is mostly to catch the falling/rolling rocks. The width of such benches vary from 65% to 70% of the height of the bench. *However, at the end of the life of the pit, it reduces to 30% to 35% of the height of the bench.* Some times they are formed by merging two benches into one.



PIT LIMITS

Safety Berm/ Windrow/Bund/Berm It is made of loose rock or ore all along the crest to protect the operating equipment from falling off the ground from edge of a *dump terrace* , *haul road* or *any bench*.

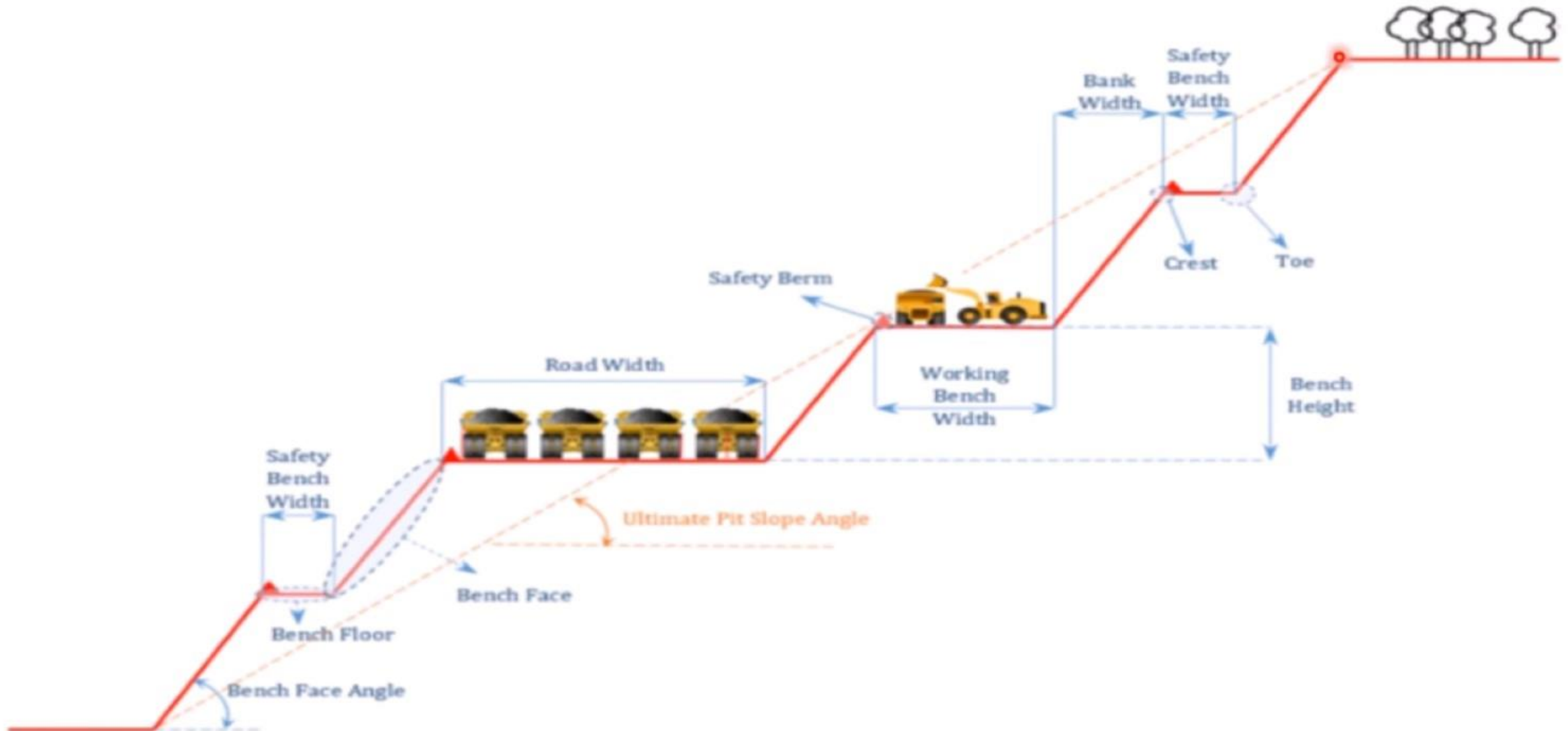
In very competent rocks

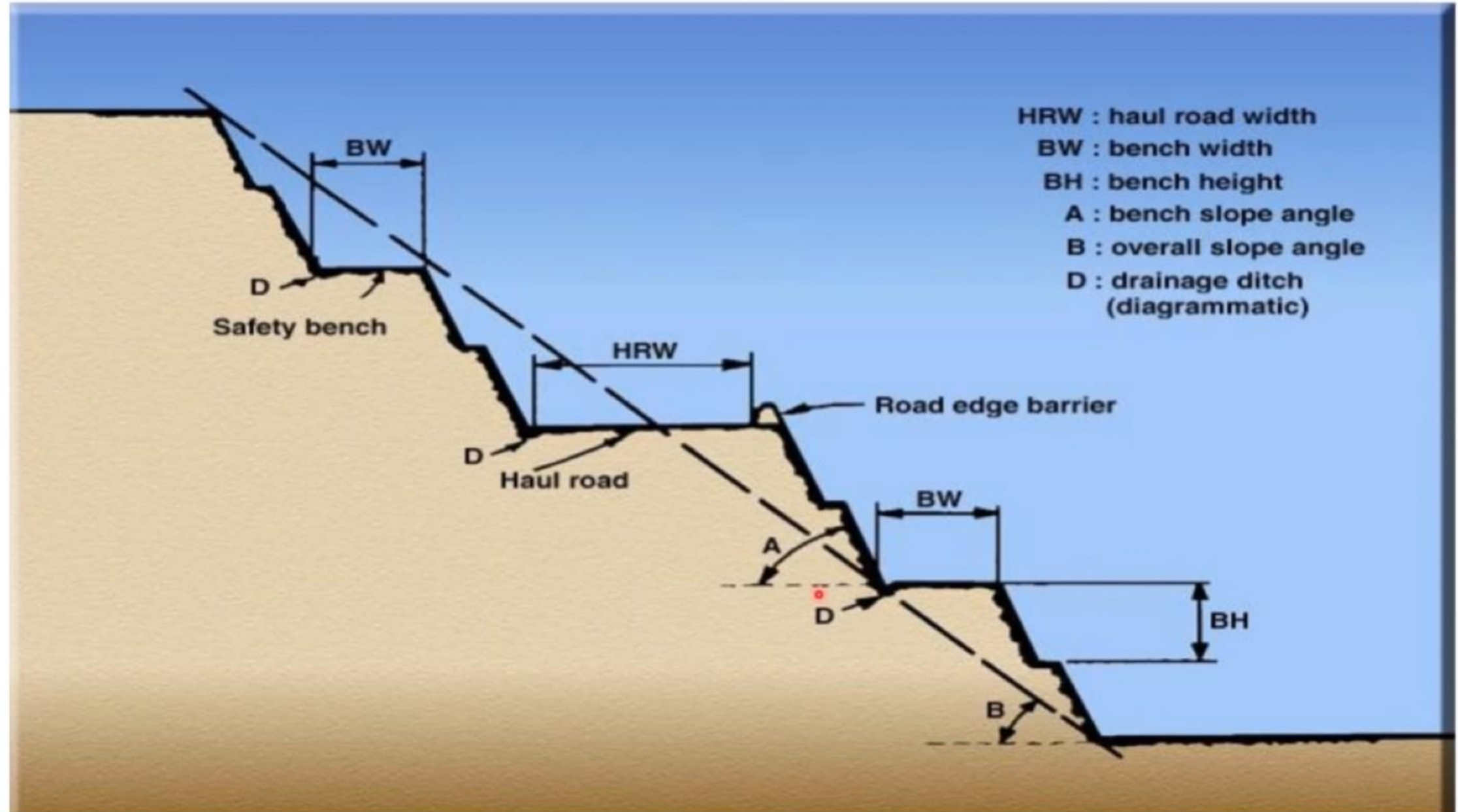


Bench height m	Impact zone m	Berm height m	Berm width m	Minimum bench width m
15	3.5	1.5	4	7.5
30	4.5	2	5.5	10
45	5	3	8	13

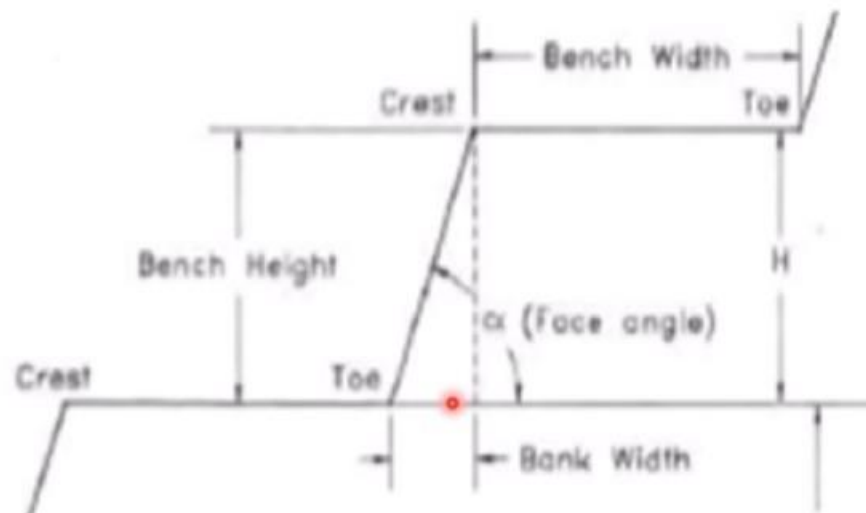
Note: **Berm** in this context is strictly a safety berm. Many mine design software uses the term “berm” as safety bench or catch bench

- Berm facts:**
- **Berm** height is usually greater than or equal to 1/2 tire radius.
 - **Berm** slope is normally 35°





Bench height is the vertical distance between the crest and toe of a bench. Height of the bench is governed by **strength** of the rock. Other governing factors are the **drilling length** of equipment, production requirement, grade variation, overall pit slope, stripping ratio, geological disturbances, hydrological condition & digging height of excavator etc. Higher the benches fewer working levels to maintain but lesser stability.



As specified by the Directorate General of Mines Safety (DGMS), it shall not be more than the -

- maximum digging height of the excavator excavating the bench in case of hard formation
- maximum digging height of the excavator excavating the bench + 3m in case of soft formation.

From the operational point of view, the bench height is decided based on the optimum digging height of the excavator for efficient **operations.**

In loose rock it is 1.5 m and harder rock it may go up to 10 m or even more with DGMS permission..

As per Regulation 98 of CMR 1957,

In alluvial soil etc.

- Sides shall be sloped at 45° or Benched – height not more than 1.5m
Width – not less than height

In hard ground

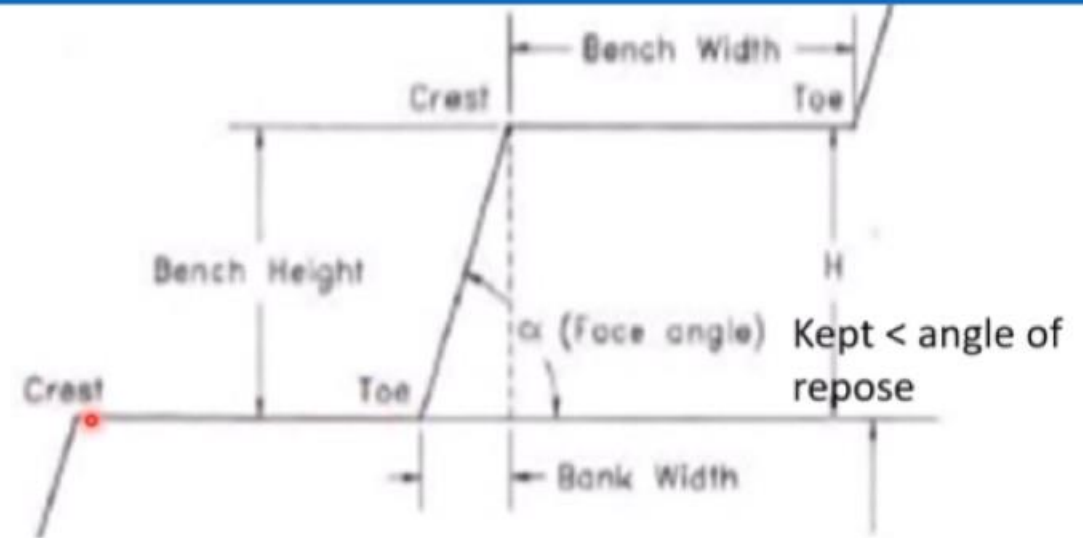
- Sides adequately benched/sloped so as to prevent danger from fall of sides

Coal

- Sides shall be sloped at 45° or Benched – height not more than 3m

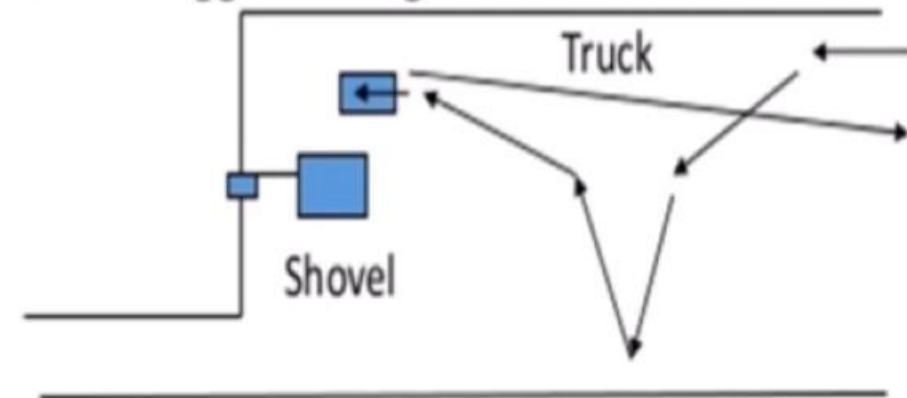
However, exemption can be sought from Inspectorate.

Bench width It is the *exposed to air* horizontal distance between the crest and the *toe* of a bench . *As specified by the DGMS working bench shall be more in width than the bench height, shall be more than the maximum width of the largest machine +4m, Shall be more than 5 times in two way traffic and more than 3 times in one way traffics of the width of the widest plying machine.*



Flat area on benches provides room for equipment to move

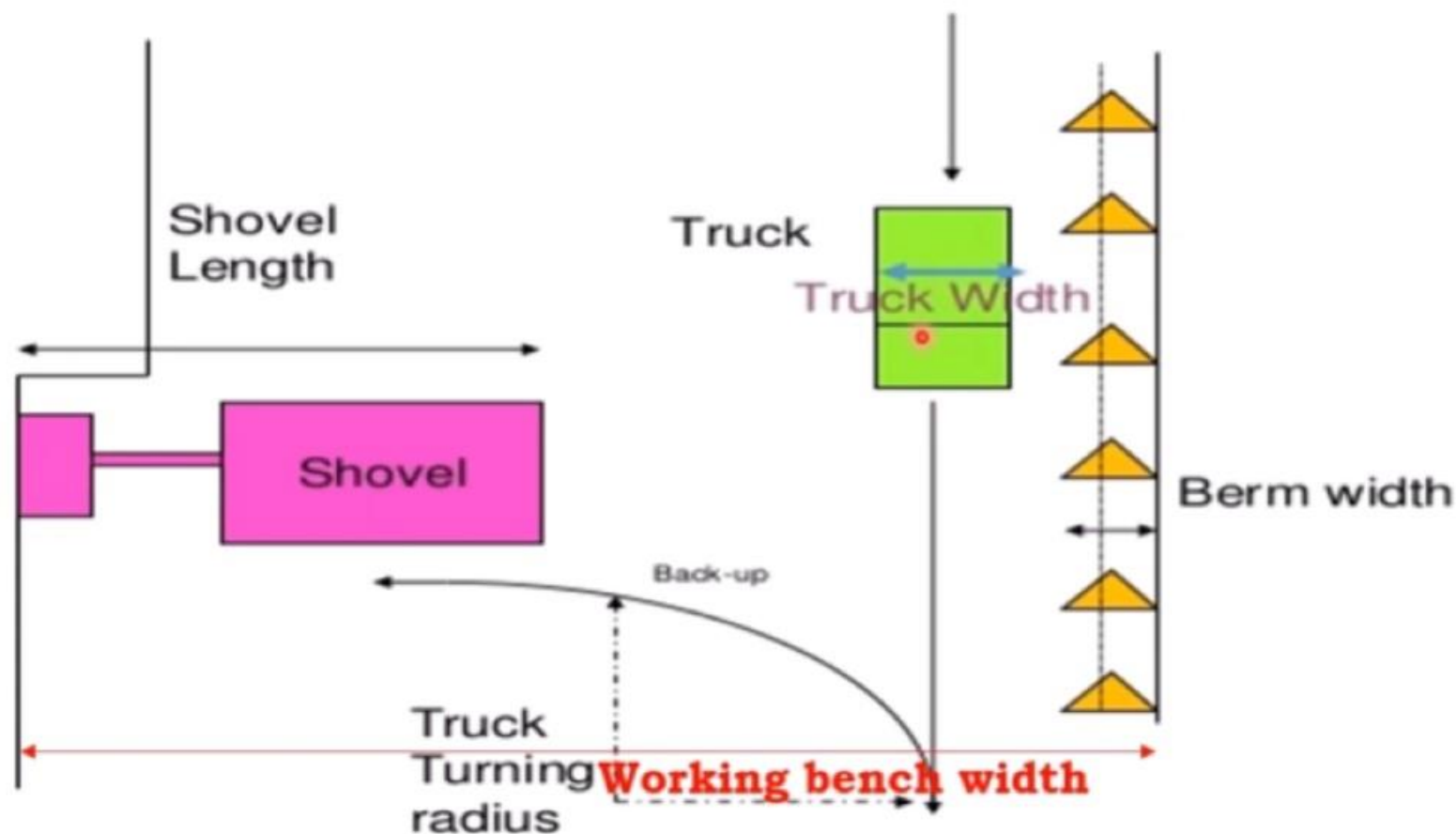
➤ Bigger trucks have bigger turning radius



Plan view of bench work area

Working bench width

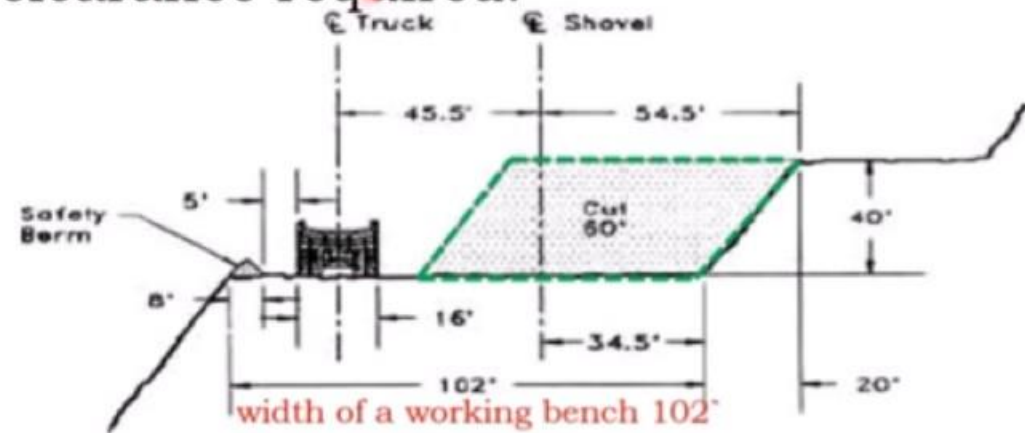
Geometry of a Working Bench



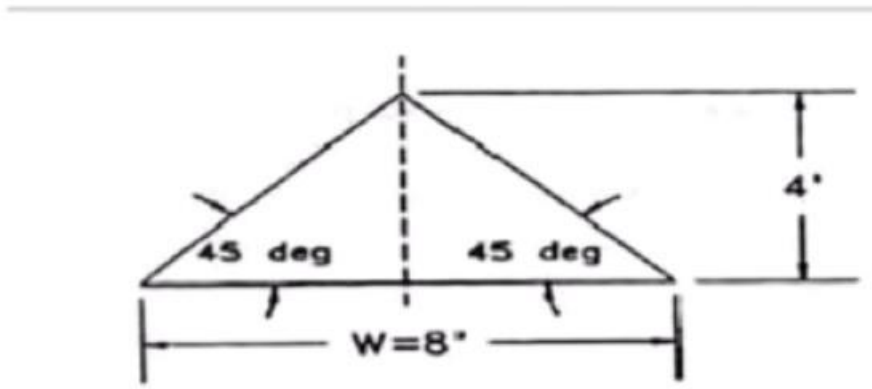
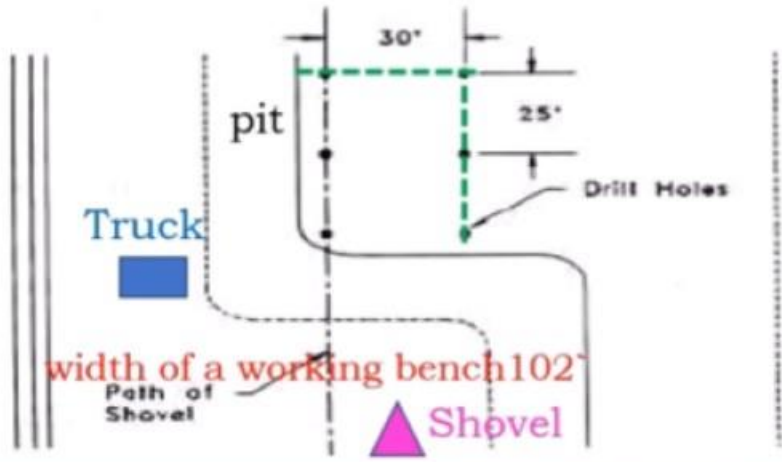
A bench big enough
To accommodate
Equipment working is
Much wider than one
Only intended to catch
Rolling rocks.

PIT LIMITS

From the **operational** point of view, minimum **width of a working bench** should be equal to **$0.8 \times \text{bench height} + \text{dumper width} + \text{shovel dimension} + 4\text{m}$** . Others factors are width of the blasted muck pile and safe/efficient clearance required.



Section and plan views through a working bench.



Simplified presentation of a safety berm.

A typical set of bench height, bench width and bench slope are given below:

Mineral	Bench Height (m)	Bench Width (m)	Bench Slope (°)
Copper	12-18	24-38	50-60
Iron	9-14	18-30	60-70
Non-metallic	12-30	18-45	50-60
Coal	15-23	15-30	60-70

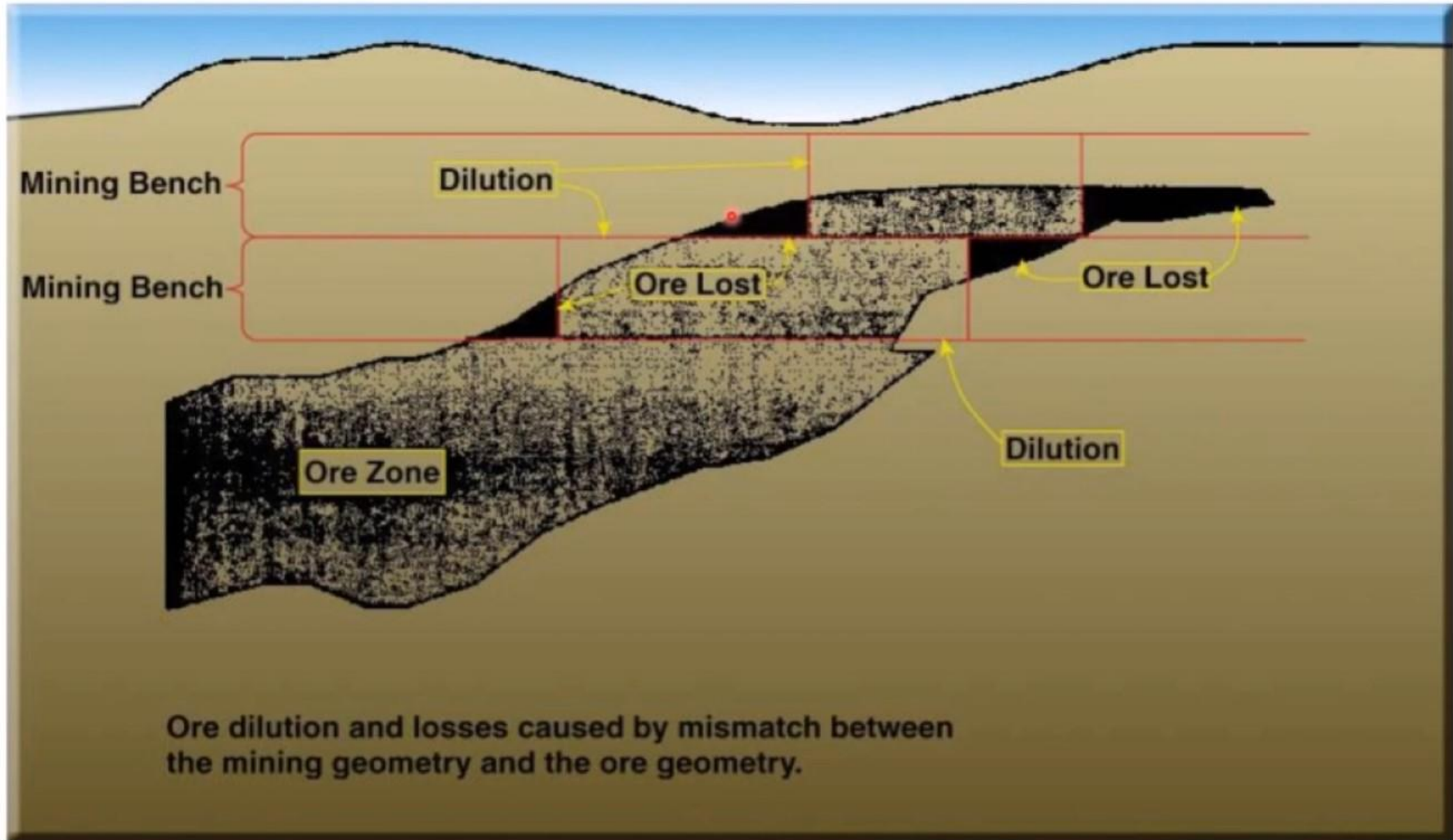
Selection of Bench parameters

Higher and wider benches offer :

1. Less selectivity (dilution and loss)
2. Fewer working places thus less flexibility
3. Flatter slopes: As large machines need wide benches
4. Fewer equipment relocations & set-ups
5. Higher productivity at lower unit cost
6. Improved control and supervision

How High Should a Bench Be ?

1. Deposit character and geology: selectivity
2. Production strategy: ore/waste ratios, blending requirements, no. of working faces, operating/capital costs, etc.
3. Slope stability considerations
4. Equipment set / equipment specific optimum geometry



Reduced Bench Heights

Advantages

1. ramp volume reduced (fill ramps)
2. contour areas easier to drill & blast
3. ore grading and selective mining is easier
4. multi-row blasts easier to blast
5. single pass drilling easier

Disadvantages

1. poor bench grade control by shovel is accentuated
2. reduced drilling yield and increased drill and blast cost
3. sub grade costs more (relatively larger)
4. reduced shovel productivity

Slope Design: Conclusions

1. Maximize the height of the benches
 - * at the expense of mining selectivity?
2. Minimize the width of working benches
 - * at the expense of productivity?
3. Minimize the number & width of catch benches
 - * at the expense of safety?
4. Minimize the width & number of haul roads
 - * at the expense of productivity & safety?

False benching

Lease boundary

**False benches made
of Rock /ore**

Q The feasibility study of an opencast mine revealed the following:

Recoverable value per tonne of ore = Rs. 900

Production cost per tonne of ore = Rs. 550

Stripping cost per tonne of waste = Rs. 45

The Breakeven stripping ratio will be

a) 7.77 b) 32.20 c) 1.78 d) 12.22

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Thank You

Queries??

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